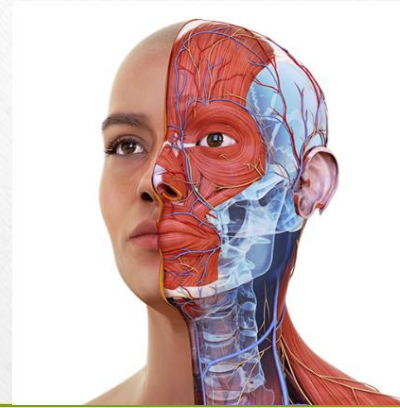
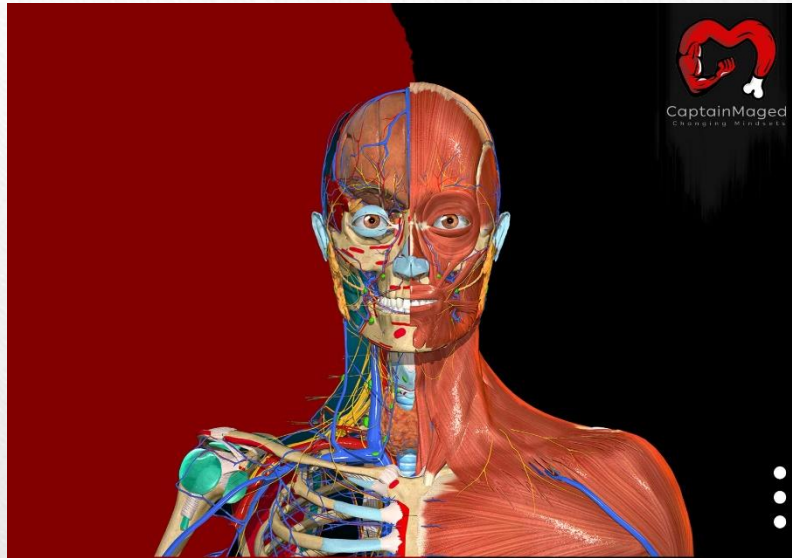
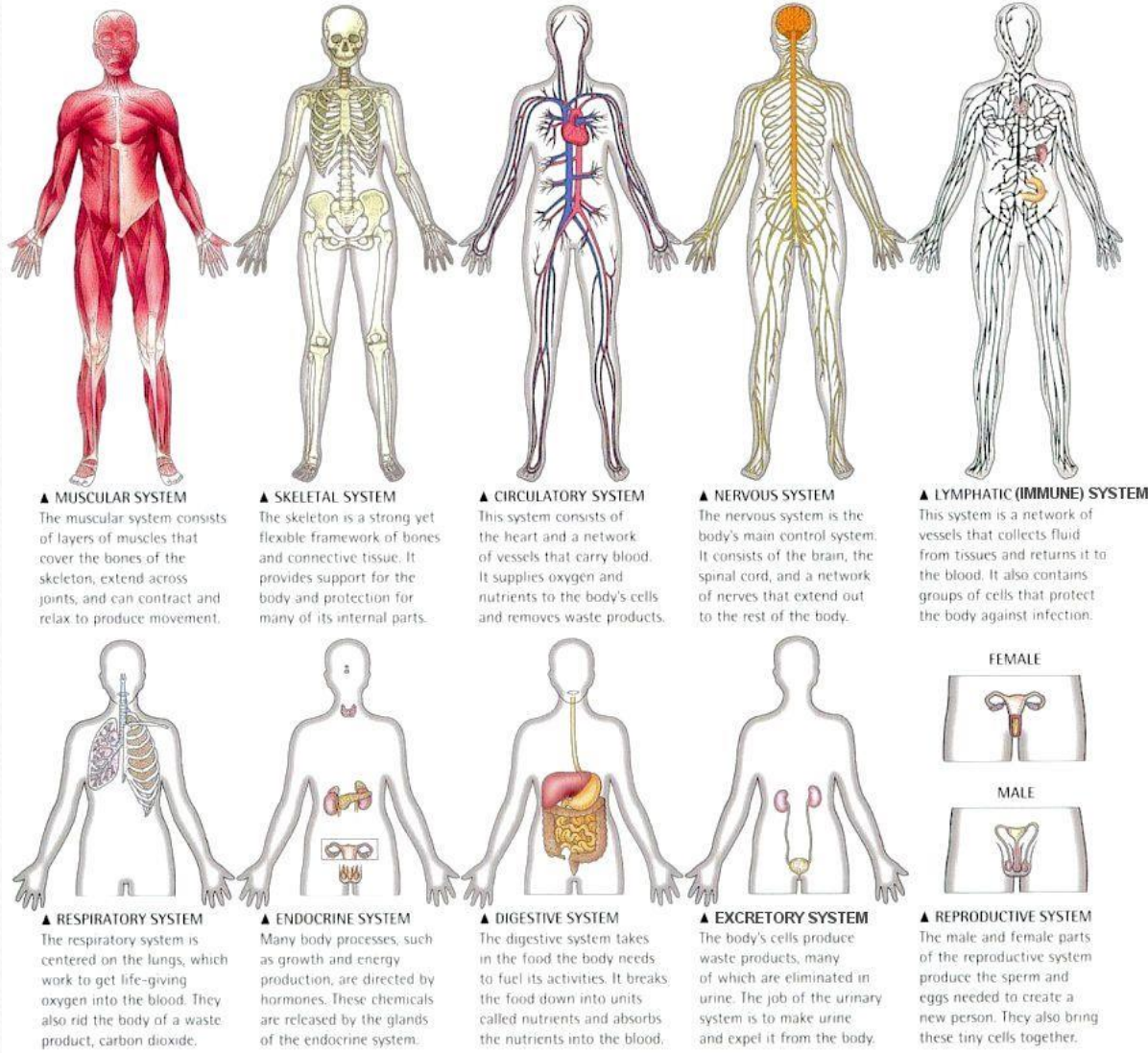


Anatomy



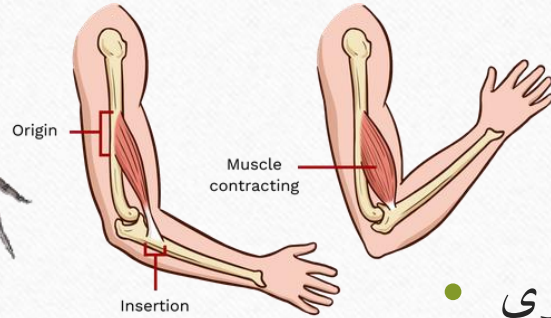
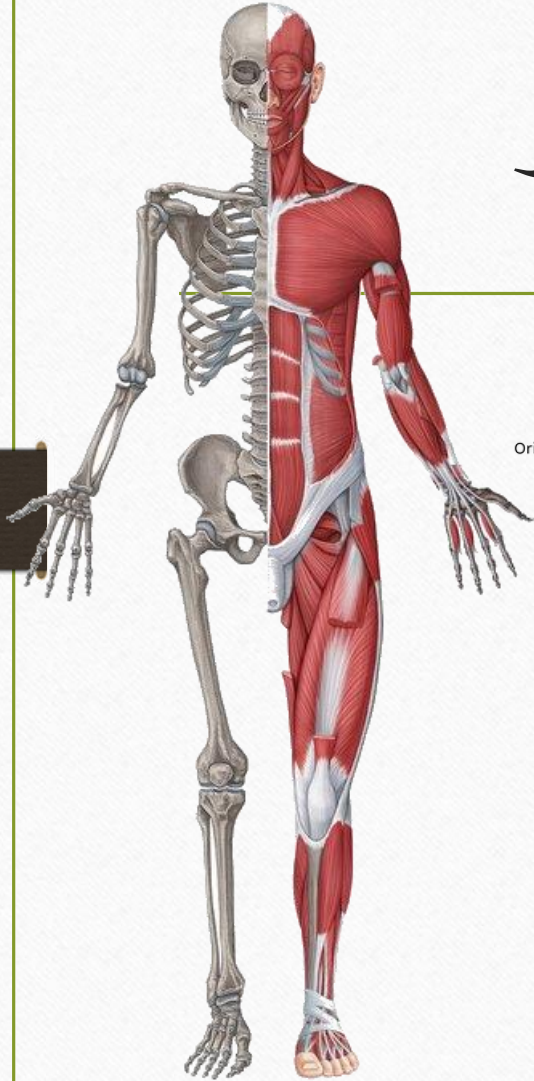
Captain Maged

انظمة الجسم Body systems



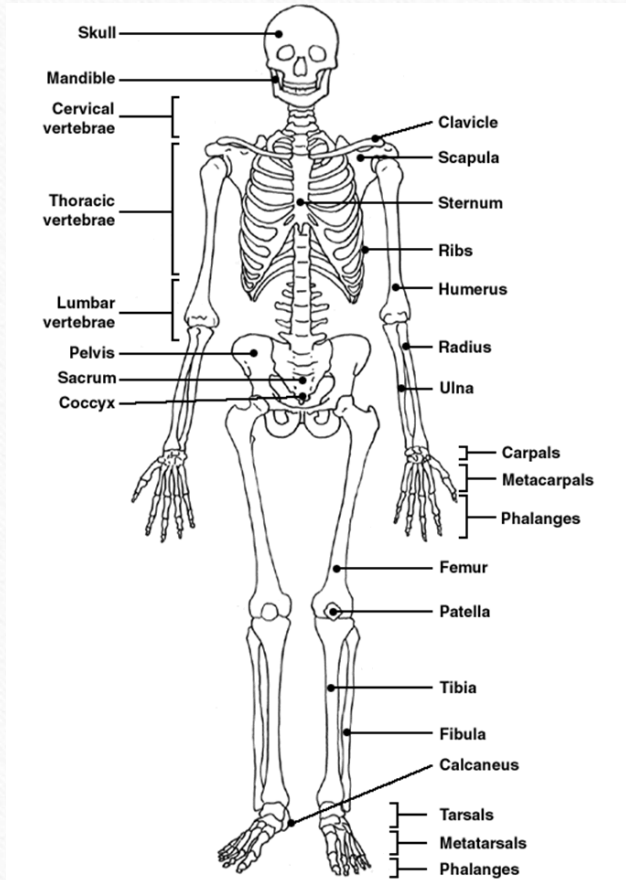
ما هي انظمة الجسم
وظيفتها و فائدتها

كل الانظمة تعتمد على بعضها الاخر



- حتى تحدث الحركة فيجب ان تتصل العضلات بالعظام
 - يجب ان يكون لكل عضلة بداية و نهاية
 - Origin & insertion
 - يجب ان تكون العضلة بدايتها في عظمة و نهايتها في عظمة اخرى
 - و لهذا يطلق على النظامين معا باسم الجهاز العضلى الهيكلى
 - Musculoskeletal system : **bone, joints, connective tissue**
Tendons, Cartilage and ligaments - Muscles
- العظام والمفاصل والانسجة والعضلات و الاوتار و الاربطة و الغضاريف

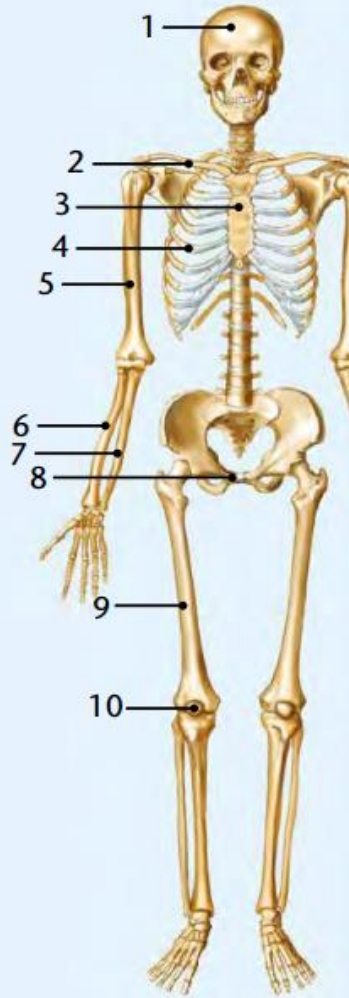
اهمية الهيكل العظمى Skeletal system



- الهيكل العظمى مكون من 206 عظمة ما هو اهميته ؟
- يعمل كالرافعات لنقل القوة
- الهيكل العظمى يحمى الاعضاء والجسم مثل المخ و قلب و الرئتين الخ.
- هو خزين للمعادن مثل الكالسيوم و الفسفور و غيره
- انتاج خلايا الدم الحمراء من نخاع العظام
- ربط الانظمة ببعضها

FRONT

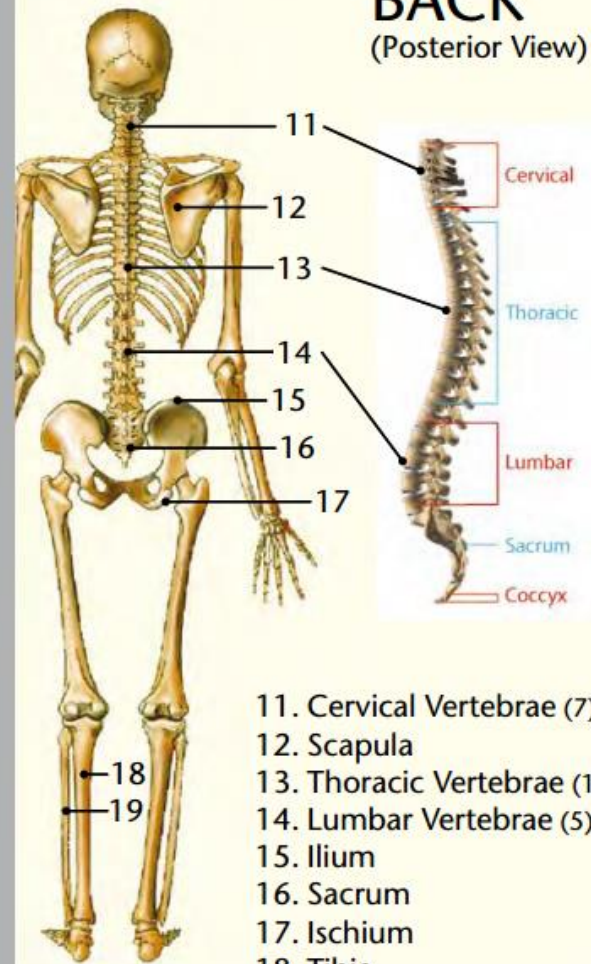
(Anterior View)



1. Cranium
2. Clavicle
3. Sternum
4. Rib
5. Humerus
6. Radius
7. Ulna
8. Pubis
9. Femur
10. Patella

BACK

(Posterior View)



11. Cervical Vertebrae (7)
12. Scapula
13. Thoracic Vertebrae (12)
14. Lumbar Vertebrae (5)
15. Ilium
16. Sacrum
17. Ischium
18. Tibia
19. Fibula

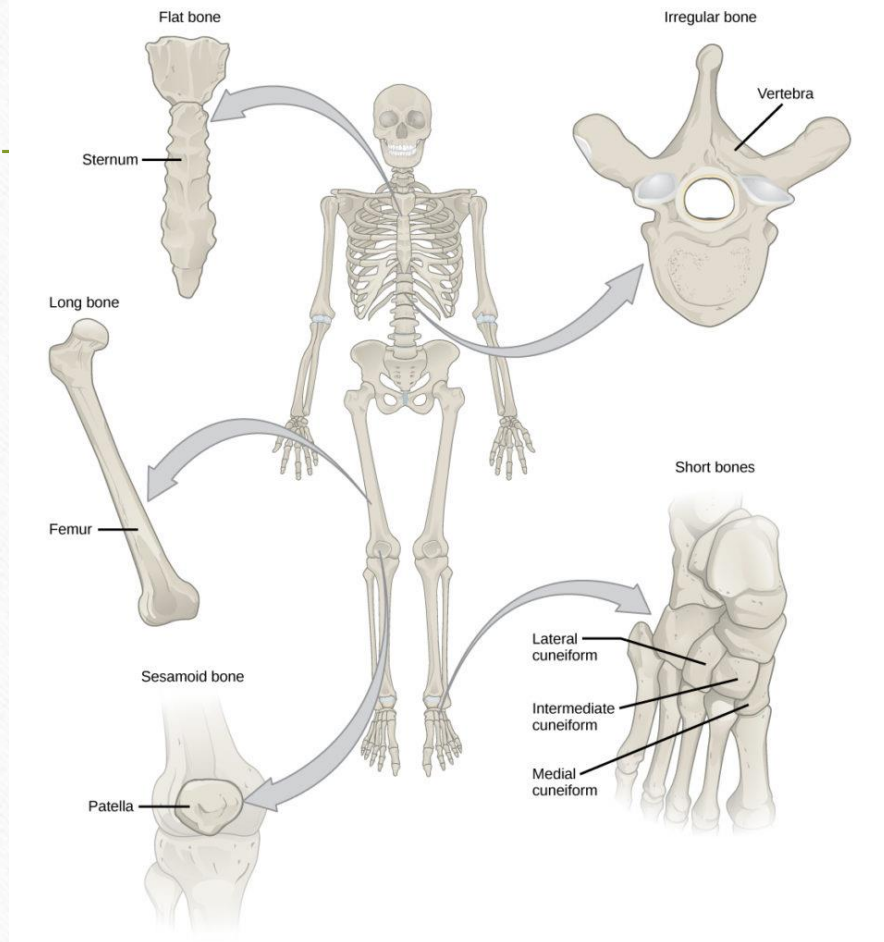
Skeleton

- حفظ جميع اسماء العظام
- عصام هيساعدنا

حد حفظ 10 منهم على الاقل ؟

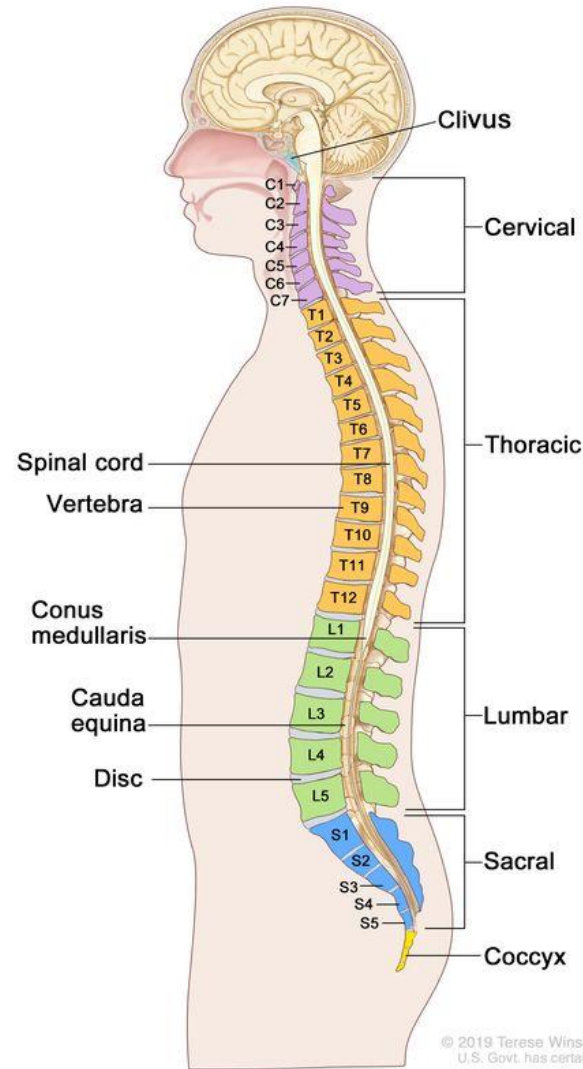


تنقسم الـ 206 عظمة الى 5 انواع مختلفين فى الشكل والحجم



- 1- Flat bone عظم مسطح
- 2- Short bones عظام قصيرة
- 3- Long bones عظام طويلة
- 4- sesamoid bones
- 5- irregular bones عظام غير منتظمة

العمود الفقري

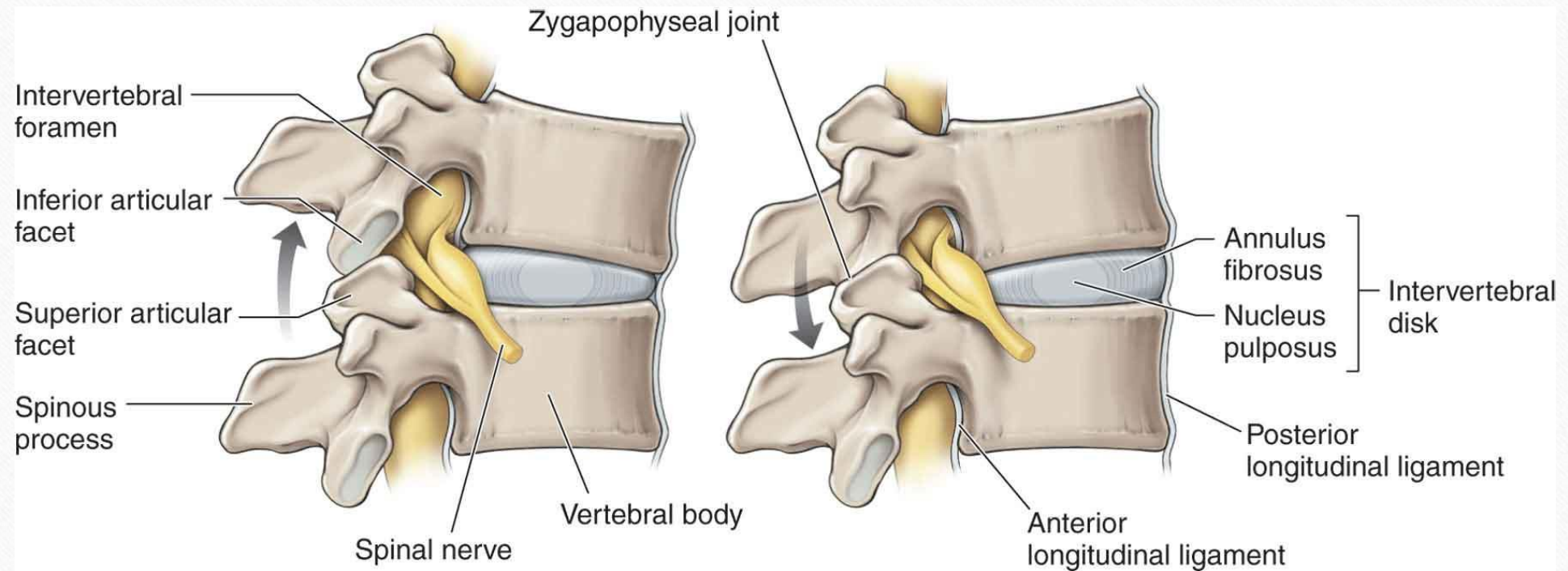


- يتكون من 33 فقرة
- مقسمين الى 5 مناطق
 - cervical spine (C1-C7),
 - thoracic spine (T1-T12),
 - lumbar spine (L1-L5),
 - sacral spine (S1-S5),
- and the tailbone (coccyx) 4 parts

منقسمين الى كام جزء و كام فقرة و اسماء الاجزاء دي ايه ؟

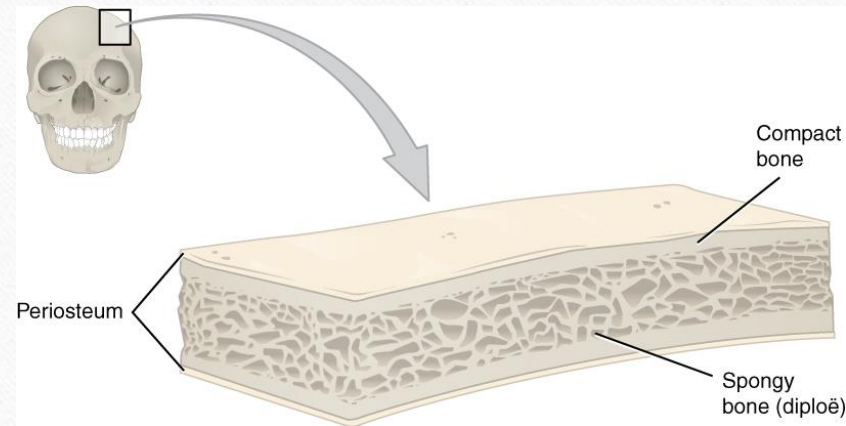
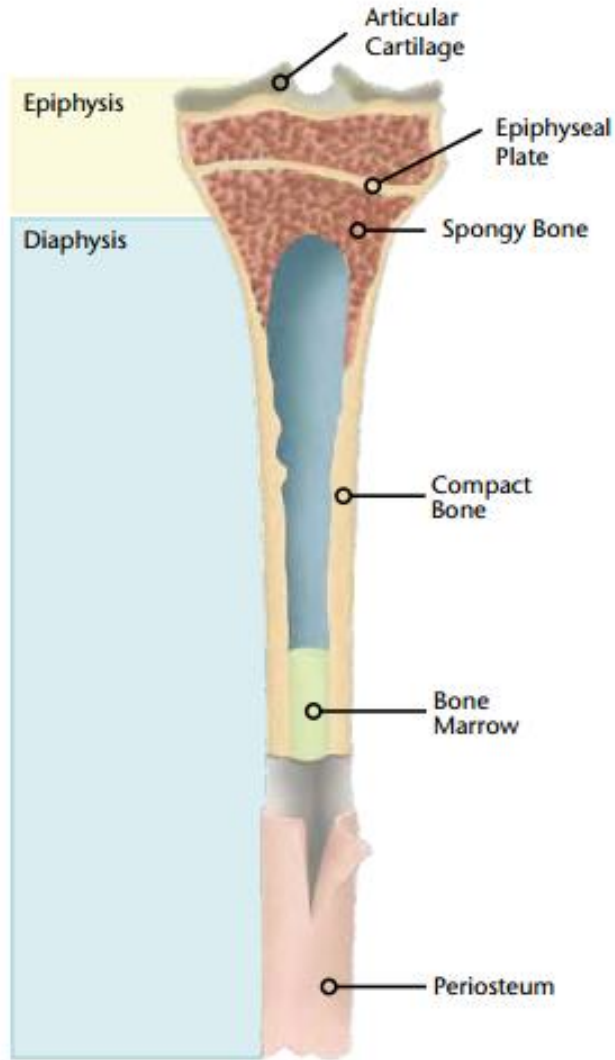


Normal spine segment

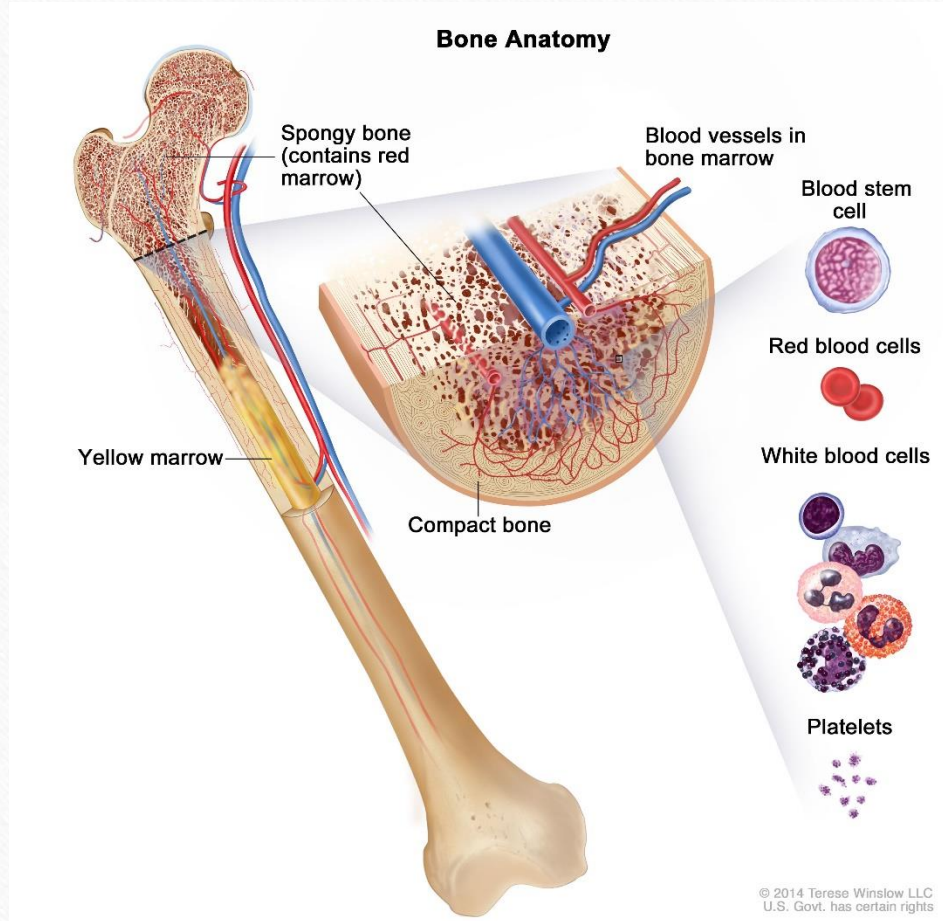


مكونات العظمة

- كل عظمة من الـ 206 عظمة مكون من 3 طبقات
 - 1- نخاع العظام Bone Marrow
 - 2- Compact bone
 - 3- Periosteum

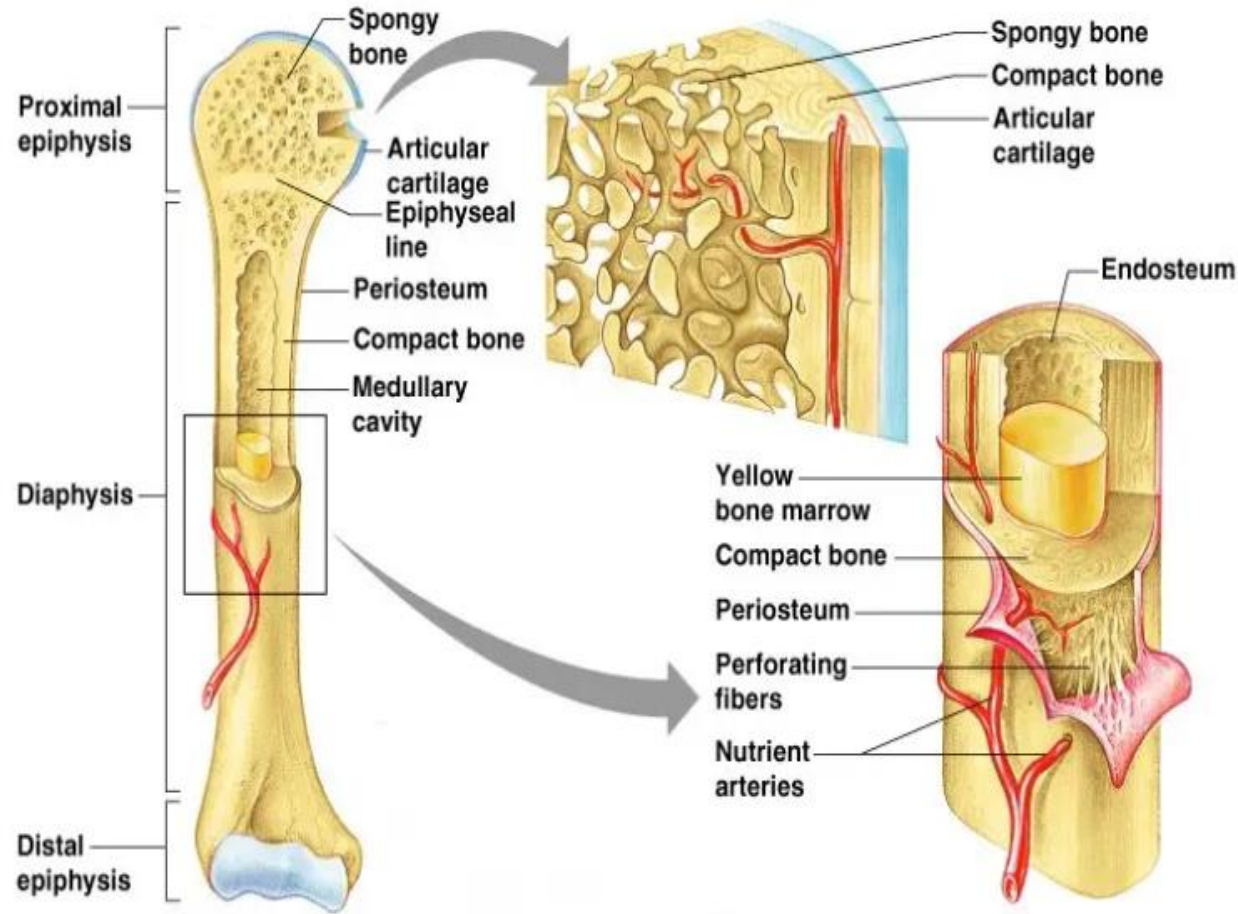


نخاع العظام Bone Marrow 1-



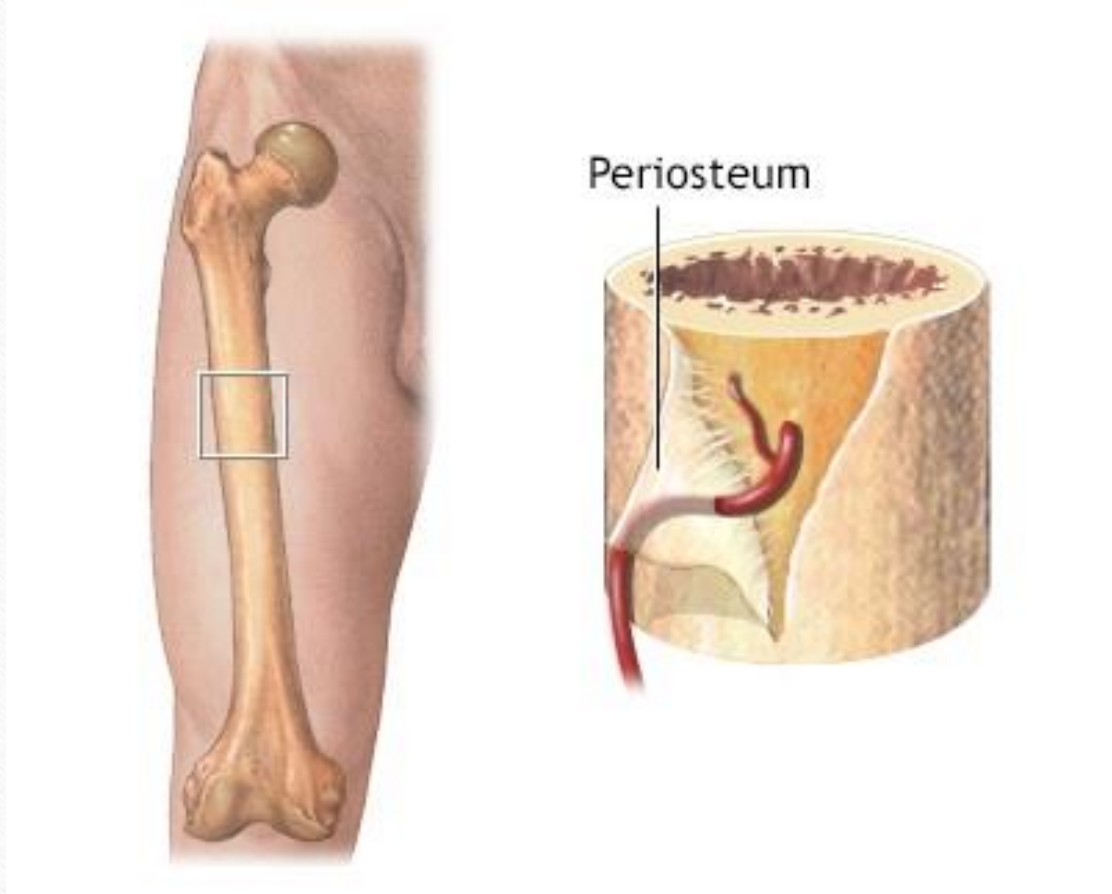
- ينتج نخاع العظام خلايا الدم الحمراء الذي يحمل Red blood cells الاوكسوجين
 - ينتج خلايا الدم البيضاء الذي يحارب العدوى White blood cells
- ينتج الصفائح الدموية التي بدورها تقف النزيف و انت تيجي تشيل القشرة زى الخنزير Platelets

2- Compact bone



- يحيط بنخاع العظام هذه الطبقة الثميكة
- تمر الاوعية الدموية و الاعصاب منها
- تمر المغذيات و الاكسوجين من خلالها
- هذه الطبقة ثميكة و تحتوى على الكالسيوم و المعادن و ايضا تعطى وزن للشخص و قوة للعظام

3- Periosteum



- تكون هذه الطبقة هي الطبقة الاخيرة
- تمثل الغطاء للعظمة مثل الجلد للجسم
- تتكون هذا الطبقة من خلايا التي تنتج العظام فتعمل على اصلاح الكسور و الشروخ فى الجسم

ايه هو ال3 طبقات و فائدة كل طبقة ؟



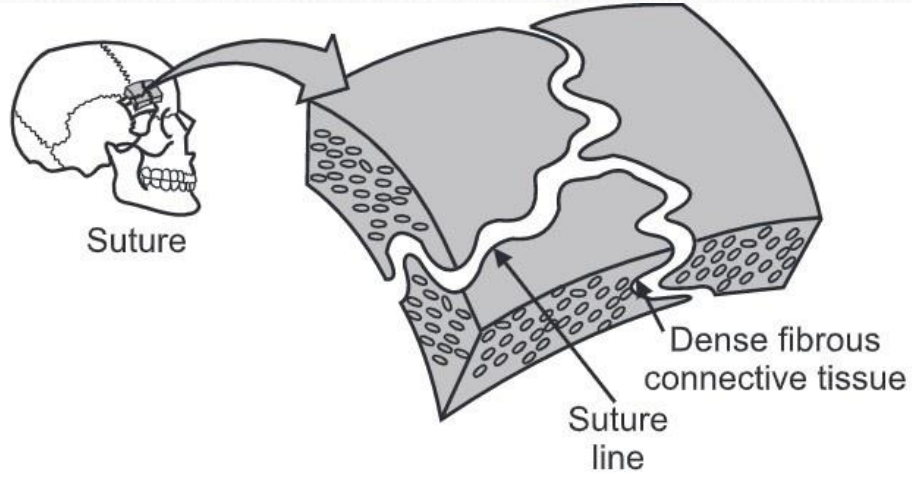
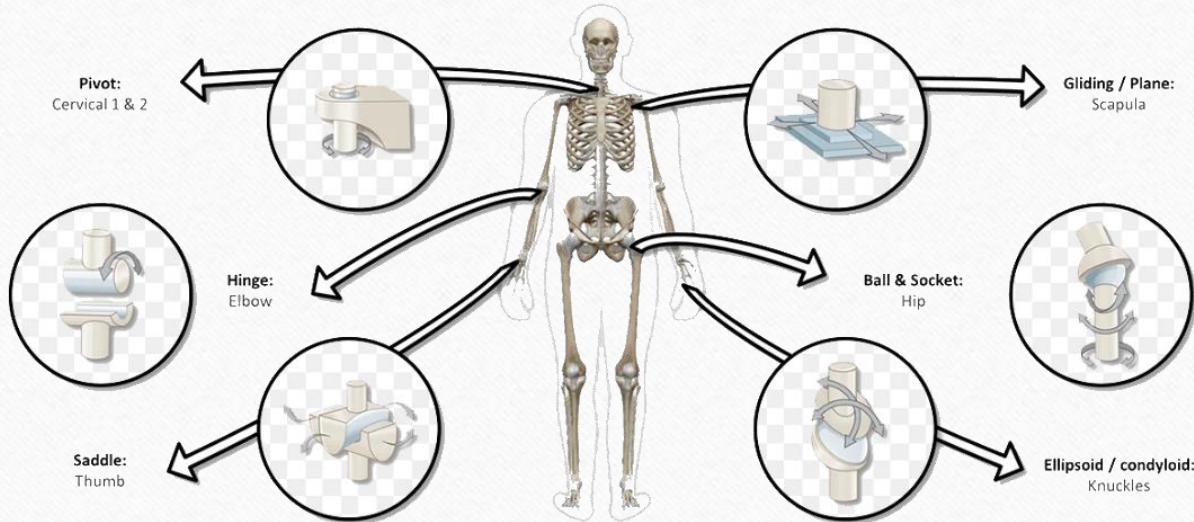


Fig. 1: Fibrous joint (Suture of the skull)

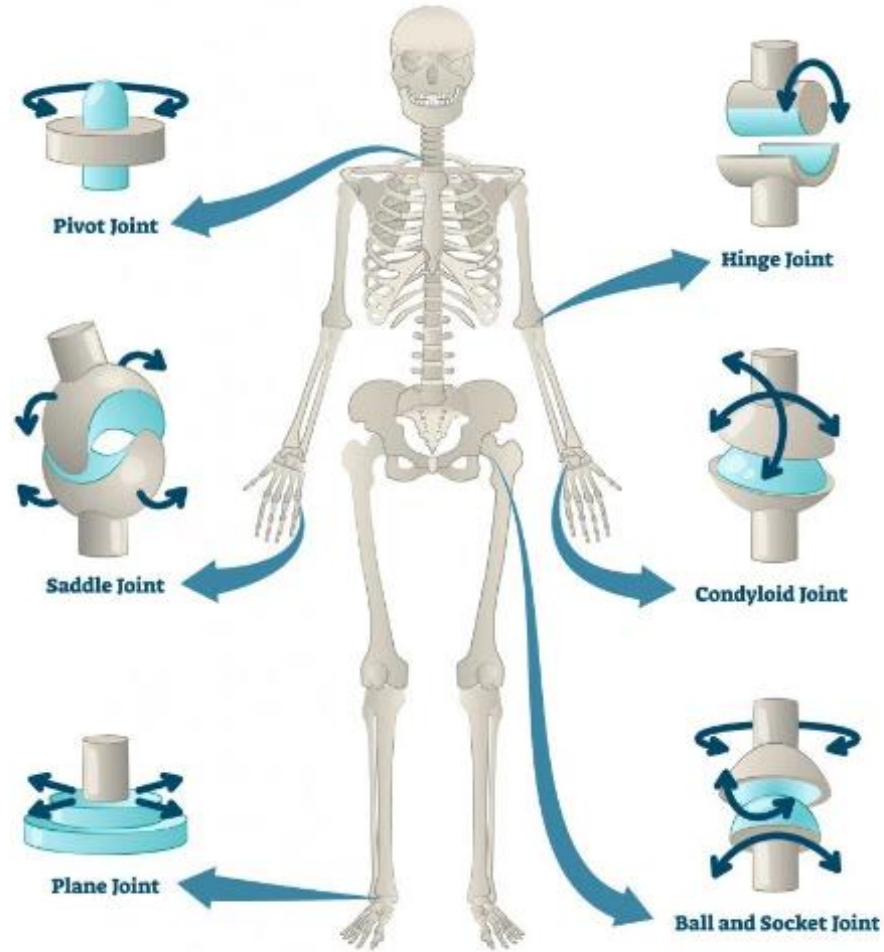
Types of synovial joints



المفاصل Joints

- توجد المفاصل عندما يتقابل العظام ببعضها الآخر
- يوجد تصنيفان للمفاصل واحد مفيش فيه فراغ زى الجمجمة
- وواحد فيه فراغات و انفصال زى باقى المفاصل و فيه السائل الزلالى و عبارة عن 6 مفاصل
- Synovial joints have synovial fluid

TYPES OF JOINTS

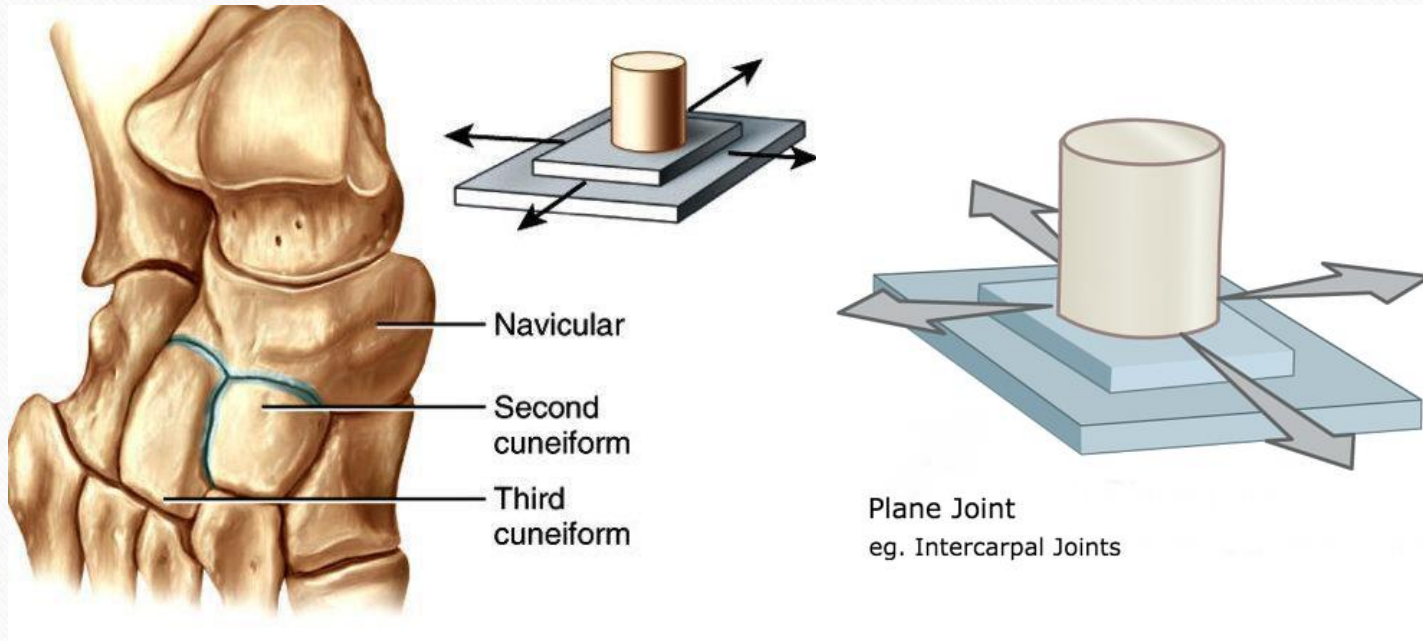


Types of Synovial joints

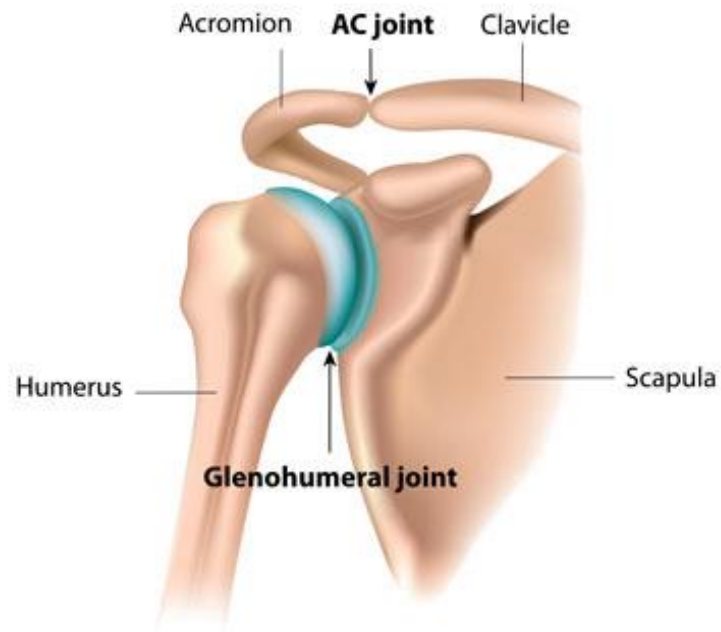
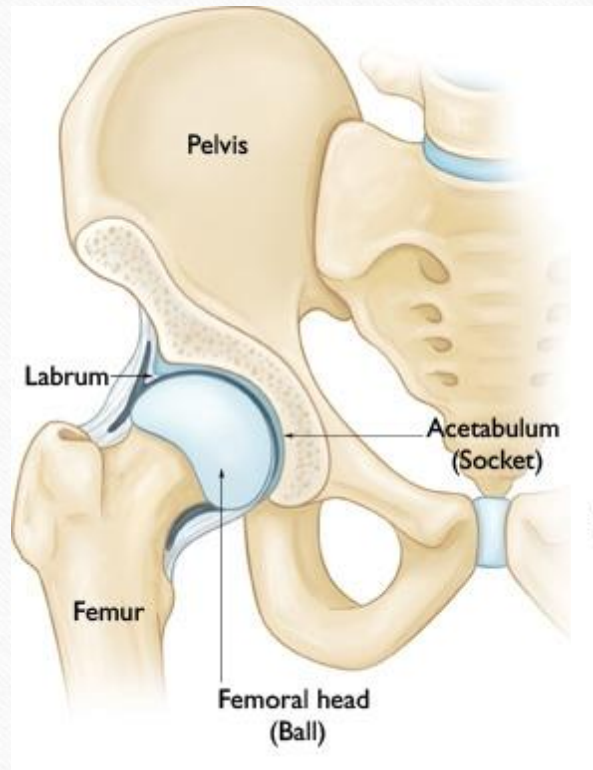
- 1- gliding joint or plane joint
 - 2- Ball and socket joint
 - 3- Hinge joint
- 4- condyloid or ellipsoid joint
 - 5- Saddle joint
 - 6- pivot joint

1- Gliding joint or plane joint

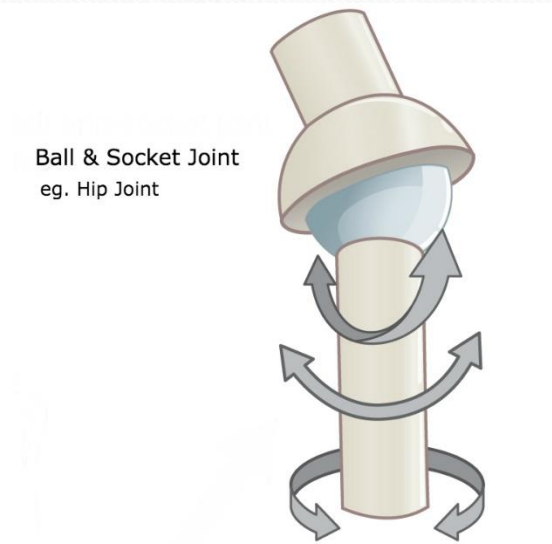
- سطحين مستويين واحد فوق الاخر يمكنهم الدوران او الزحقة
- موجودين فى اليد و الرجل



2- Ball and socket joint

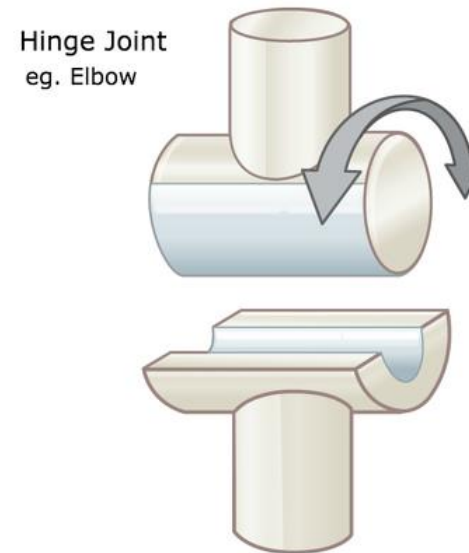
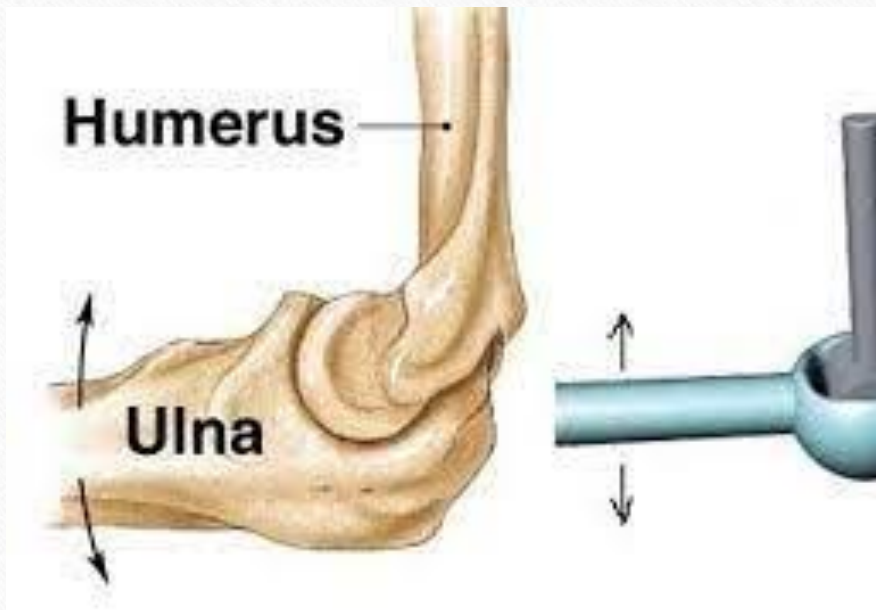


- مسموح له بالحركة في جميع الاتجاهات
- مثل مفصل الكتف و مفصل الحوض



3- Hinge joint

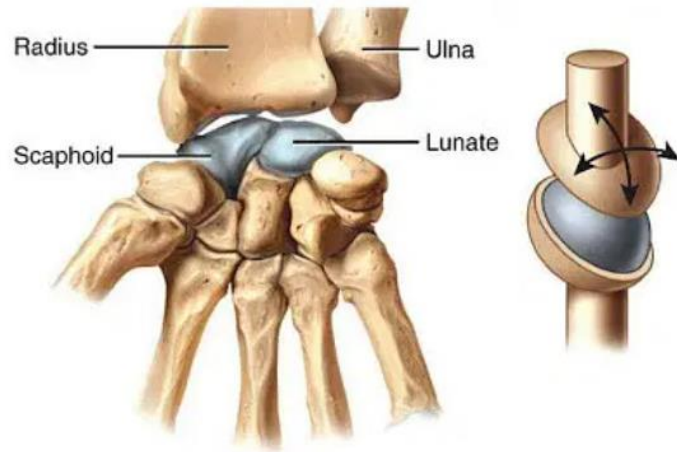
- لديه الحركة في اتجاه واحد للأعلى وللأسفل
Flexion & Extension



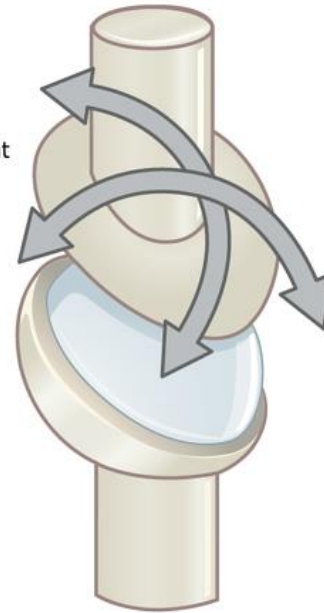
4- condyloidal or ellipsoid joint

- شبيه جدا لمفصل الكتف و لكن لديه حركة في اتجاهين فقط

Bi-axial joints Ellipsoidal joints

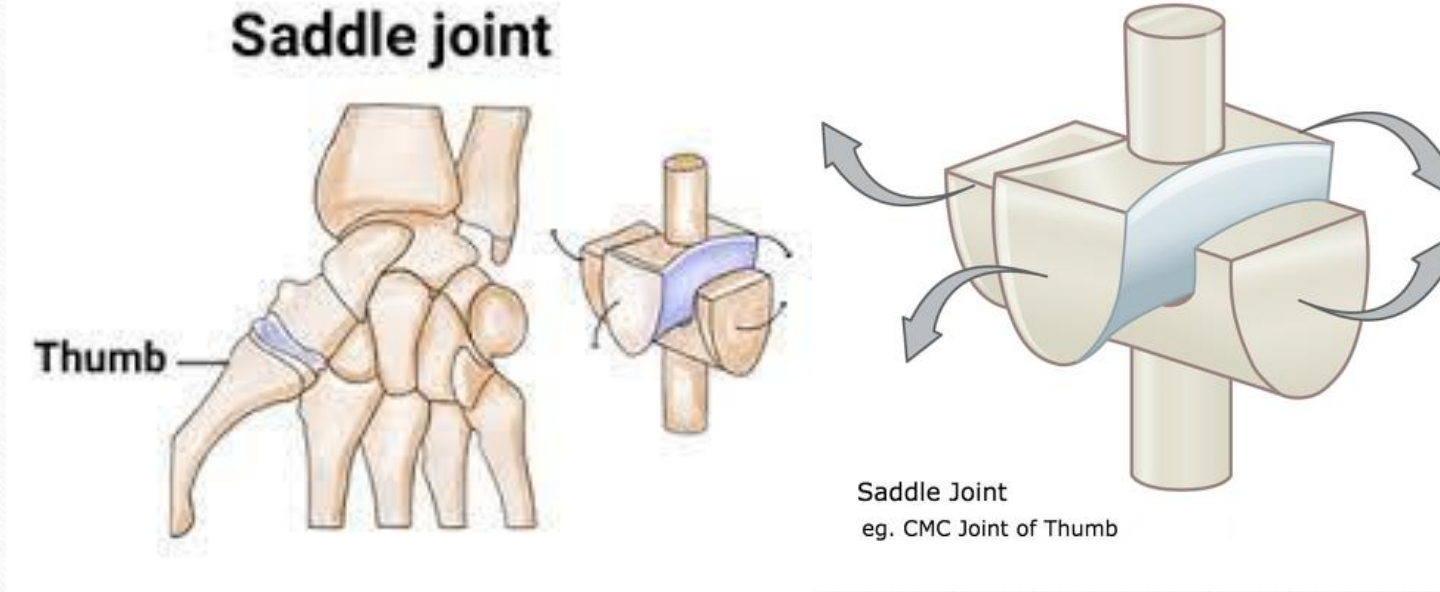


Condyloid Joint
eg. Radiocarpal Joint



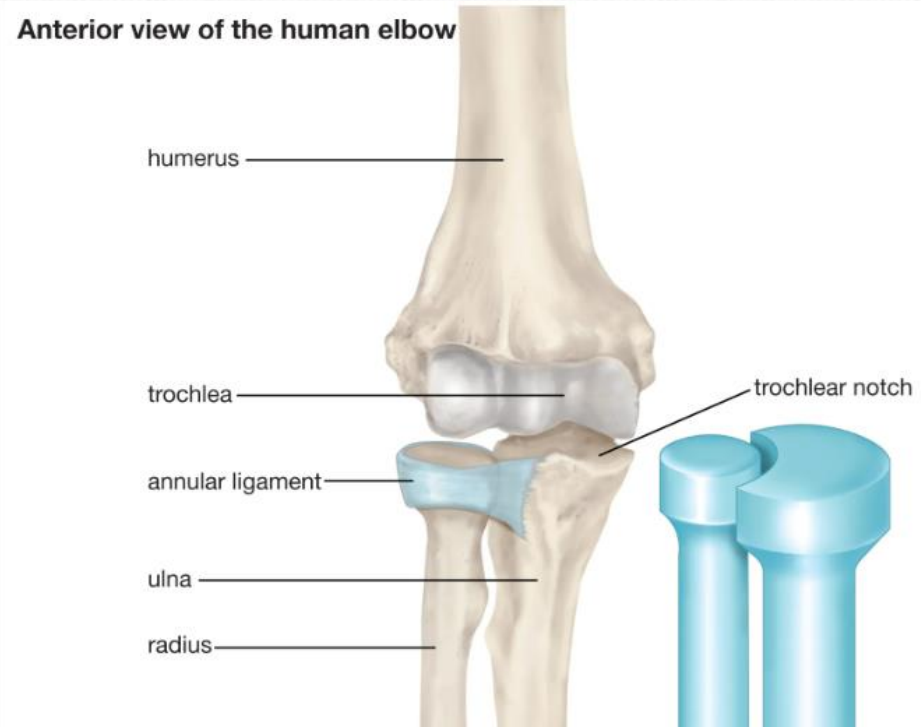
- ليس لديه القدرة على الدوران
- Flexion & Extension
- Adduction & abduction
- Wrist joint مثل مفصل اليد

5- Saddle joint

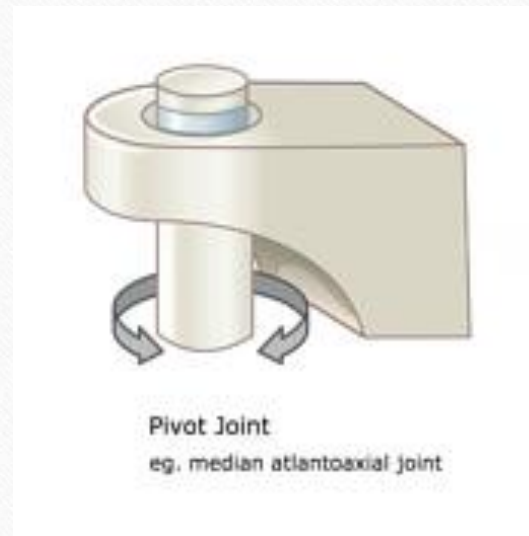


- يمكنه الحركة في جميع الاتجاهات ما عدا الدوران
- يوجد مفصل واحد فقط و هو مفصل الابهام الصباع الكبير

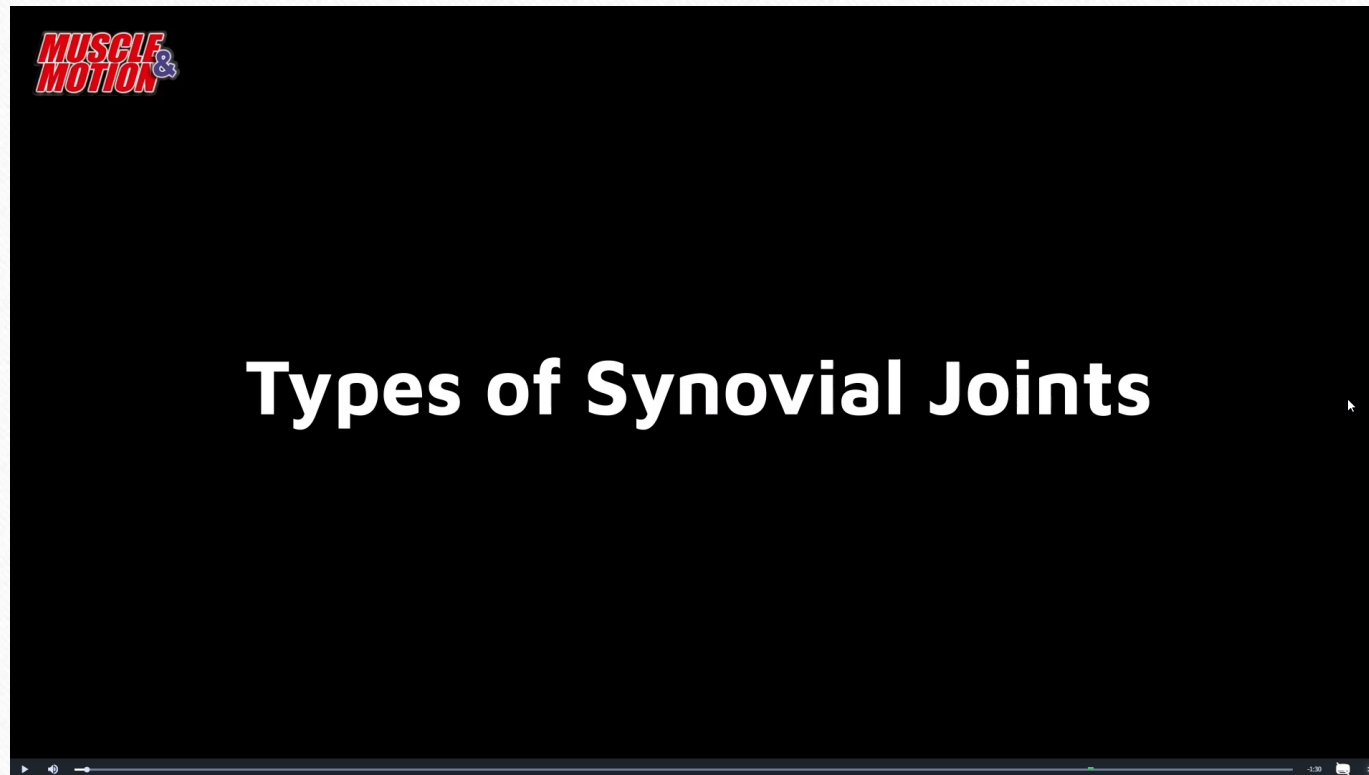
6- pivot joint



- يسمح بالحركة حول المحور مثل دوران عظام الساعد
- Radius rotation with Ulna



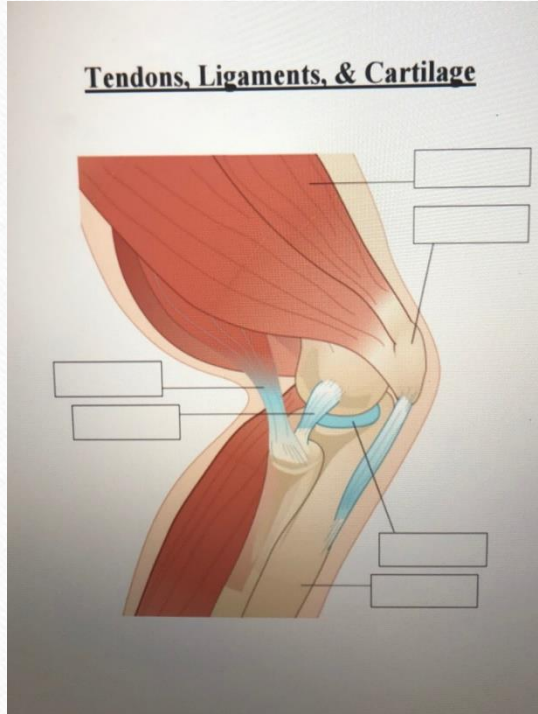
Types of synovial joints



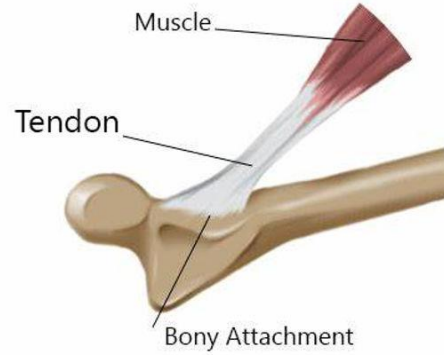
عدد المفاصل و اذكر 4 منهم بامثلة



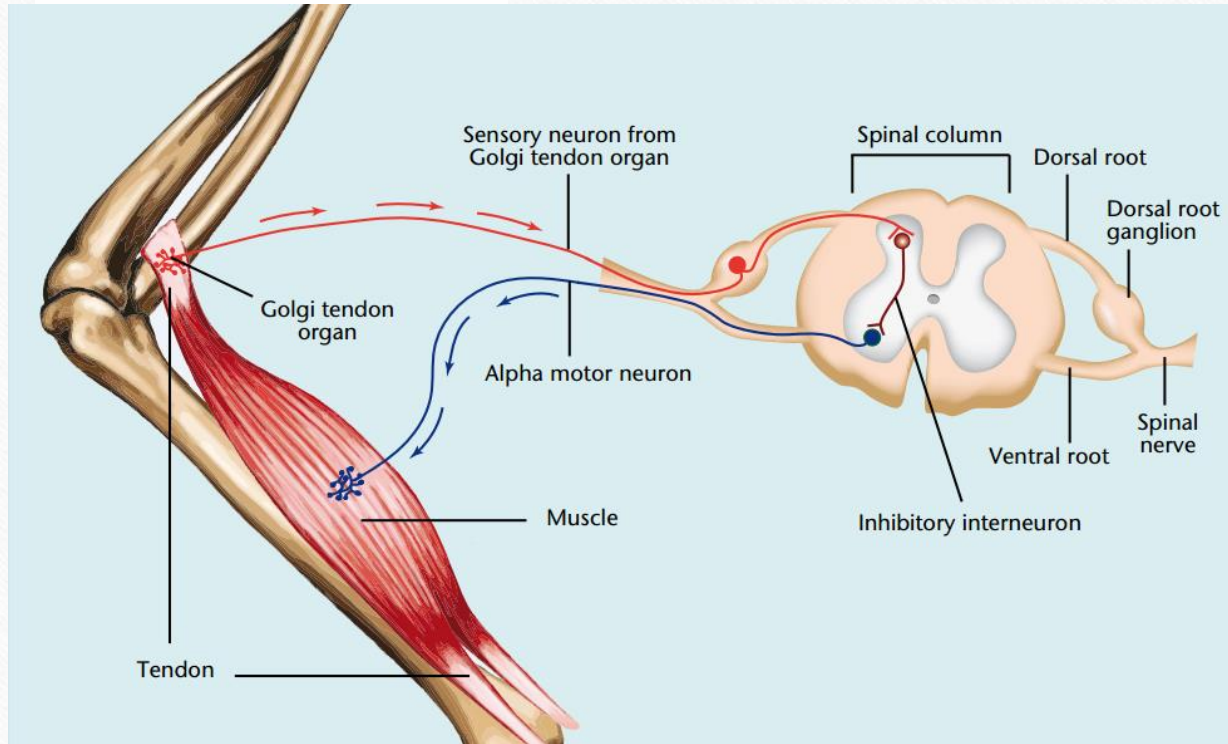
الأنسجة Connective tissue



- النسيج الضام يتكون من الكولوجين
- الكولوجين هو نوع من انواع البروتين المتوفر بكثرة فى الجسم
- الأوتار و الاربطة و الغضاريف تتكون من الانسجة
 - يعمل على ربط العضلات بالعظم
 - يعمل على ربط المفاصل ببعضها
- لا يحتاج الى اوكسوجين و مغذيات كثيرة

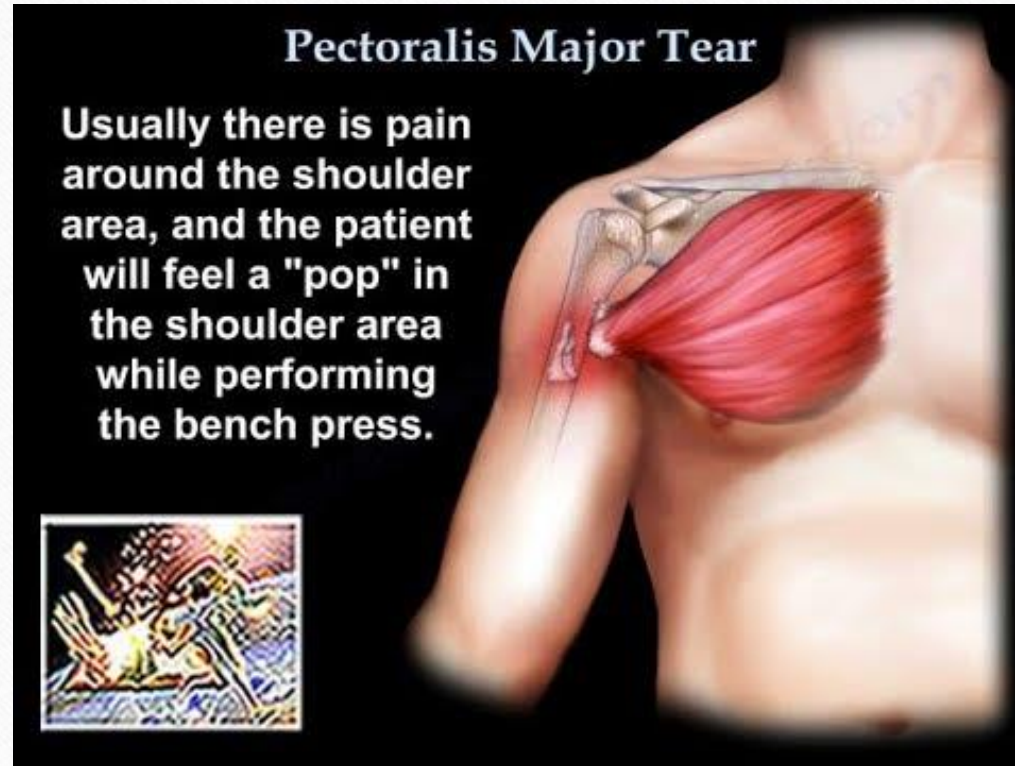


الاورتار Tendons



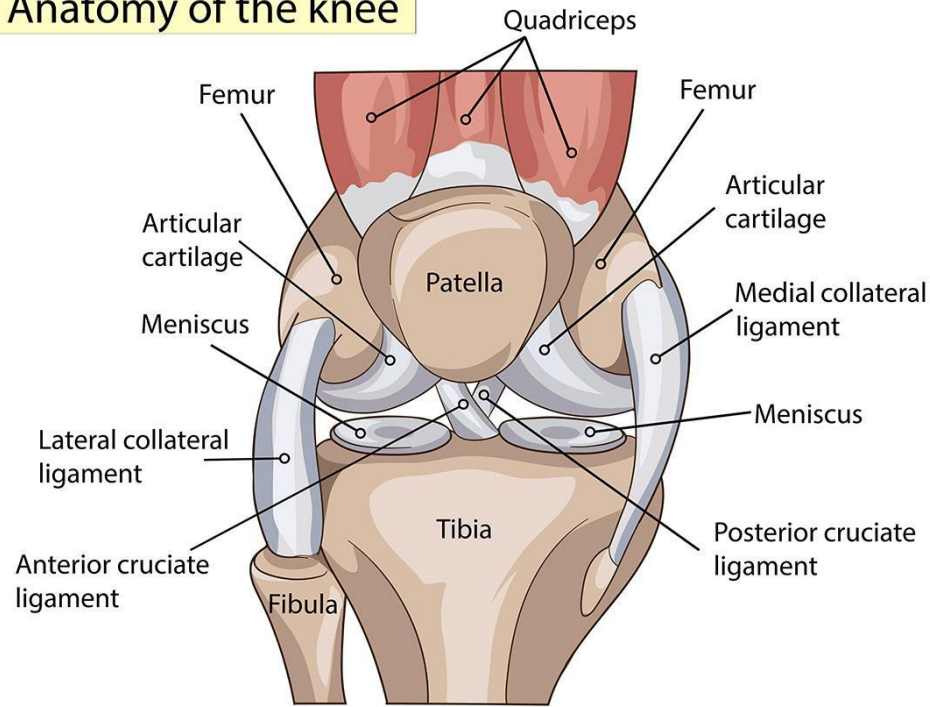
- تعمل على اتصال العضلات بالعظم
- الاوتار هي اكثر مرونة من الاربطة
- Golgi tendon organ
- Safety valve يكون مثل صمام الامان حتى لا تحدث اصابة

تدخل الاستيرويد فى زيادة العضلات و قطع الوتر



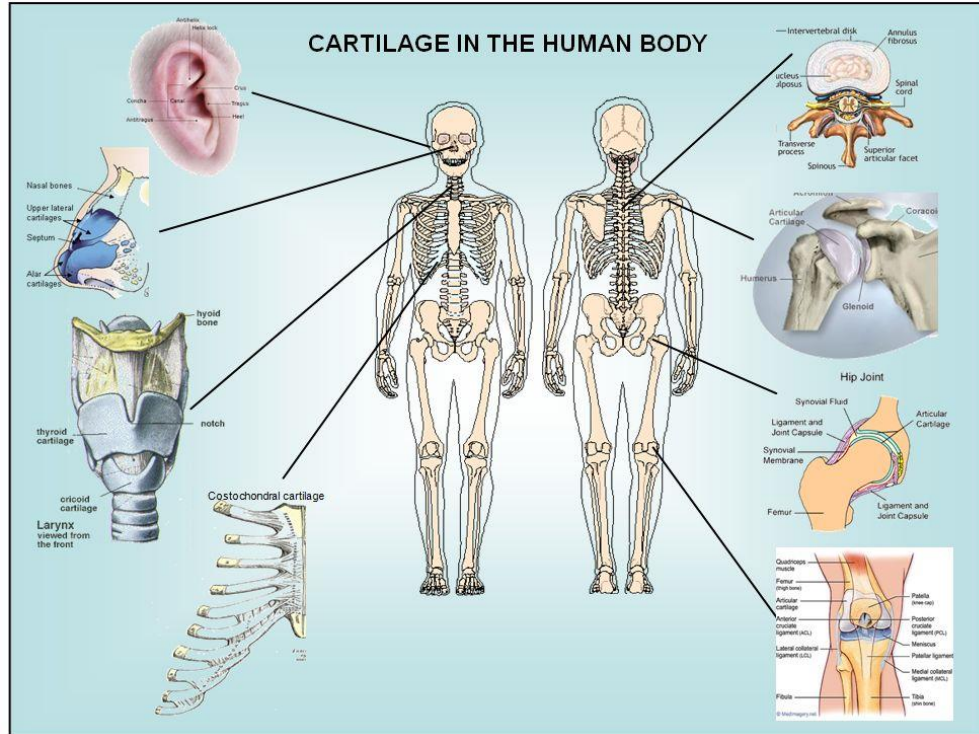
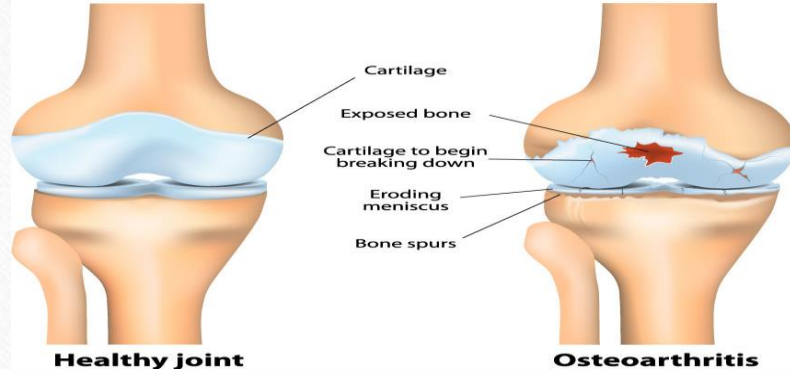
الاربطة Ligaments

Anatomy of the knee



- تعمل على اتصال العظم ببعضه
- امثلة على الركبة
- الرباط الصليبي الامامي و الخلفي وباقي الاربطة

OSTEOARTHRITIS



الغضاريف Cartilage

- ناشفة و طرية و مطاطة و مرنة
- تعمل لامتصاص الصدمات و عدم احتكاك العظام
- توجد في نهاية الضلوع في القفص الصدري
- توجد بين فقرات العمود الفقري
- توجد على سطح المفاصل
- توجد في الاذن و الانف

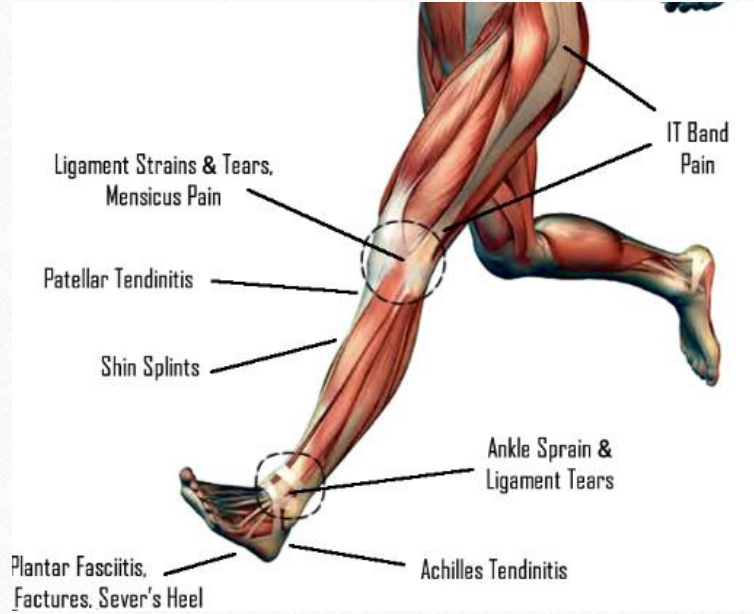
Effect of Exercise on Connective tissue

تأثير التمرين على الانسجة

- عندما تتمرّن فان الجسم يتأقلم على المجهود و يقوى جميع الانسجة و ليس النسيج العضلى فقط

- فيصبح الجسم اقل عرضة للاصابة

- تزداد الانسجة حجم و قوة و تتأقلم على عدم التعب و شدة المقاومة



- مثال : البشرة و الشمس

- الارض و الجلد

- الاربطة و قوتها

- العضلات و زيادتها

فائدة كل من الاوتار و الاربطة و الغضاريف ؟



Muscles

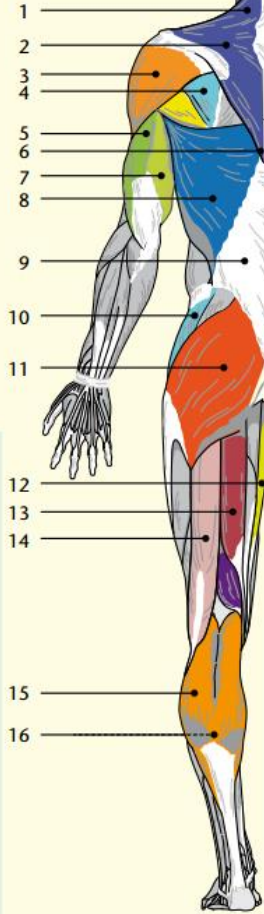
BACK (Posterior View)

1. Trapezius (upper)
2. Trapezius (middle)
3. Posterior deltoid
4. Teres minor
5. Triceps brachii (lateral head)
6. Rhomboids (underlying)
7. Triceps brachii (long head)
8. Latissimus dorsi
9. Erector spinae
10. Gluteus medius
11. Gluteus maximus
12. Gracilis
13. Semitendinosus
14. Biceps femoris
15. Gastrocnemius
16. Soleus (deep to gastrocnemius)

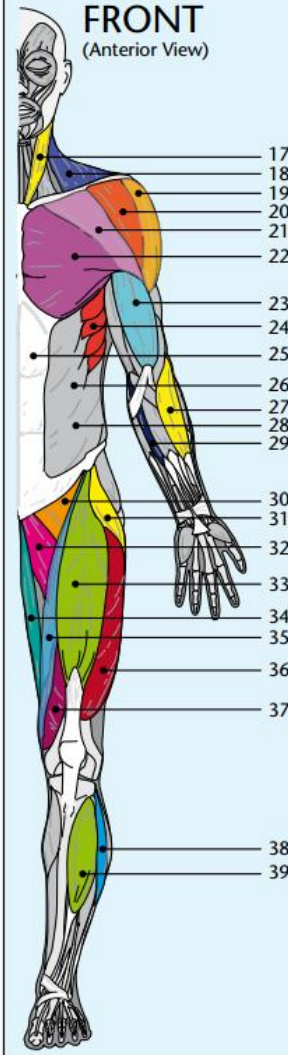
FRONT (Anterior View)

17. Sternocleidomastoid
18. Trapezius (upper)
19. Medial deltoid
20. Anterior deltoid
21. Clavicular pectoralis major
22. Sternal pectoralis major
23. Biceps brachii
24. Serratus Anterior
25. Rectus abdominis
26. Internal oblique (underlying)
27. Brachioradialis
28. External oblique
29. Palmaris longus
30. Pectineus
31. Tensor fasciae latae
32. Adductor longus
33. Rectus femoris
34. Gracilis
35. Sartorius
36. Vastus lateralis
37. Vastus medialis
38. Peroneus longus
39. Tibialis anterior

BACK (Posterior View)



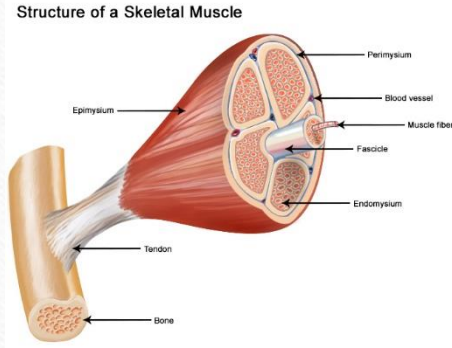
FRONT (Anterior View)



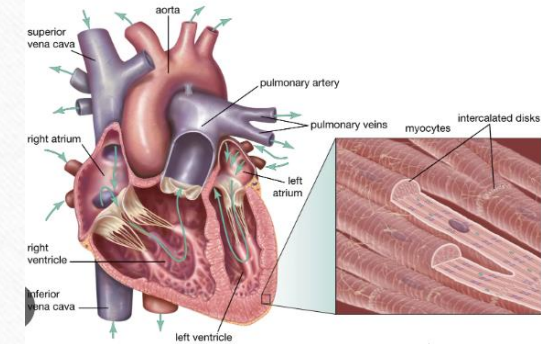
- الجسم مكون من اكثر من 600 عضلة
- عضلات مسئولة عن الضحك
- حركة الاكل فى الجهاز الهضمى
- عضلات مسئولة عن الدورة الدموية
- و لكن سنتحدث عن ال49 عضلة الأهم فى الجيم
- نحفظ سوا بقى و نعرف الاسماء و النطق
- شرح الاسماء و الاماكن من الصور و الابلكيشن

اذكر 20 اسم عضلة منهم

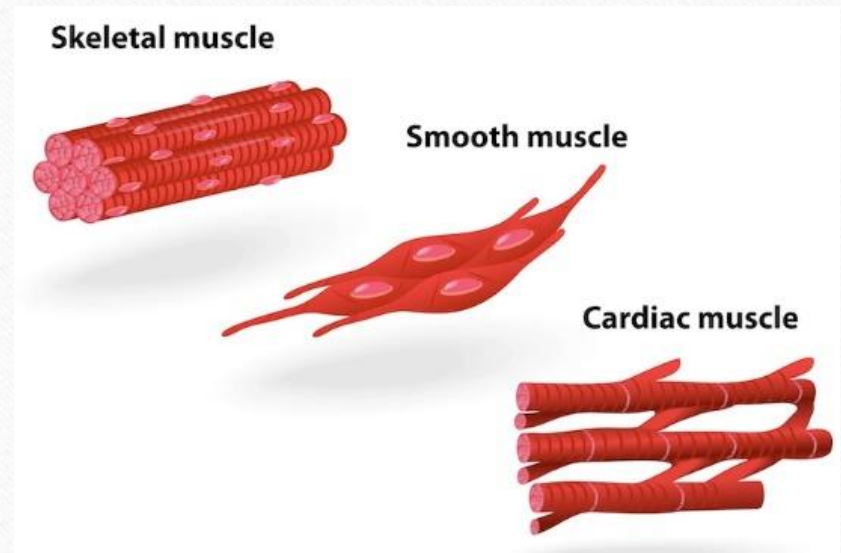




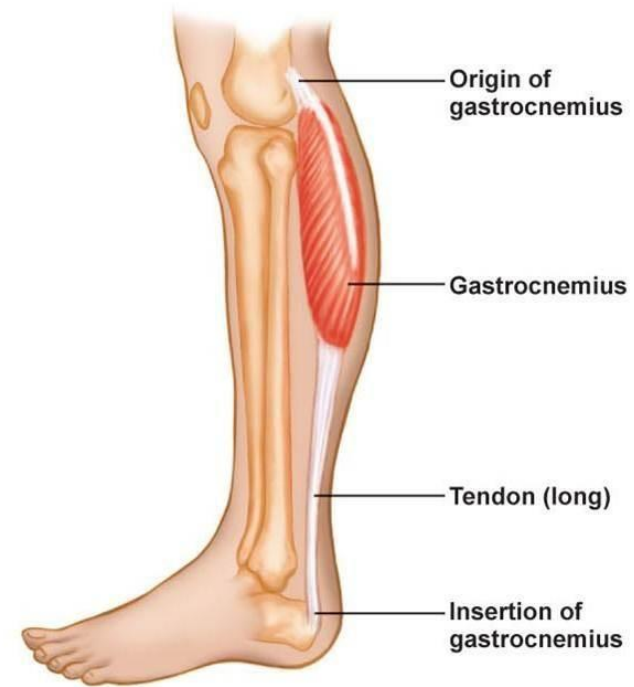
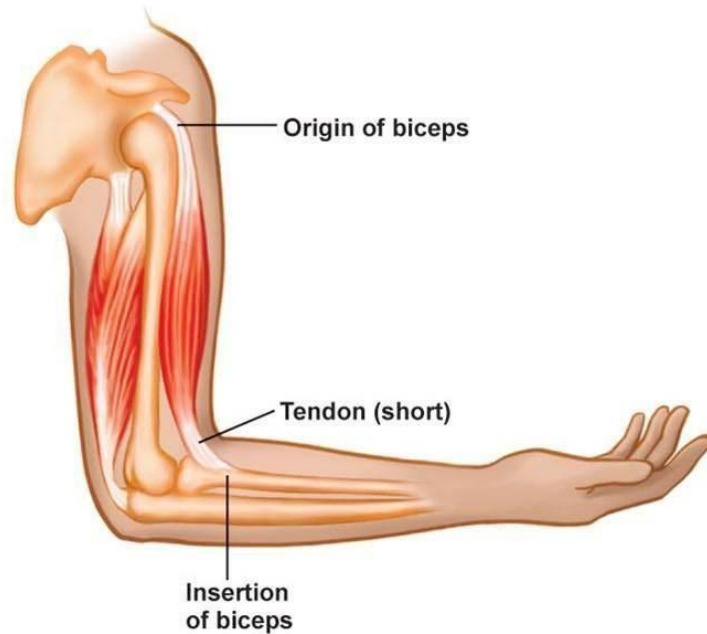
Types of muscles tissue



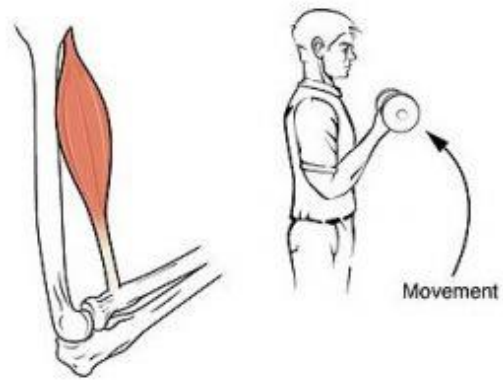
- Cardiac muscle tissue: contract the heart and pump the blood , doesn't fatigue easily, the rest periods between contractions
- Smooth muscle tissue: wall of the blood vessel
Digestive, respiratory track, lymphatics
It change the diameter of the vessels
- Skeletal muscle tissue: muscle attached to bones



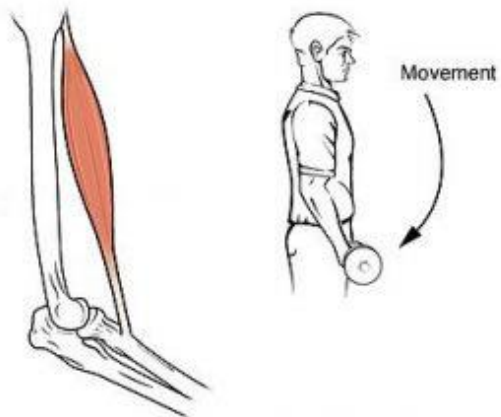
What is Origin & Insertion ?



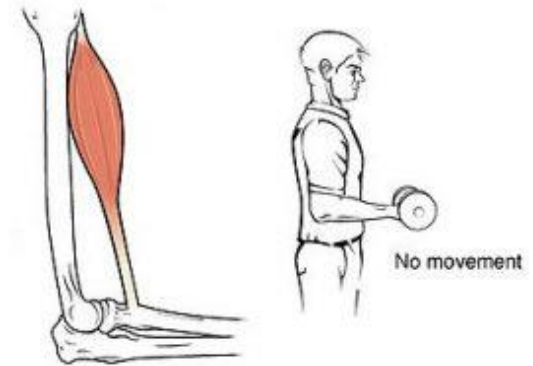
Types of contractions



CONCENTRIC
(shortening)



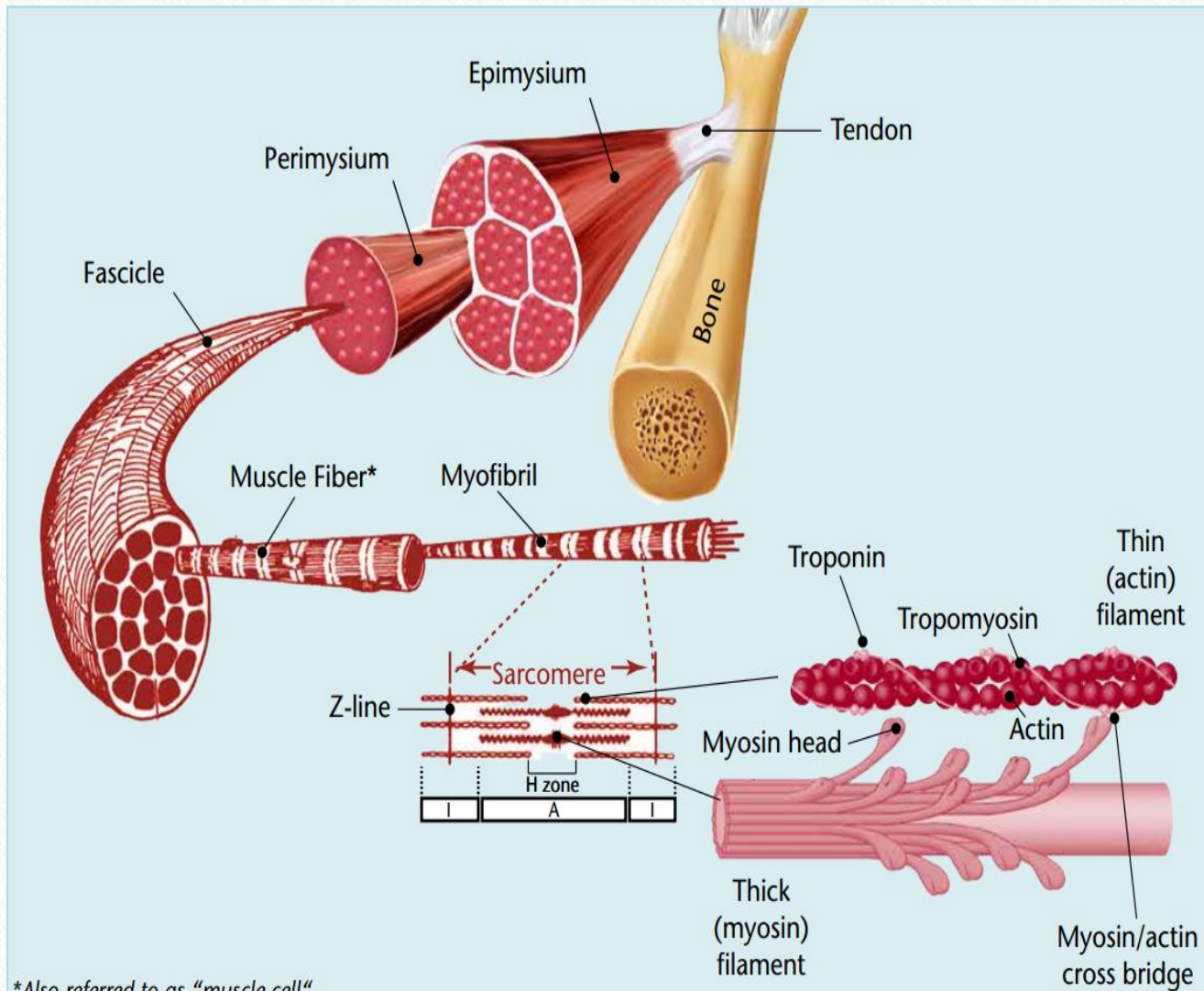
ECCENTRIC
(lengthening)



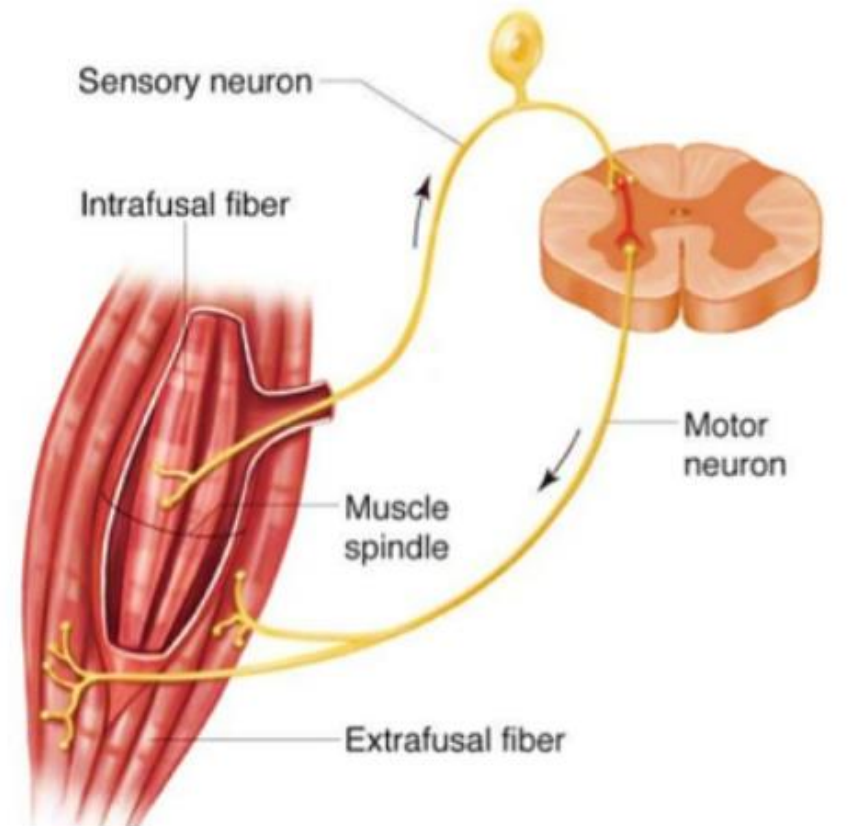
ISOMETRIC
(no movement)

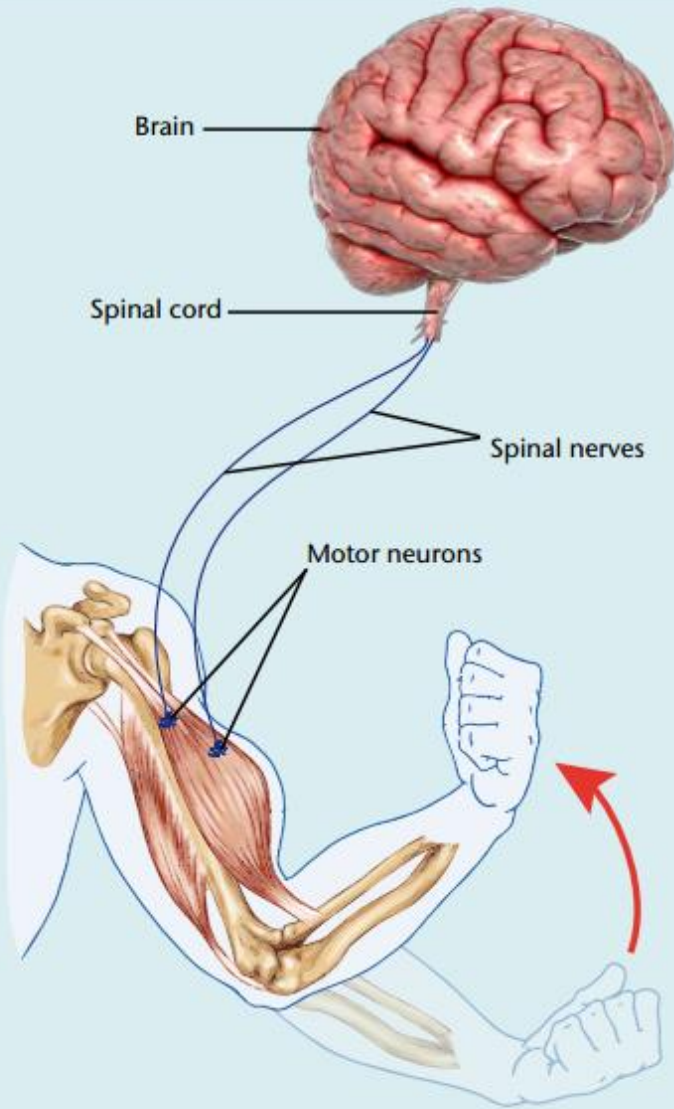
Muscle and nerve supply

Muscle contraction



*Also referred to as "muscle cell"

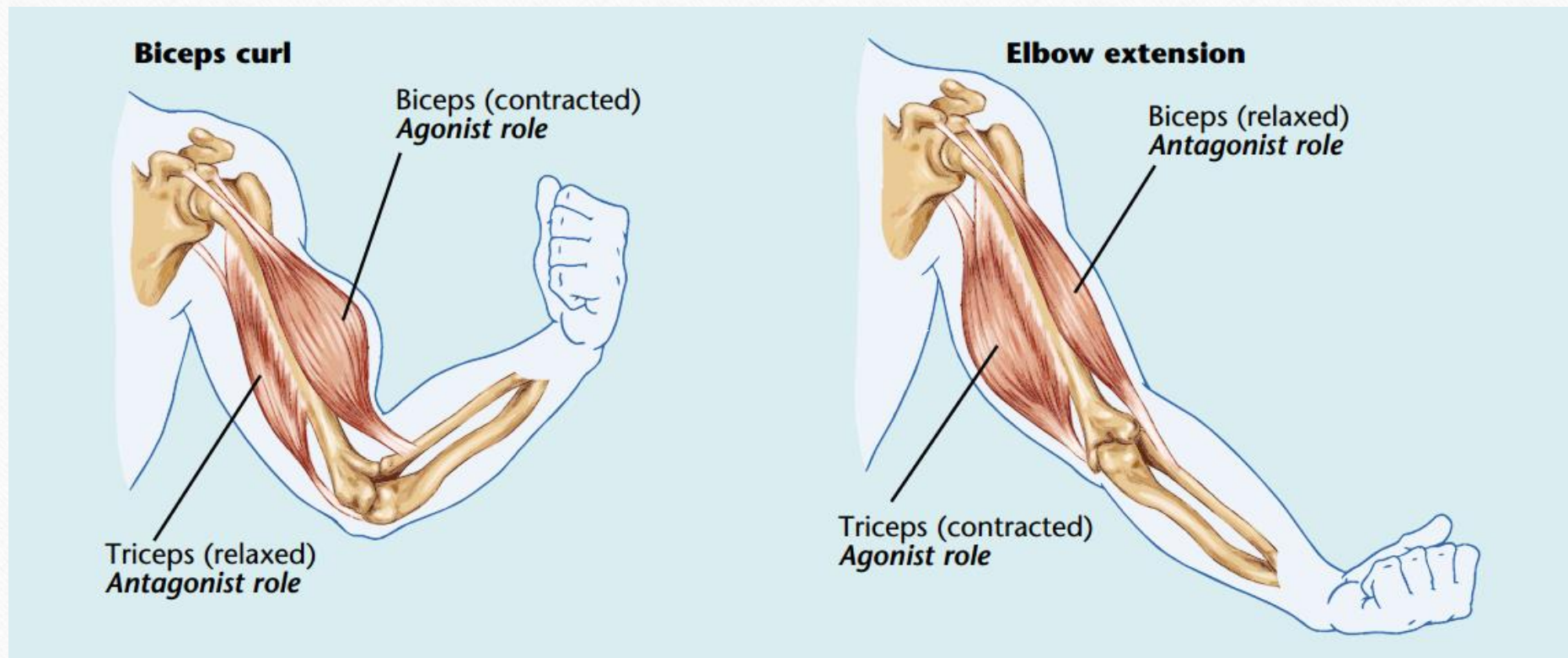




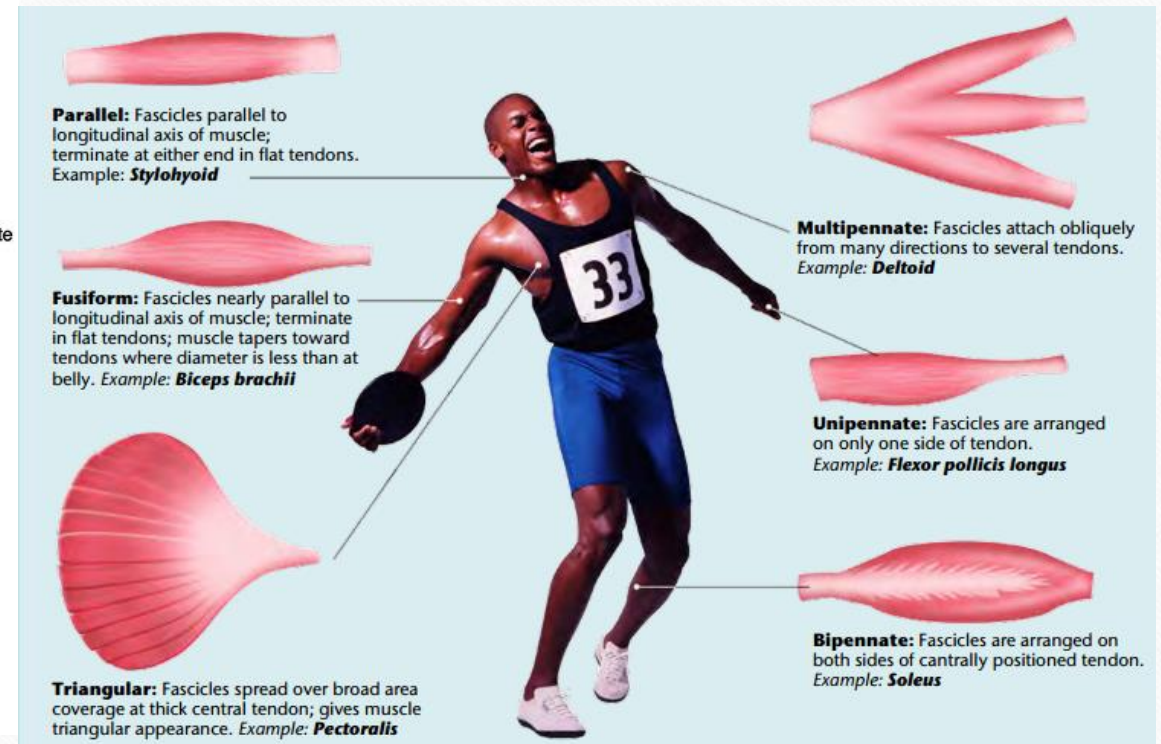
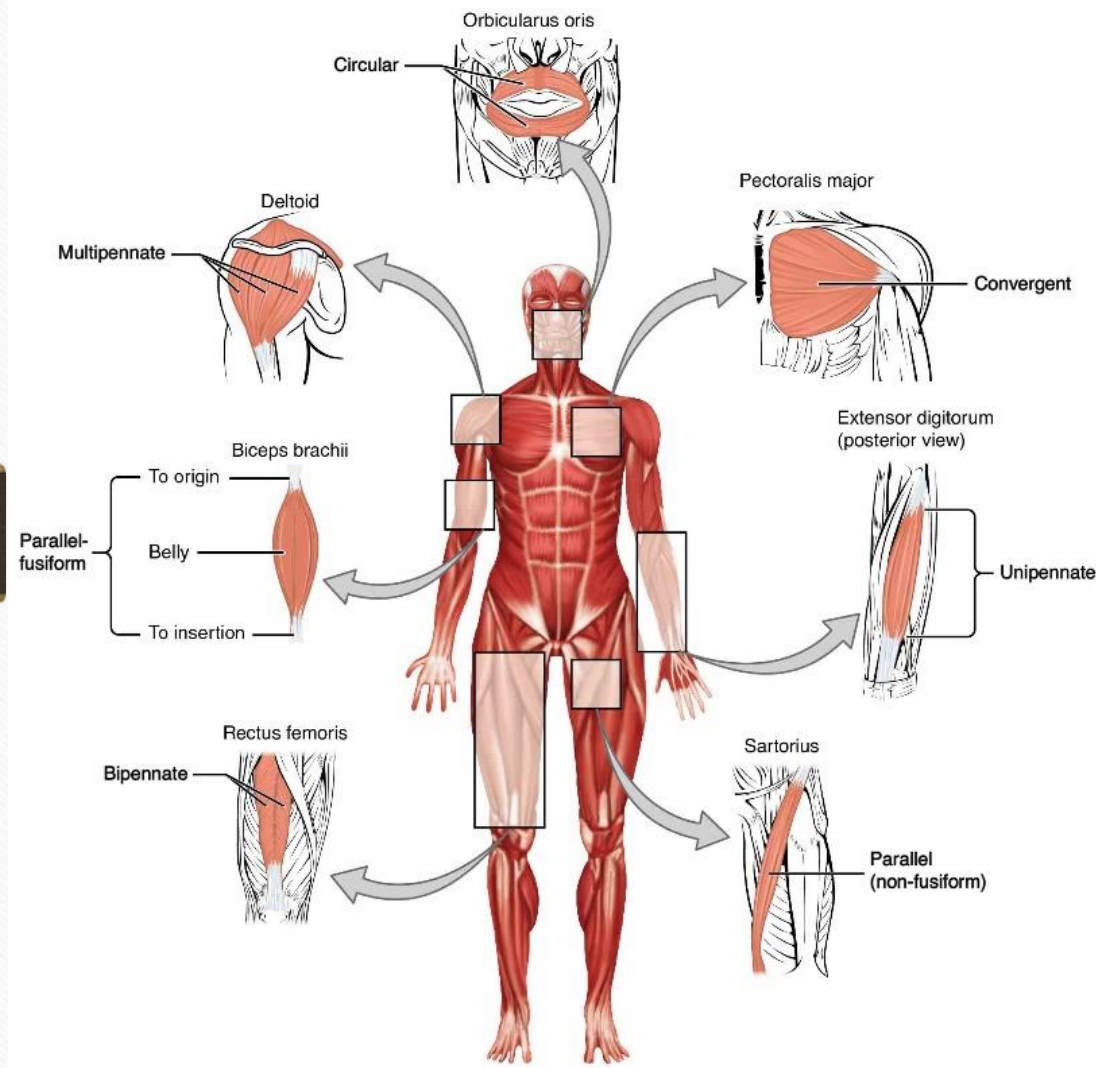
1. Brain sends out electrical signal.
2. Signal travels through the spinal cord.
3. To the spinal nerves.
4. To the motor neurons.
5. Resulting in the propagation of an electrical current through the muscle fiber.
6. Electrical signal triggers the release of calcium inside the muscle fiber.
7. The released calcium binds to the contractile protein **ACTIN**.
8. This permits its interaction with the **MYOSIN** contractile protein.
9. ATP provides the energy that permits the "walking" of **MYOSIN** across the **ACTIN**.
10. This pulling action of the **MYOSIN** across the **ACTIN** results in the shortening of the muscle fiber during **MUSCLE CONTRACTION**.

How does the motion happen

Agonist & Antagonist role



Arrangement of muscle fibers



Skeletal Muscle Fiber Types

Type I (Slow-Twitch Oxidative, SO)

- Slow twitch (slow speed of muscle fiber contraction), low force capacity, high oxidative metabolism capacity, highly fatigue resistant, with little capacity for exercise induced hypertrophy, high resistance to exercise induced structural damage. High density of mitochondria per fiber, which increases with aerobic endurance training. Small diameter, high capillary density.

Type II a (Fast-Twitch Oxidative-Glycolytic, FOG)

- Fast twitch (fast speed of muscle fiber contraction), medium force capacity, oxidative-glycolytic metabolism, highly fatigue resistant, high capacity for exercise induced hypertrophy, moderate resistance to exercise induced structural damage. Medium density of mitochondria per fiber. Important for sustained power endurance. Medium diameter, intermediate capillary density, intermediate glycogen content.

Type II b (Fast-Twitch Glycolytic, FG)

- Fast twitch (fast speed of muscle fiber contraction), high force capacity, high capacity for glycolytic metabolism, low oxidative capacity, highly susceptible to fatigue, great capacity for exercise induced hypertrophy, great susceptibility to exercise induced damage. Low mitochondria density per fiber. Generates explosive strength and power. Larger diameter, high glycogen content, low capillary density.

Types of Muscle fibers

Kinesiology

the study of movement

العضلات زى الاستيك و كل عضلة بتعمل حركة و تشد الجسم لمكان معين

- و الجسم عشان يرجع تانى لمكانه الى كان عليه فى الاول
- لازم يبقى فى عضلة مقابلة ليها بتعمل عكس الوظيفة دى
- زى الرجل عشان تعمل ابعاد للخارج او ادخال للداخل
 - Abduction & adduction
 - Gluteus Medius and adductor muscles
- Extension, hyperextension and Flexion
- Back muscles (erector spinae muscles)
- and abdominal muscles (Rectus abdominis)
- Neck muscles in their opposite directions

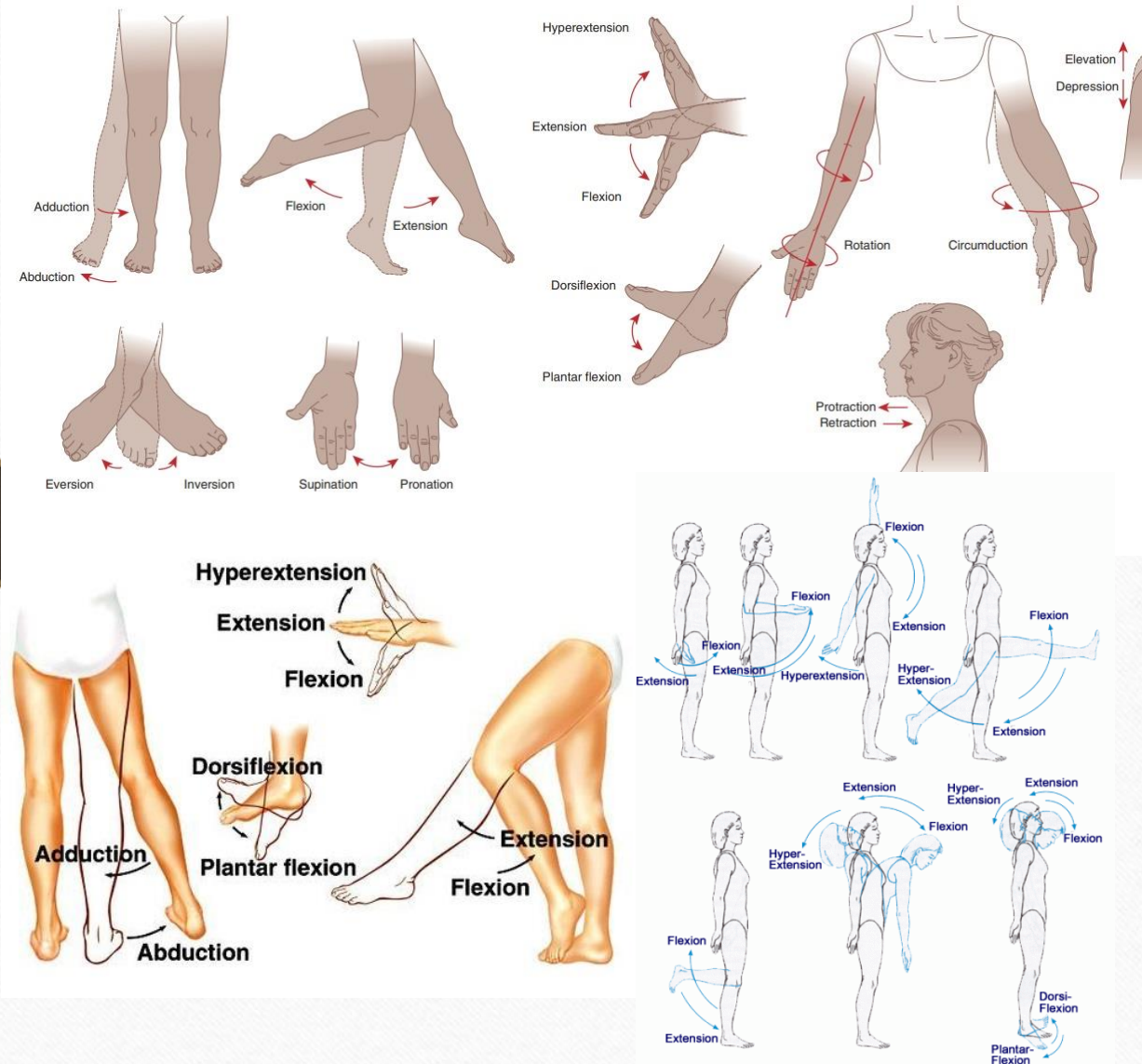
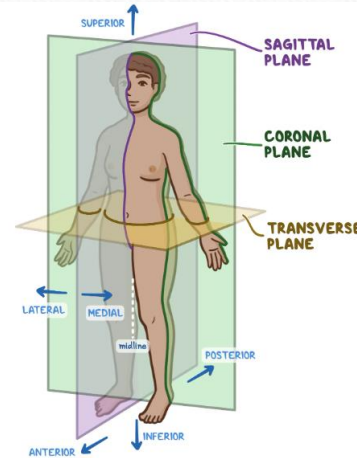


Table 4.1

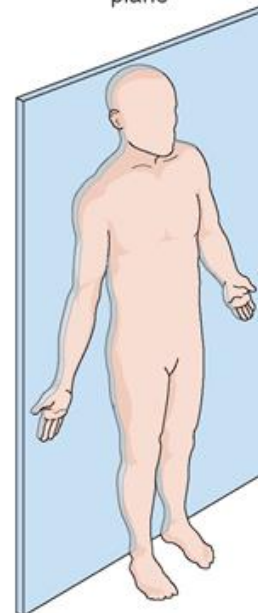
Fundamental Anatomical Movements

PLANE OF MOTION	ACTION	DEFINITION
Frontal	Abduction	Movement away from the midline of the body
	Adduction	Movement toward the midline of the body
	Elevation	Moving to a superior position at the scapula
	Depression	Moving to an inferior position at the scapula
	Inversion	Lifting the medial border of the foot (insole)
	Eversion	Lifting the lateral border of the foot
Sagittal	Flexion	Decreasing the angle between two bones
	Extension	Increasing the angle between two bones
	Dorsiflexion	Moving the top of the foot toward the shin at the ankle joint
	Plantarflexion	Moving the top of the foot away from the shin at the ankle
Transverse	Rotation	Internal or external turning about the vertical axis of a bone
	Pronation	Rotating the hand and wrist medially from the elbow
	Supination	Rotating the hand and wrist laterally from the elbow
	Horizontal adduction	From a 90° abduction arm position, the humerus is flexed toward the midline of the body in the transverse plane
	Horizontal abduction	From a 90° adduction arm position, the humerus is extended away from the midline of the body in the transverse plane
Multiplanar	Circumduction	Combination of flexion, abduction, extension, and adduction in a sequence



Frontal, Sagittal & Transverse plan

Frontal (coronal) plane



Sagittal plane



Transverse (horizontal) plane

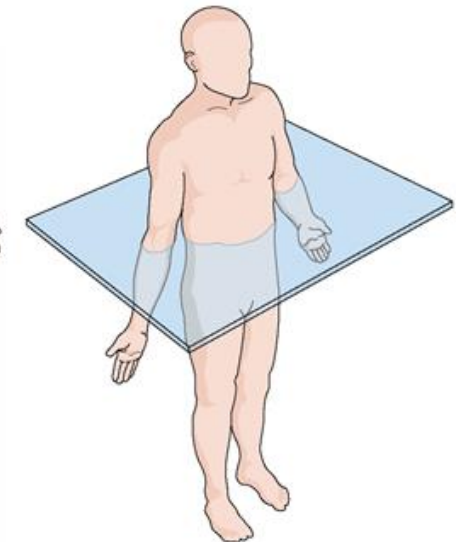
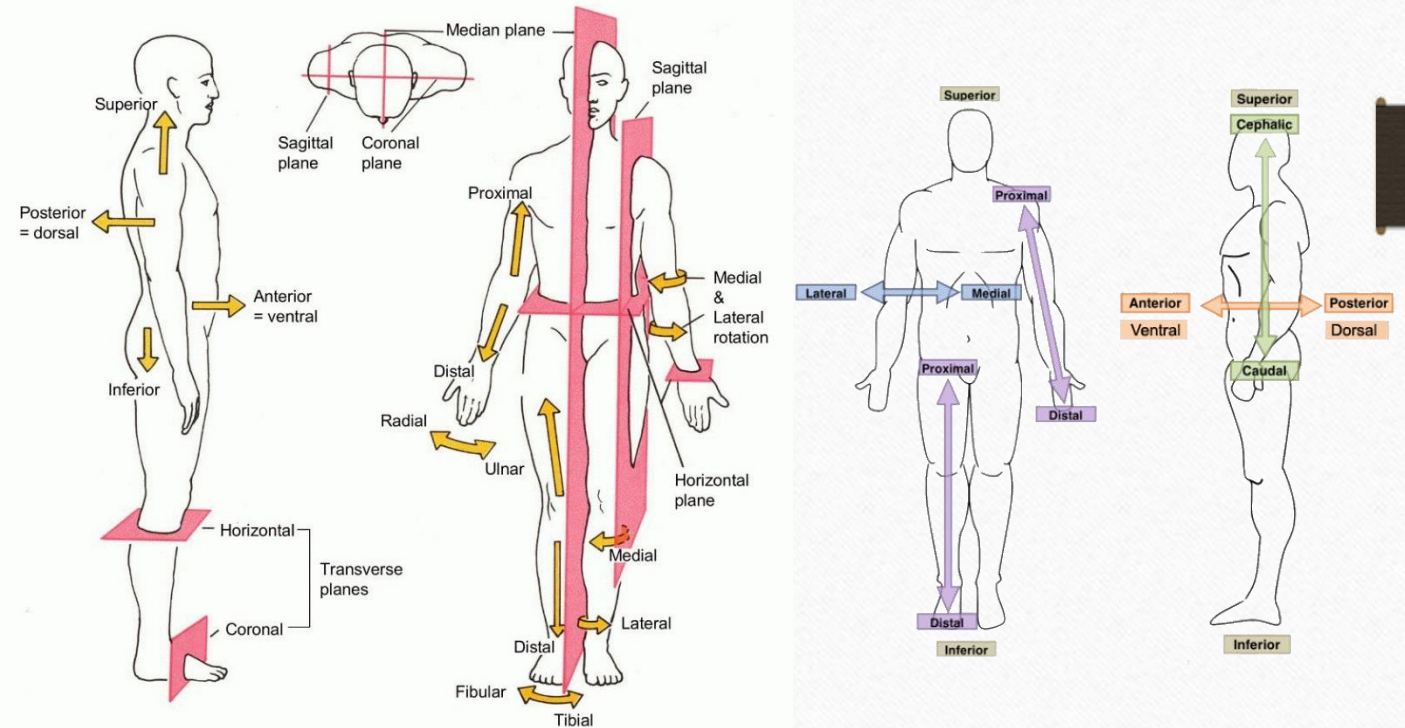


Table 3.1**Anatomical Directional Terminology**

Anterior	In front or in the front part
Anteroinferior	In front and below
Anterolateral	In front and to the side, especially the outside
Anteromedial	In front and toward the inner side or midline
Anteroposterior	Relating to both front and rear
Caudal	Below in relation to another structure; inferior
Cephalic	Above in relation to another structure; higher, superior
Contralateral	Pertaining or relating to the opposite side
Deep	Beneath or below the surface; used to describe relative depth or location of muscles or tissue
Distal	Situated away from the center or midline of the body, or away from the point of origin
Dorsal	Relating to the back; posterior
Inferior (infra)	Below in relation to another structure; caudal
Ipsilateral	On the same side
Lateral	On or to the side; outside, farther from the median or midsagittal plane
Medial	Relating to the middle or center; nearer to the medial or midsagittal plan
Posterior	Behind, in back, or in the rear
Posteroinferior	Behind and below; in back and below
Posterolateral	Behind and to one side, specifically to the outside
Posteromedial	Behind and to the inner side
Posterosuperior	Behind and at the upper part
Prone	The body lying face downward; stomach lying
Proximal	Nearest the trunk or the point of origin
Superficial	Near the surface; used to describe relative depth or location of muscles or tissue
Superior (supra)	Above in relation to another structure; higher, cephalic
Supine	Lying on the back; face upward position of the body
Ventral	Relating to the belly or abdomen
Volar	Relation to palm of the hand or sole of the foot

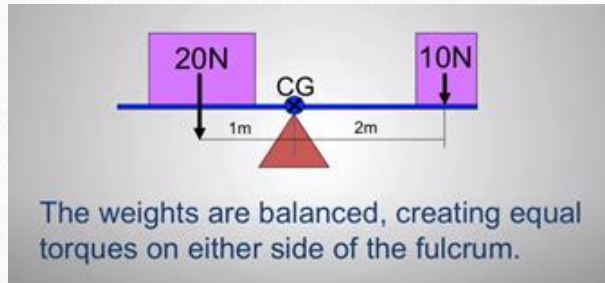
Anatomical directional terms



Biomechanics

Levers

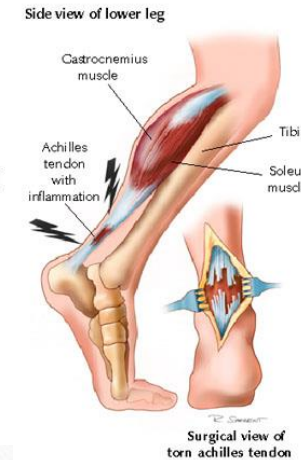
Engineering terms



- Moment : distance* weight
- Density : kg/m³ bones
- Compression: spine on the vertebrae or meniscus
- Tension: tendon out from the bone
- Shear: ACL tear and vertebrae moving on each other
- Bending: bone of Femur get broken
- Torsion: twisting the body, wrestling the arm

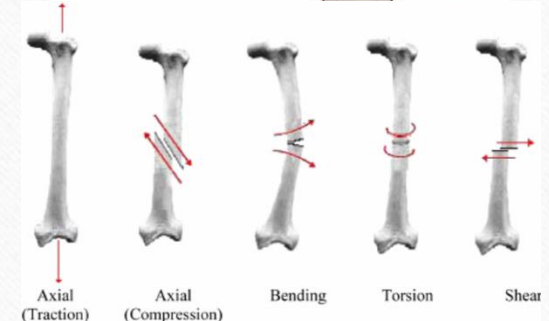
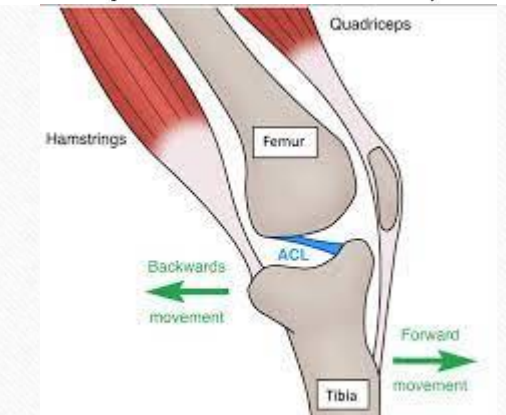


Compression Fracture



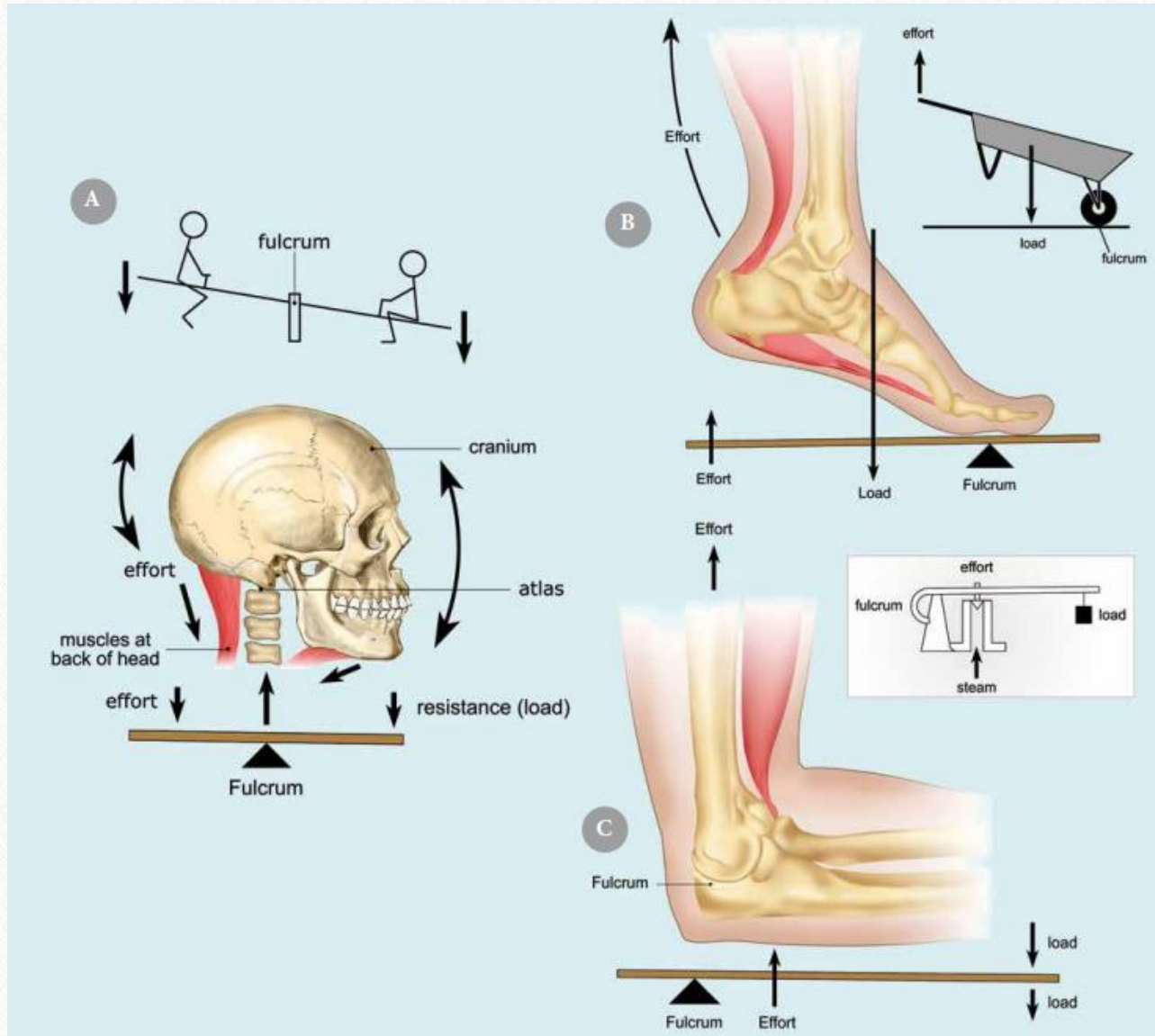
Healthy bone

Osteoporosis



Biomechanics

Levers



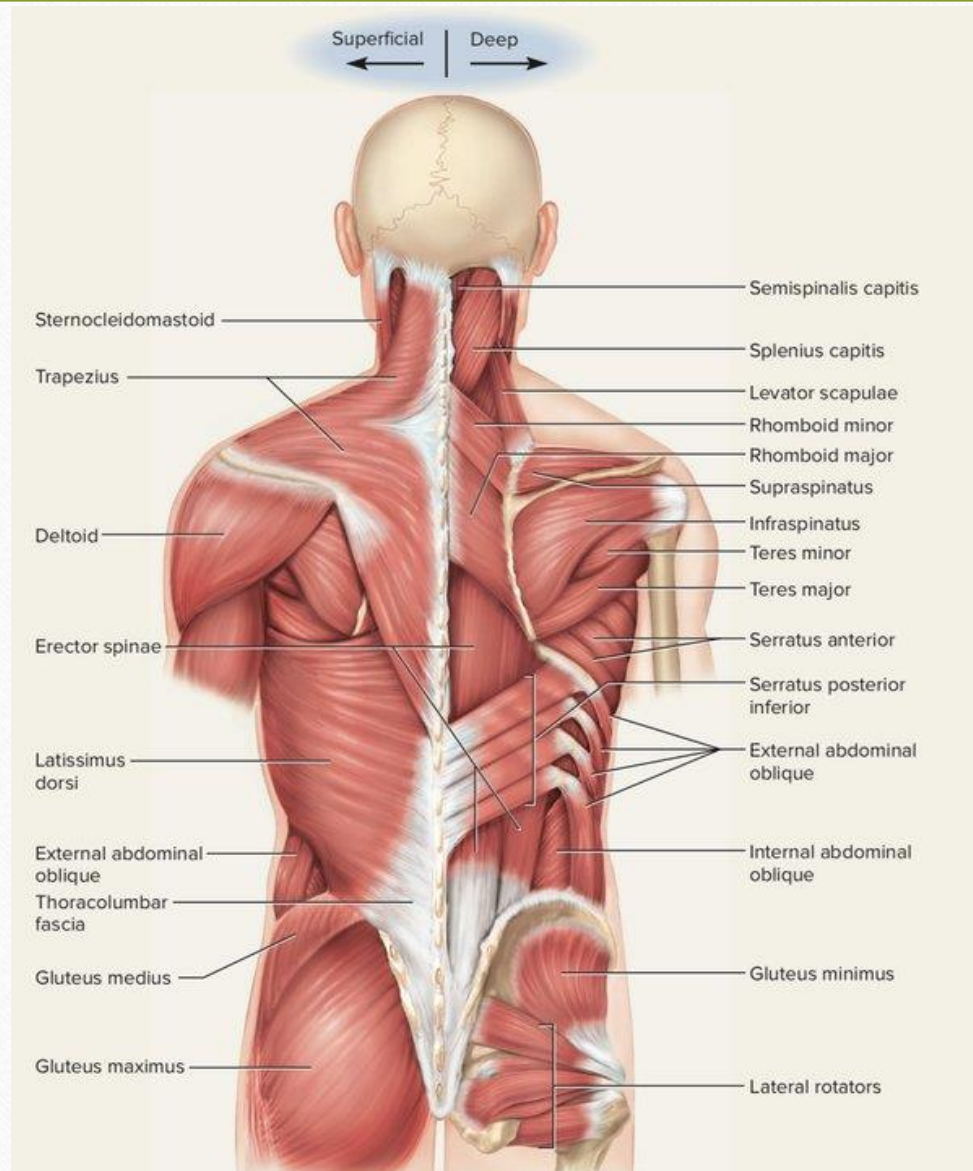
Function and motion of the muscles with photos & videos by Complete anatomy 3D



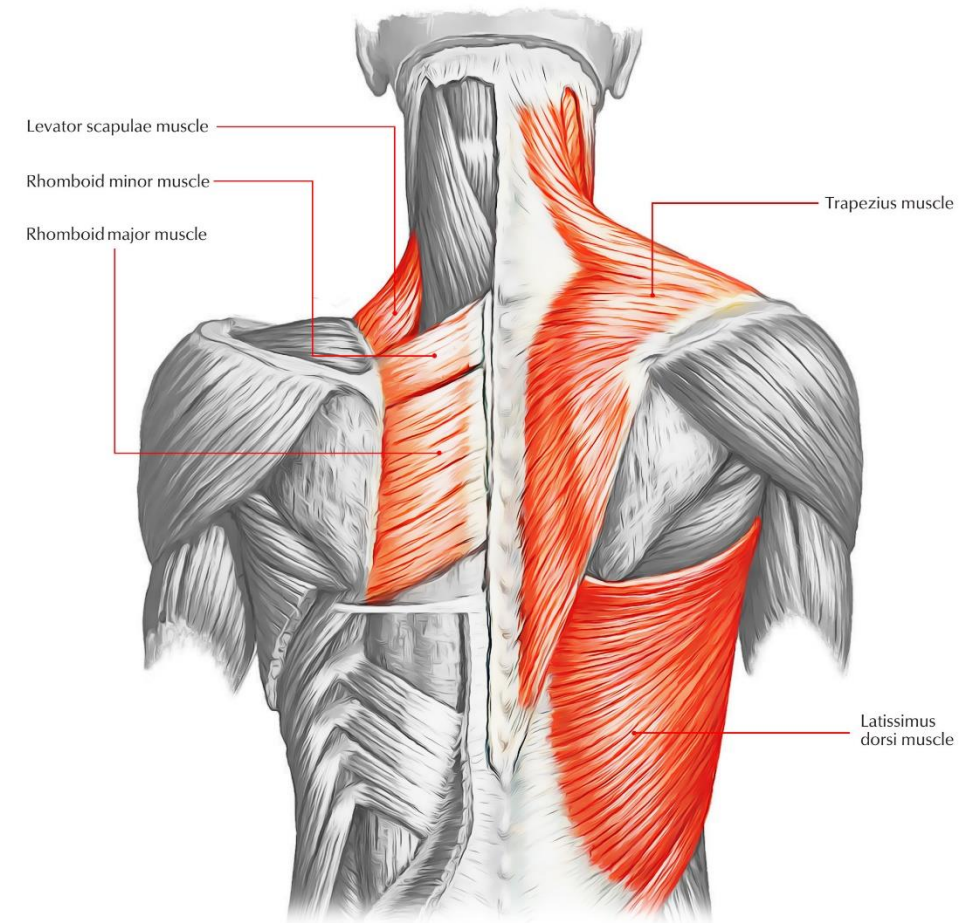
3D Anatomy of muscle at gym

Muscle & Motion 3D

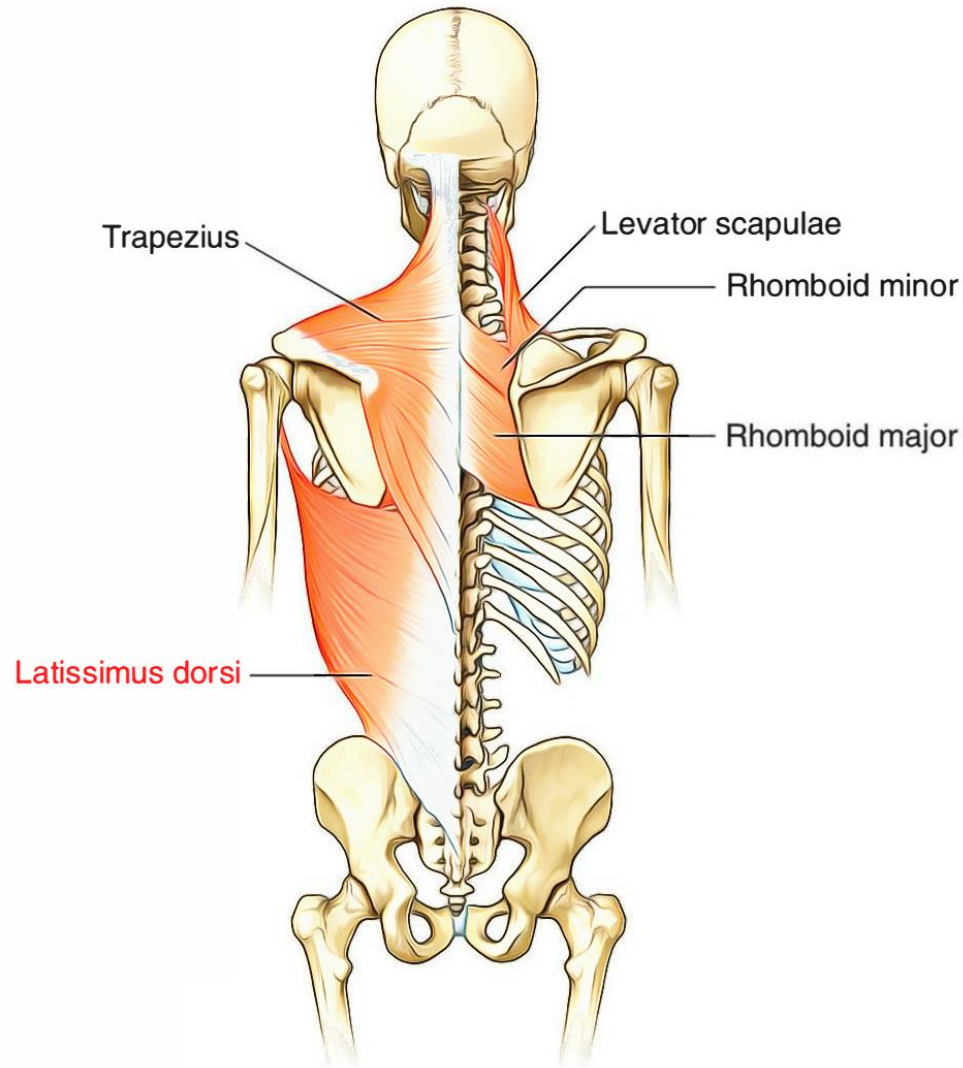




Muscles of the Back

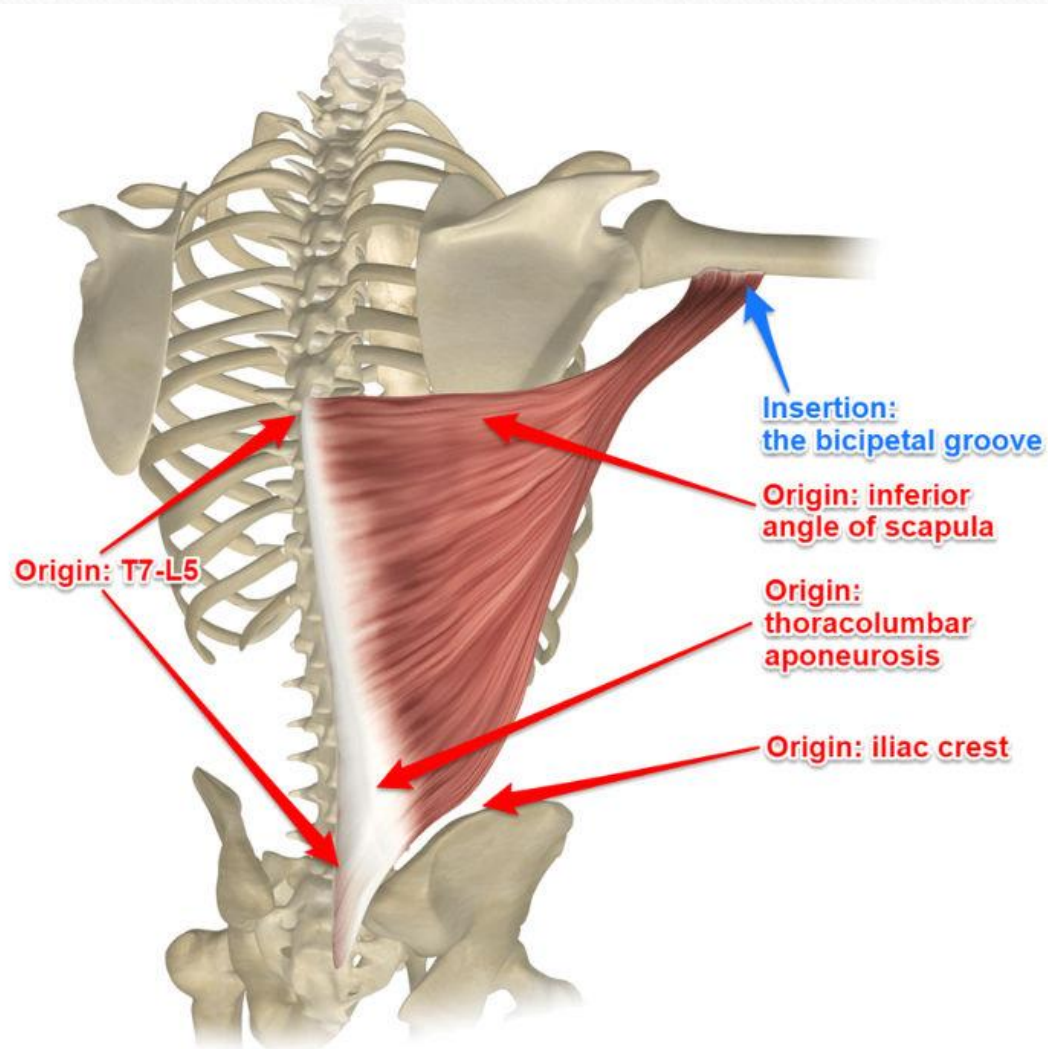


8- Latissimus dorsi



- عضلة المجنص
- Origin and Insertion
- بداية العضلة من العمود الفقري و الحوض و الضلوع
- نهاية العضلة في عظمة الذراع العضضية

Latissimus dorsi



- **Origin :**

vertebrae T7-T12, Thoracolumbar fascia
Iliac part: Posterior third of crest of ilium
Costal part: Ribs 9-12 Scapular part:
Inferior angle of scapula

- **Insertion :**

Humerus

Motion of the muscle

- Shoulder retraction
- Arm medial rotation
- Shoulder depression

Motion of the muscle

- Shoulder adduction
- Shoulder Extention
& hyperextention

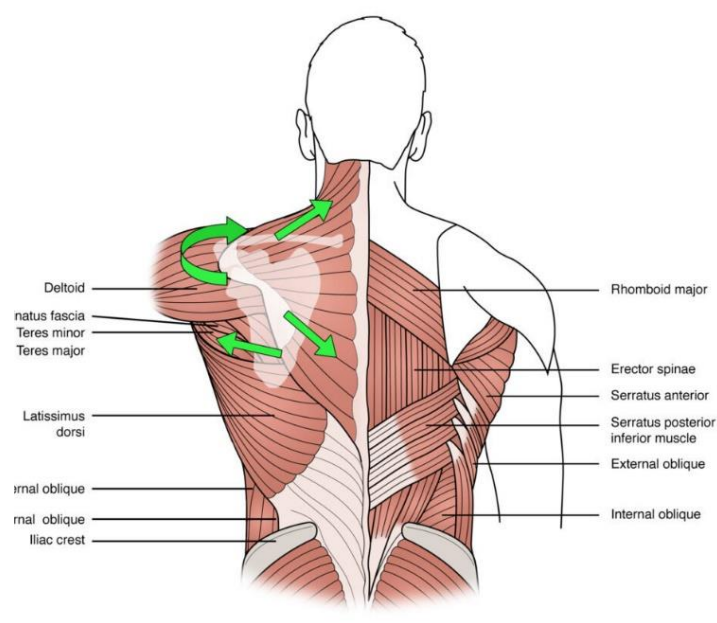
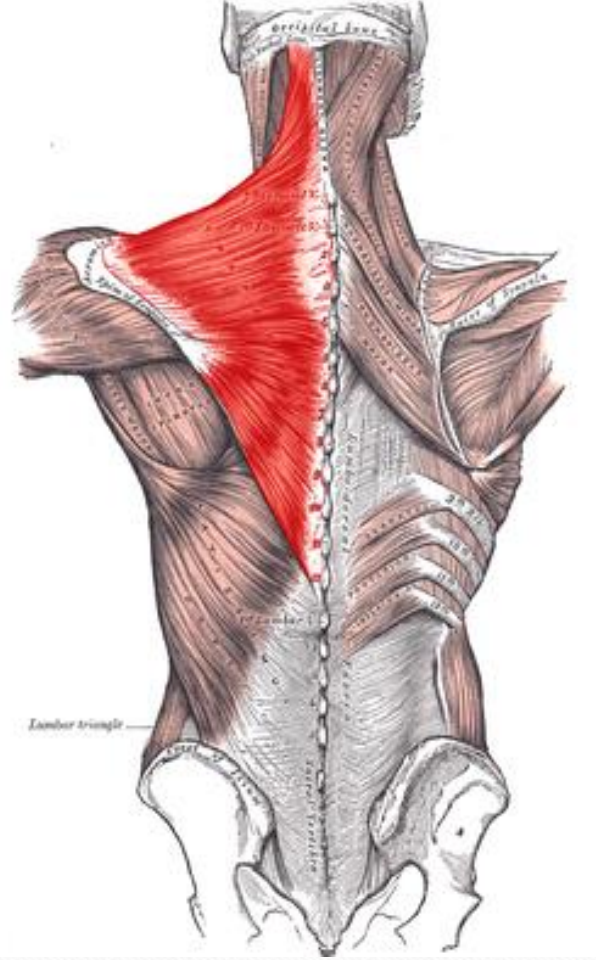
Training anatomy of lats shoulder adduction

Lat pulldown wide grip

Training anatomy of lats Shoulder Extension

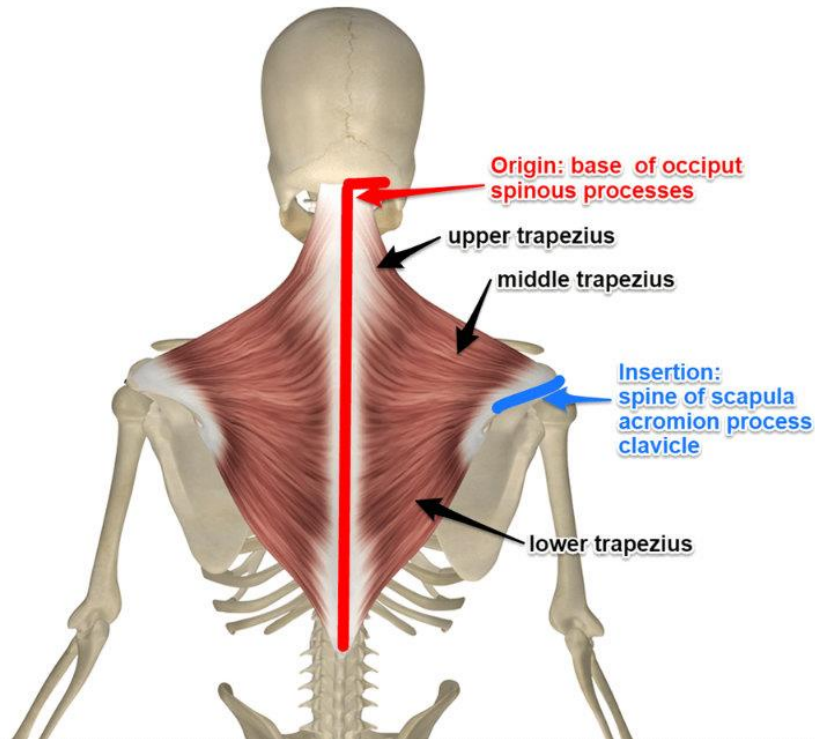
Seated rows close grip

1-2-18- Trapezius muscle:



- عضلة الترابيس منقسمة الى 3 اجزاء
- الجزء العلوى و الاوسط و السفلى
- Origin and Insertion
- البداية من العمود الفقرى و اسفل الجمجمة
- النهاية فى لوح الكتف و التركوة

Trapezius muscle: upper, middle, & lower



- **Origin :**

Base of the **skull**

spinous processes of **vertebrae C7-T12**

- **Insertion :** lateral third of **clavicle**, **acromion**, and spine of **scapula**.

Motion of the muscle

- Cervical spine extension
& hyperextension
- Cervical spine lateral flexion

Motion of the muscle

- Shoulder retraction
- Shoulder elevation
- Shoulder depression

Training anatomy of upper traps shoulder elevation

Dumbbell shrugs

Training anatomy traps Shoulder retraction
Seated machine rows wide grip

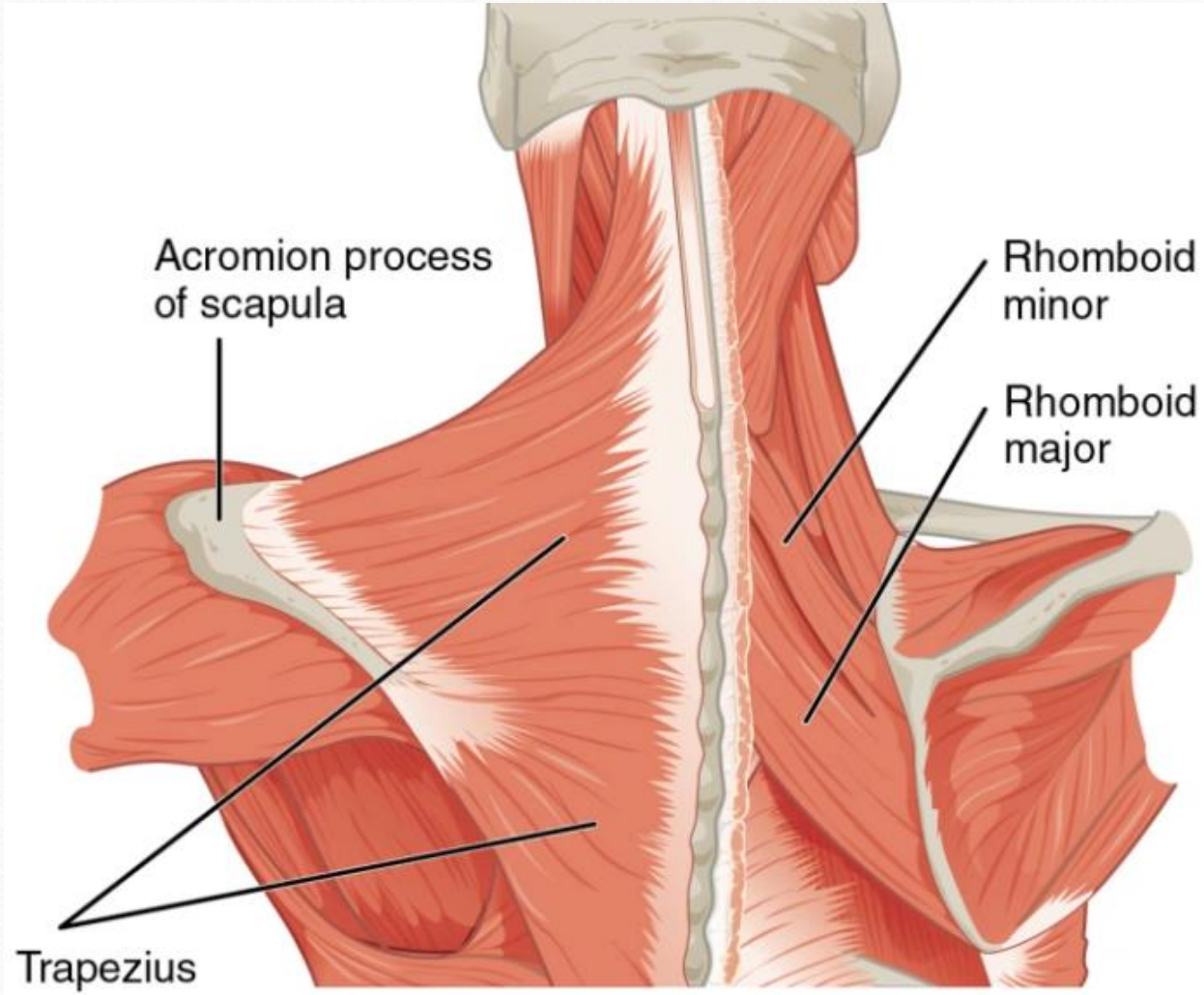
Seated rows wide grip
Shoulder retraction

Training anatomy traps Shoulder retraction
Seated machine rows wide grip (lower back injury)

بداية ونهاية العضلات

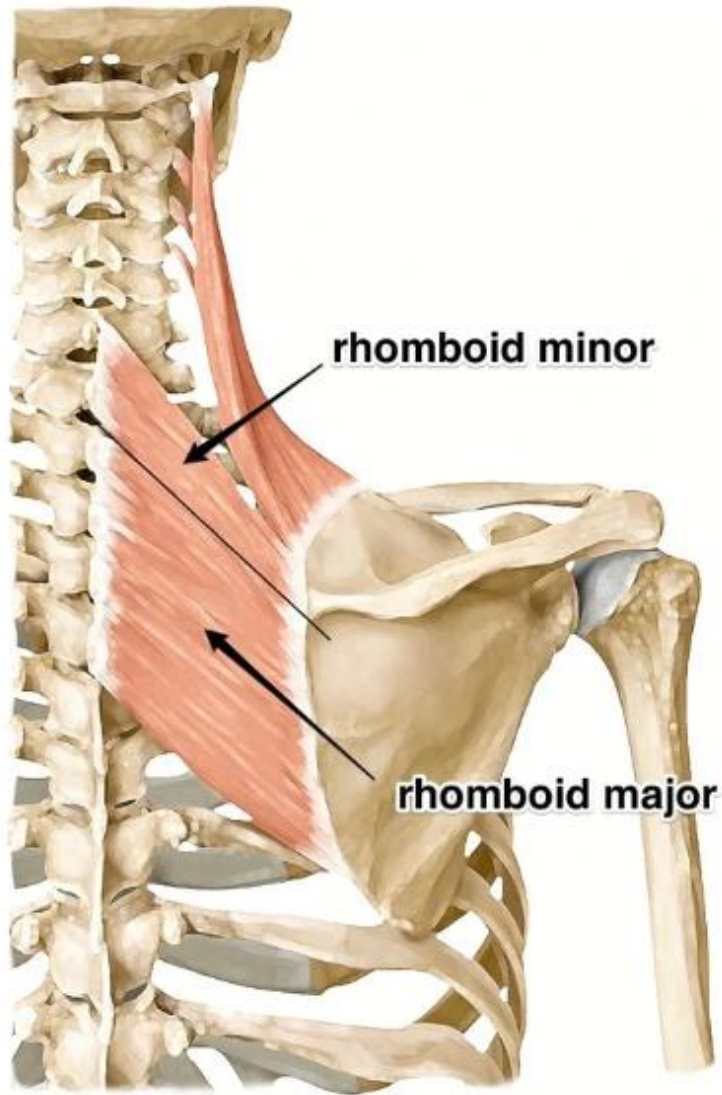


6- Rhomboid



- تتكون من عضلتين
 - Origin and Insertion
 - بداية العضلات من **العمود الفقري**
 - نهاية العضلات في **لوح الكتف**
- مكان هذه العضلات تحت عضلة الترايبس

Rhomboid major & minor



- **Origin :**

From **C7-T1** vertebrate

From **T2-T5** vertebrate

- **Insertion :**

Medial border of **Scapula**

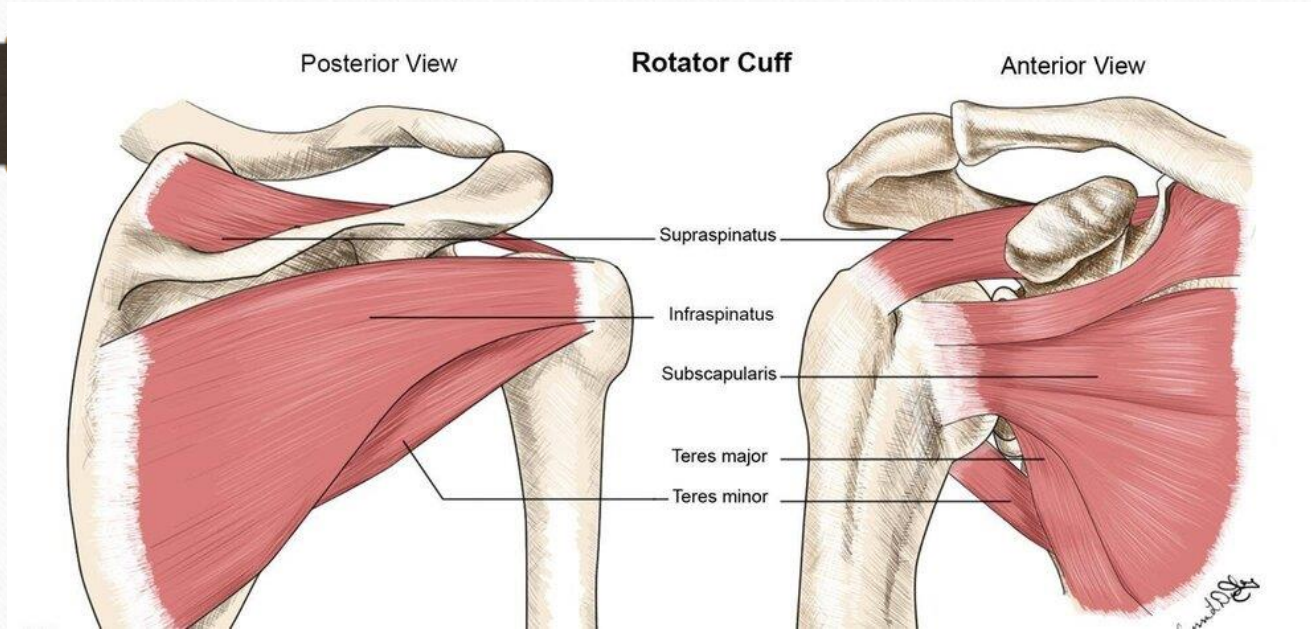
Motion of the muscle

- Shoulder retraction

Training anatomy Rhomboid Shoulder retraction
Seated machine rows wide grip

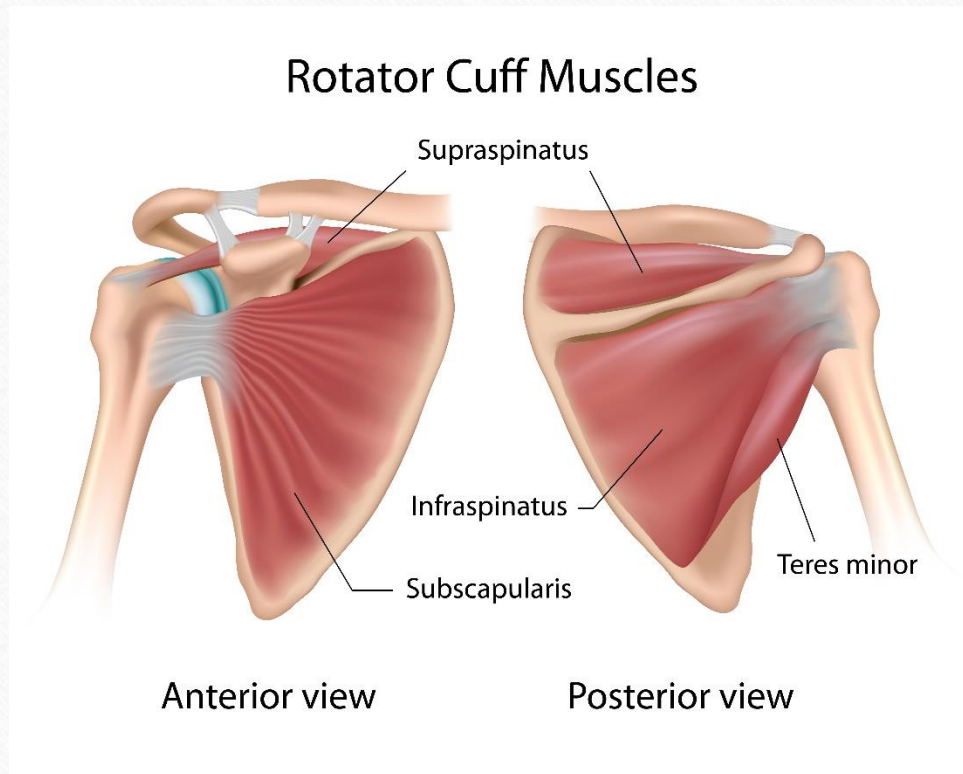
4- Rotator cuff muscles

- عضلات الكفة المدورة تتكون من 4 عضلات



- Origin and Insertion
- بداية العضلة من **لوح الكتف**
- **النهاية في عظمة الذراع العضضية**
- يوجد 3 عضلات في خلف لوح الكتف
 - و واحدة تمسك من الامام

RCM Supraspinatus, Infraspinatus, Subscapularis & Teres minor



- **Origin :**
Fossa of **scapula**, lateral border of **scapula**
- **Insertion : **Humerus****

Supraspinatus

- Shoulder abduction

Lateral raises Shoulder abduction

Infraspinatus

- Arm lateral rotation External rotation
- Shoulder transverse abduction

Reverse fly Shoulder transverse abduction

Shoulder external rotation bands

Teres minor

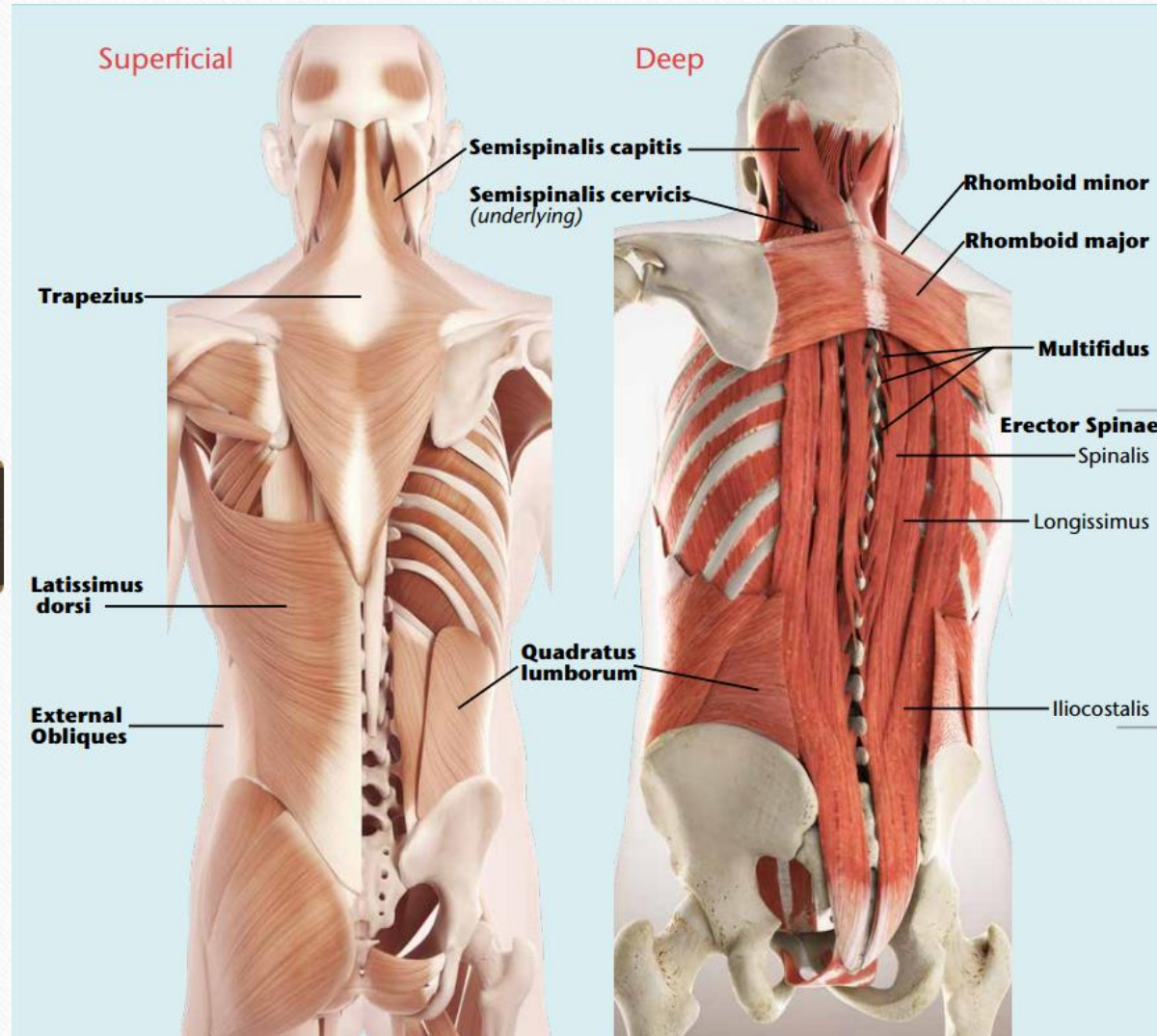
- Arm lateral rotation

Shoulder external rotation

Subscapularis

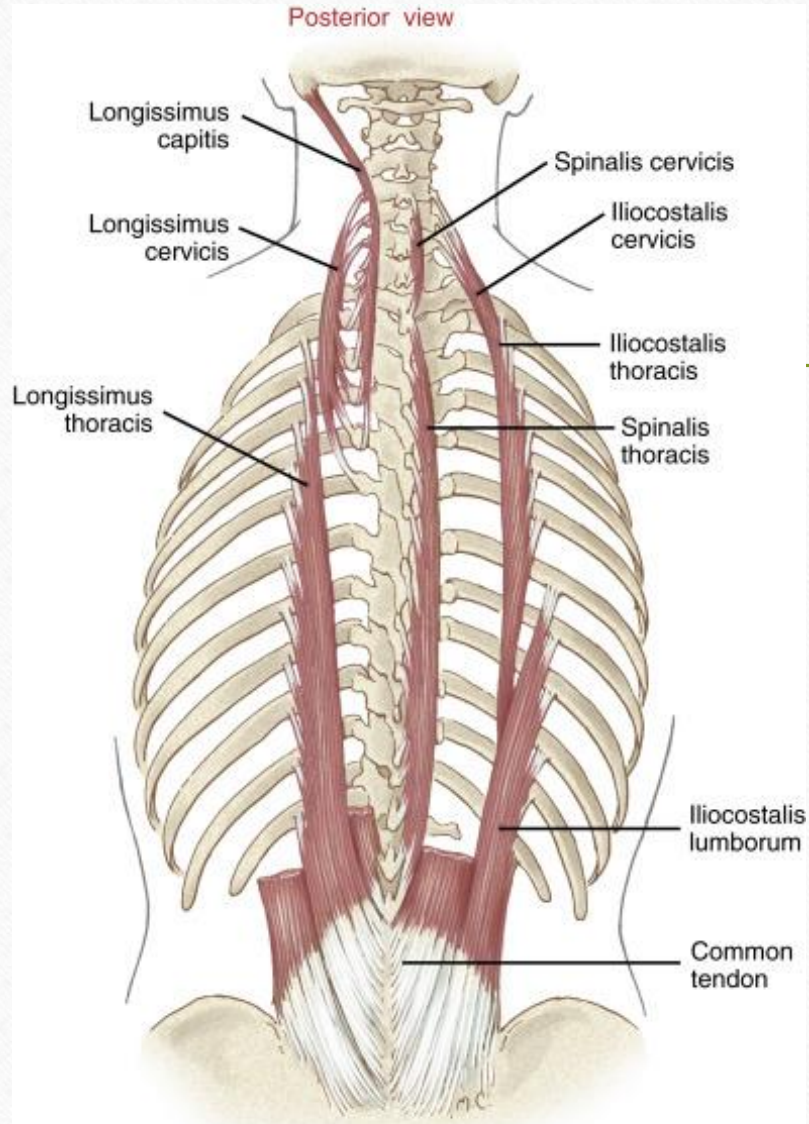
- Arm medial rotation Internal rotation

Internal and external rotation



9- Erector spinae muscles

- عضلات القطنية تنقسم الى 3 اجزاء
 - هذه العضلات طويلة و رفيعة
 - Origin and Insertion
- البداية من العمود الفقري و الضلوع و الحوض
- النهاية فى العمود الفقري و الضلوع و الجمجمة



Erector spinae muscles

- **Origin :**

Pelvic, vertebral, and rib bones located in the lower-to-mid back

- **Insertion :**

Vertebral and rib bones located higher up in the back, neck, and skull

Erector spinae muscles

- Extension and hyperextension
- Spine lateral flexion

Back extensions

Super man exercise

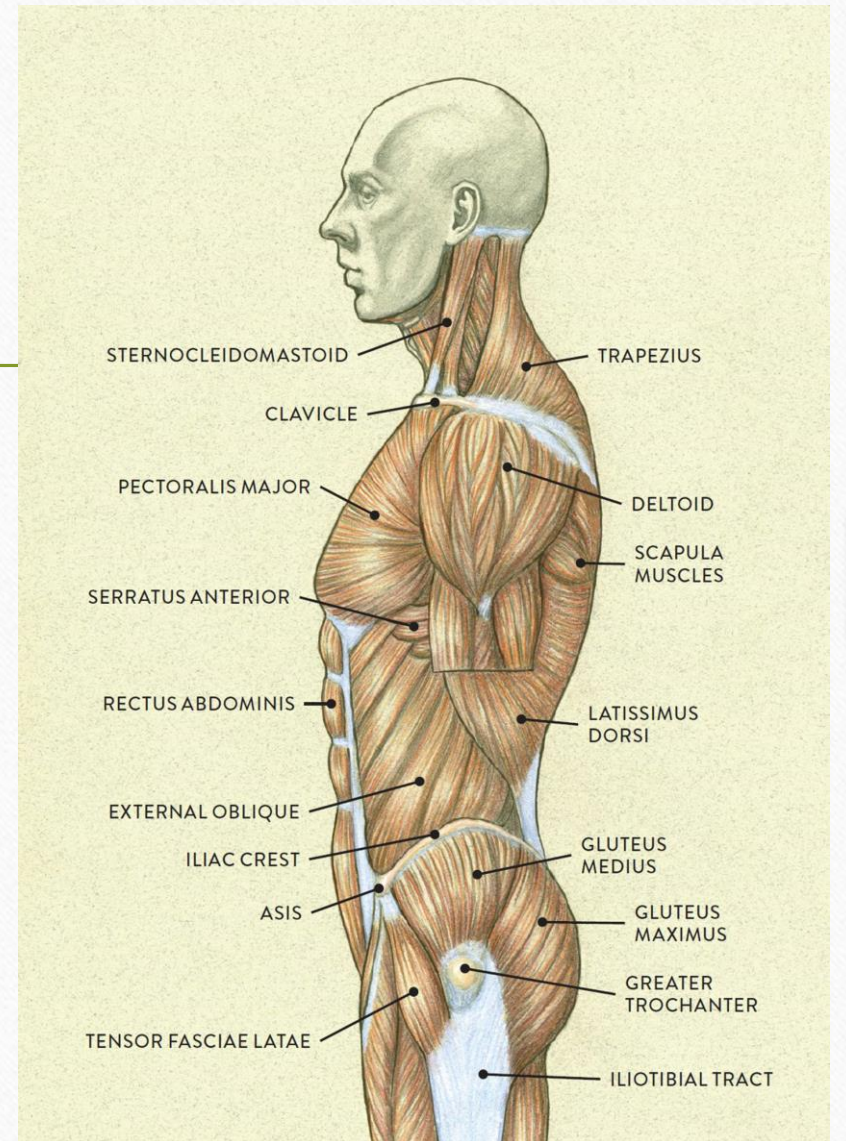
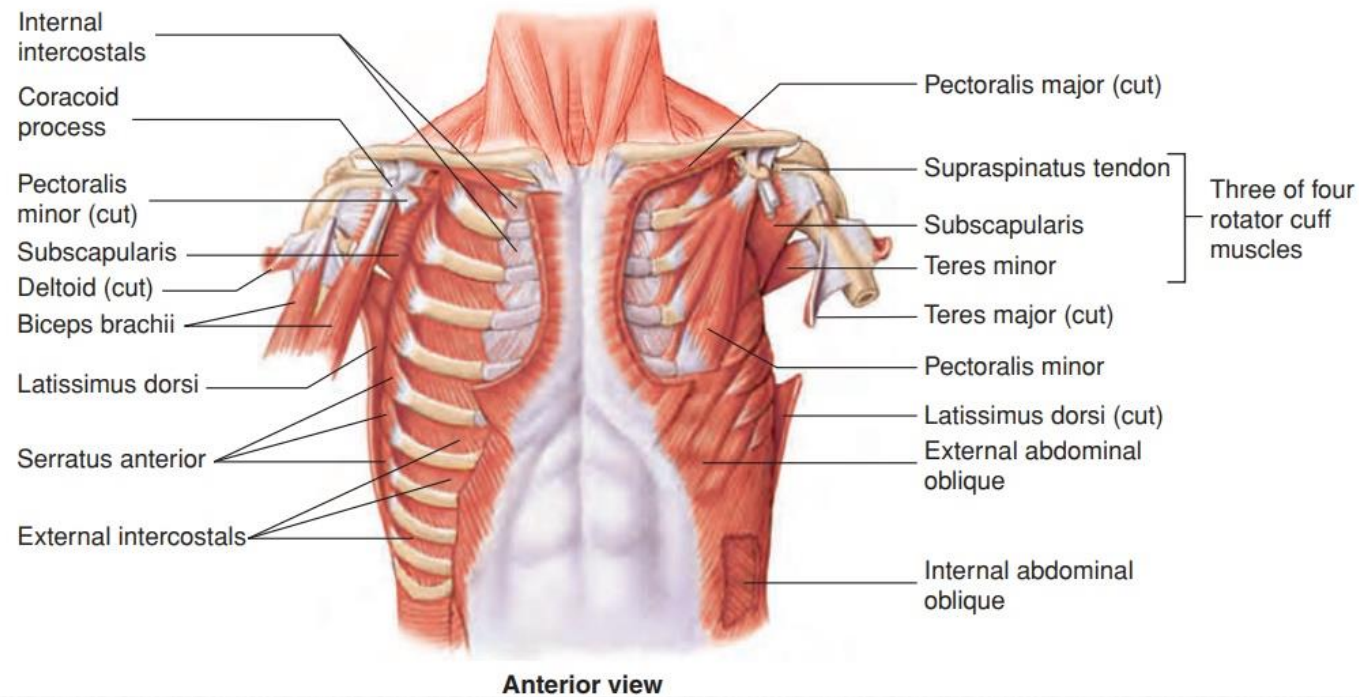
Bird Dog exercise

Back lever

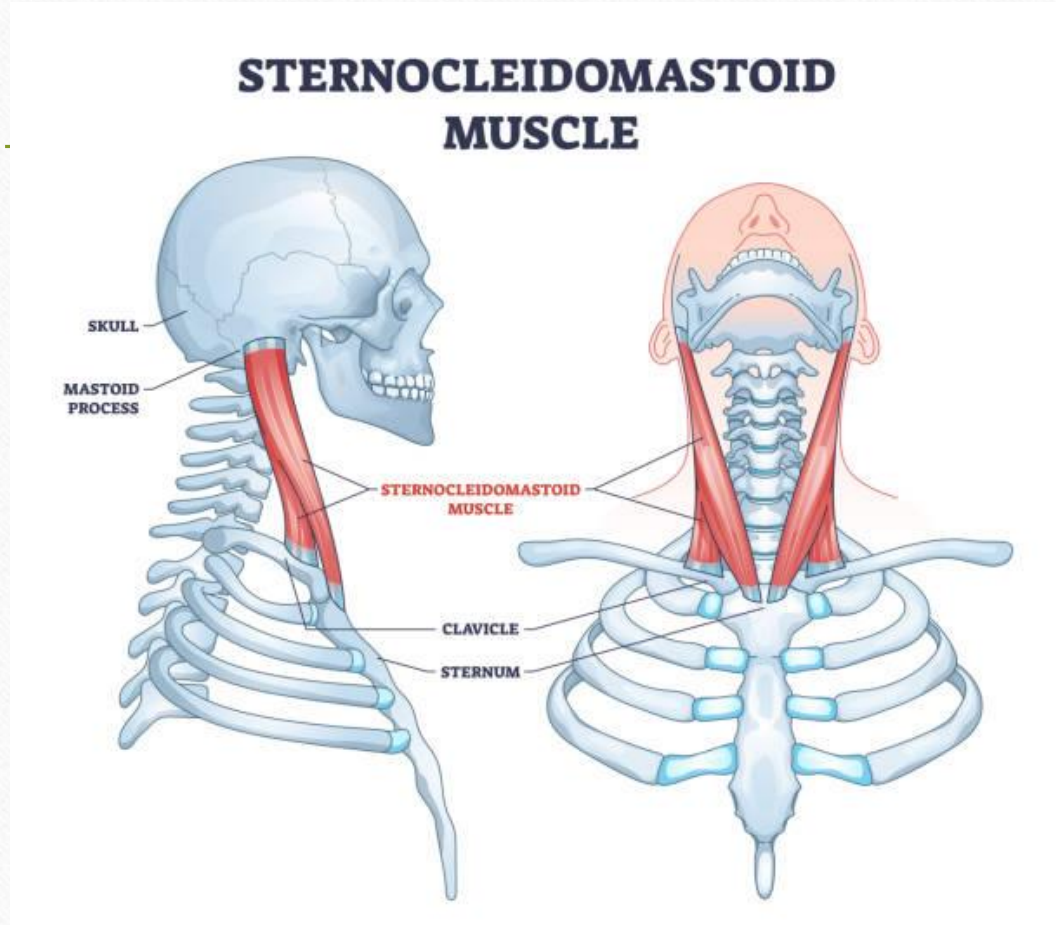
Back line

Side Bends

Front muscles of the upper body

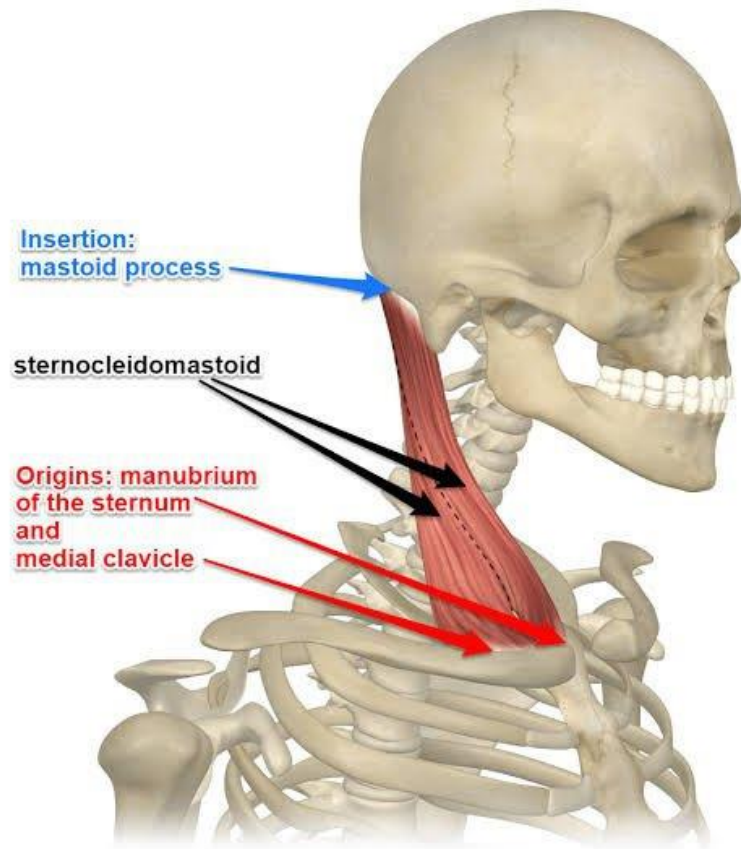


17- Sternocleidomastoid



- هذه العضلة
- Origin and Insertion
- بدايتها من عظمة التركوة
و عظمة القفص الصدري
- نهايتها فى الجمجمة من الجانب

Sternocleidomastoid



- **Origin :**
Sternum and the Clavicle
- **Insertion :**
Calcaneus

Motion of the muscle

- Spine lateral flexion

- Head & Cervical

Spine rotation

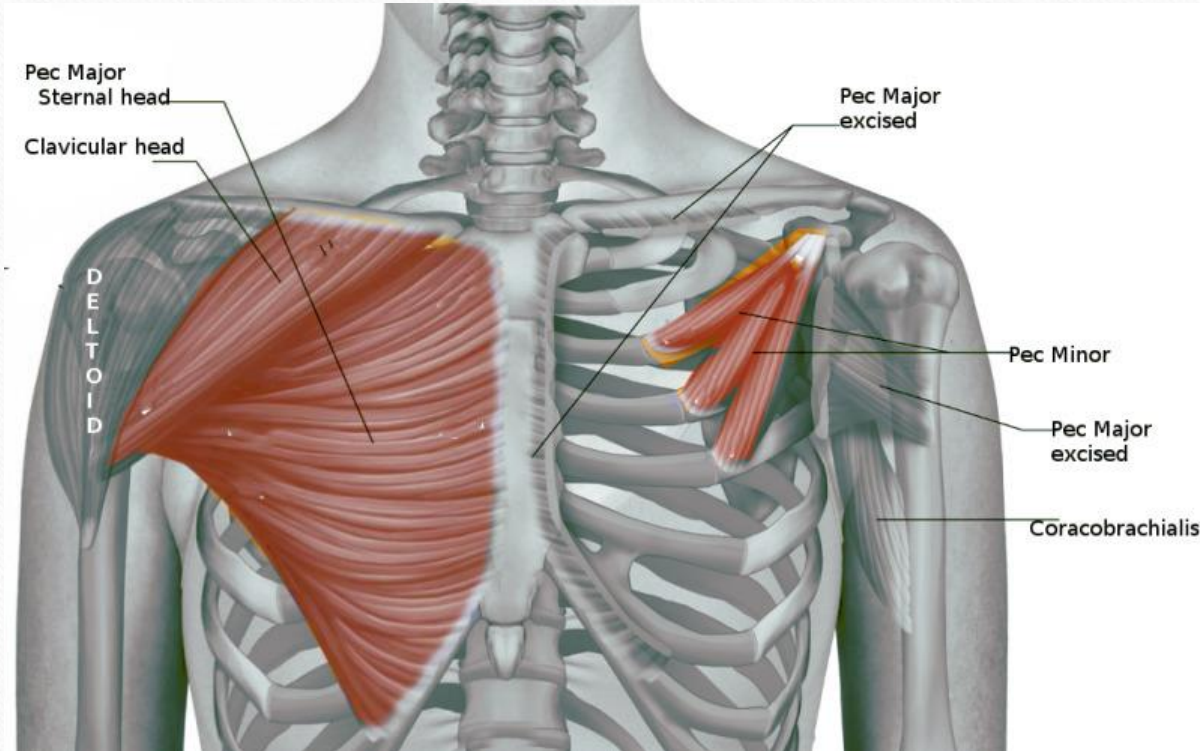
- Cervical spine flexion

Neck Flexion (Bodyweight)

Neck Lateral Flexion

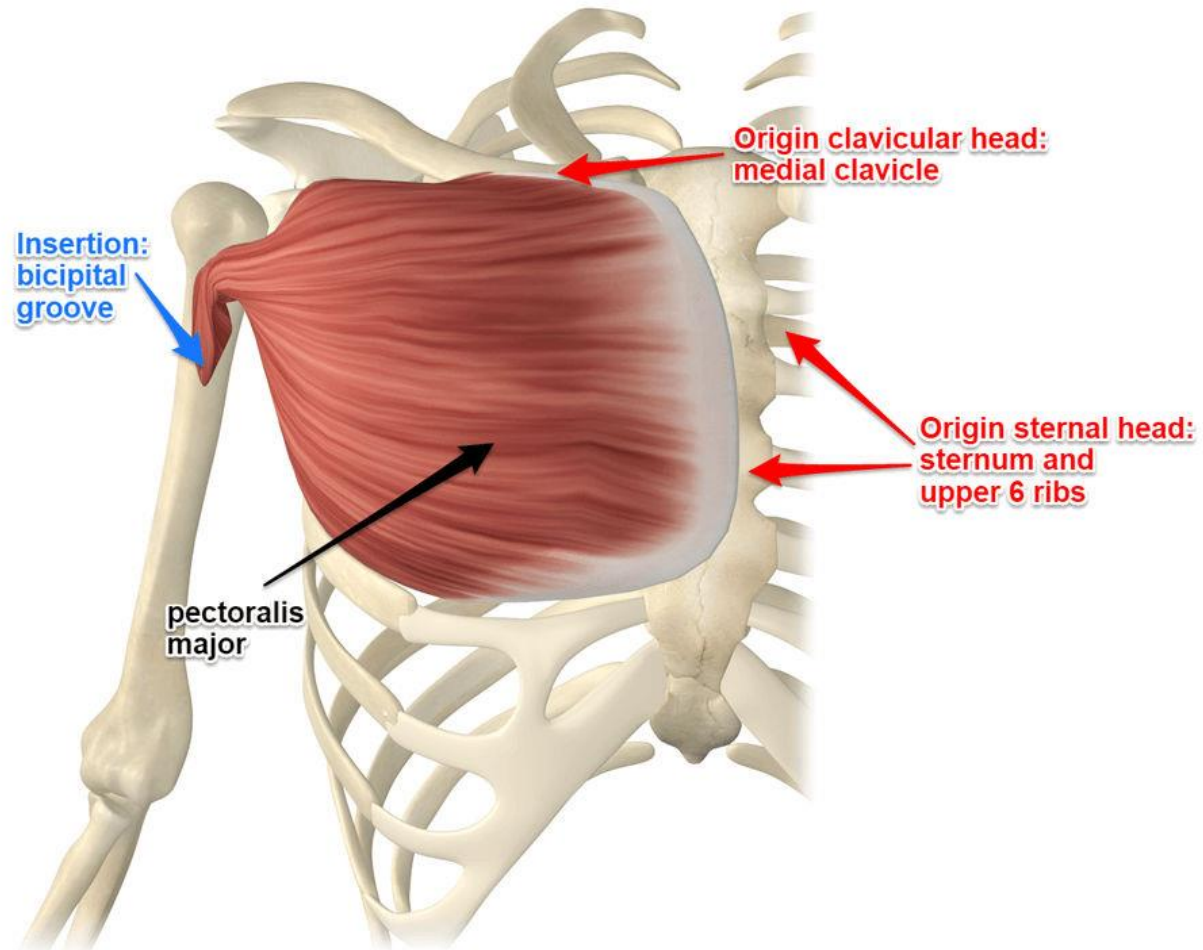
Neck Rotation (Cable)

21-22- Pectoralis muscle



- عضلات الصدر تنقسم الى عضلتين
 - عضلة الصدر الكبرى
 - عضلة الصدر الصغيرة
- Origin and Insertion
 - بداية العضلات من عظمة القفص الصدرى والتركوة و الضلوع
 - نهاية العضلات فى عظمة الدراع العضضية و فى لوح الكتف

pectoralis major



- Origin :

Sternum & Clavicle bone

- Insertion :

Humerus

Motion of the muscle

- Shoulder transverse adduction
- Shoulder adduction
- Shoulder flexion

Motion of the muscle

- Shoulder depression
- Arm medial rotation

Dumbbell flys Sternal head

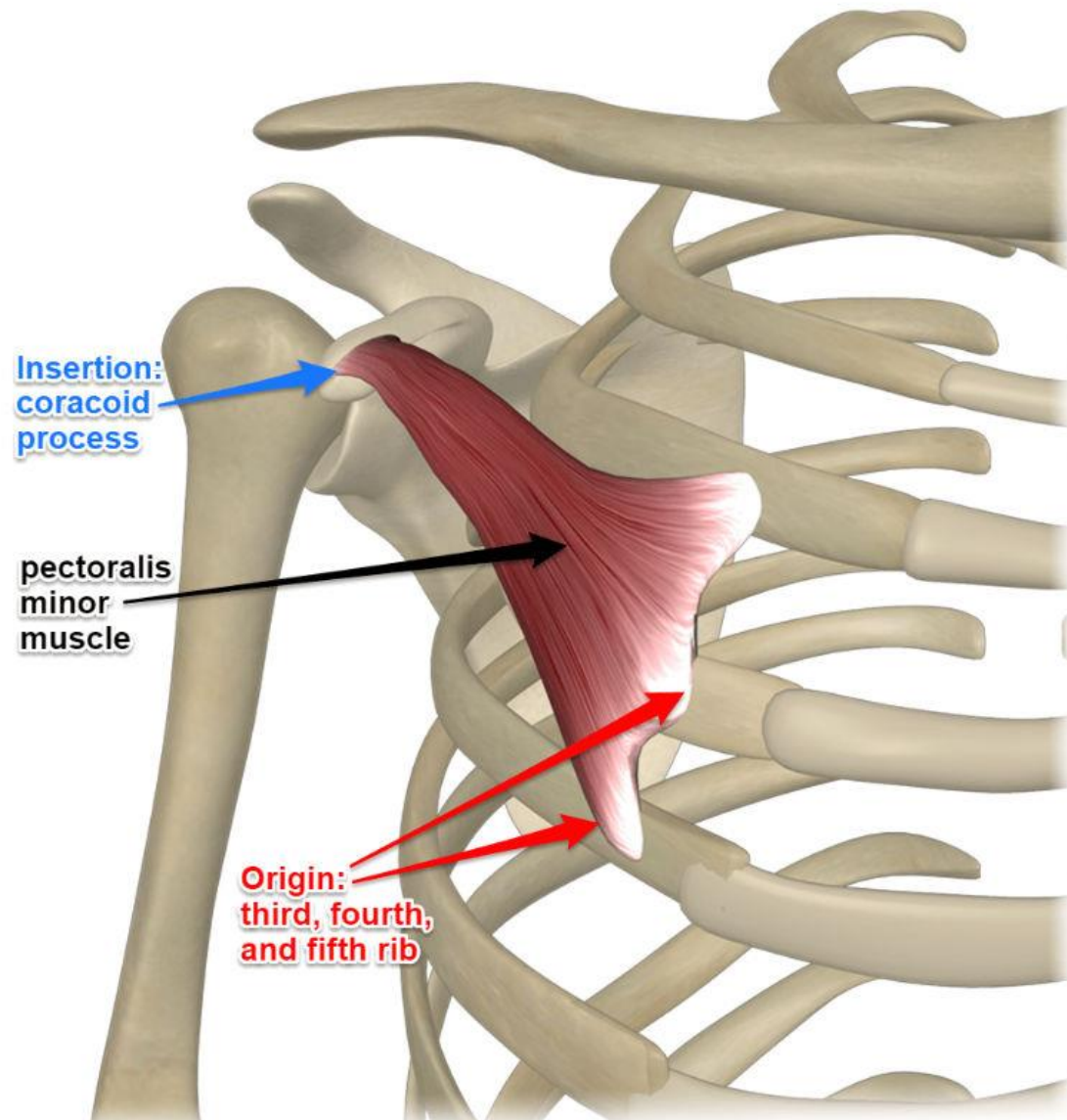
Lower cable flys lower fibers

Upper cable flys clavicular head

Barbell presses

Dumbbell presses

pectoralis minor



- **Origin :**

3rd, 4th & 5th rib

- **Insertion :**

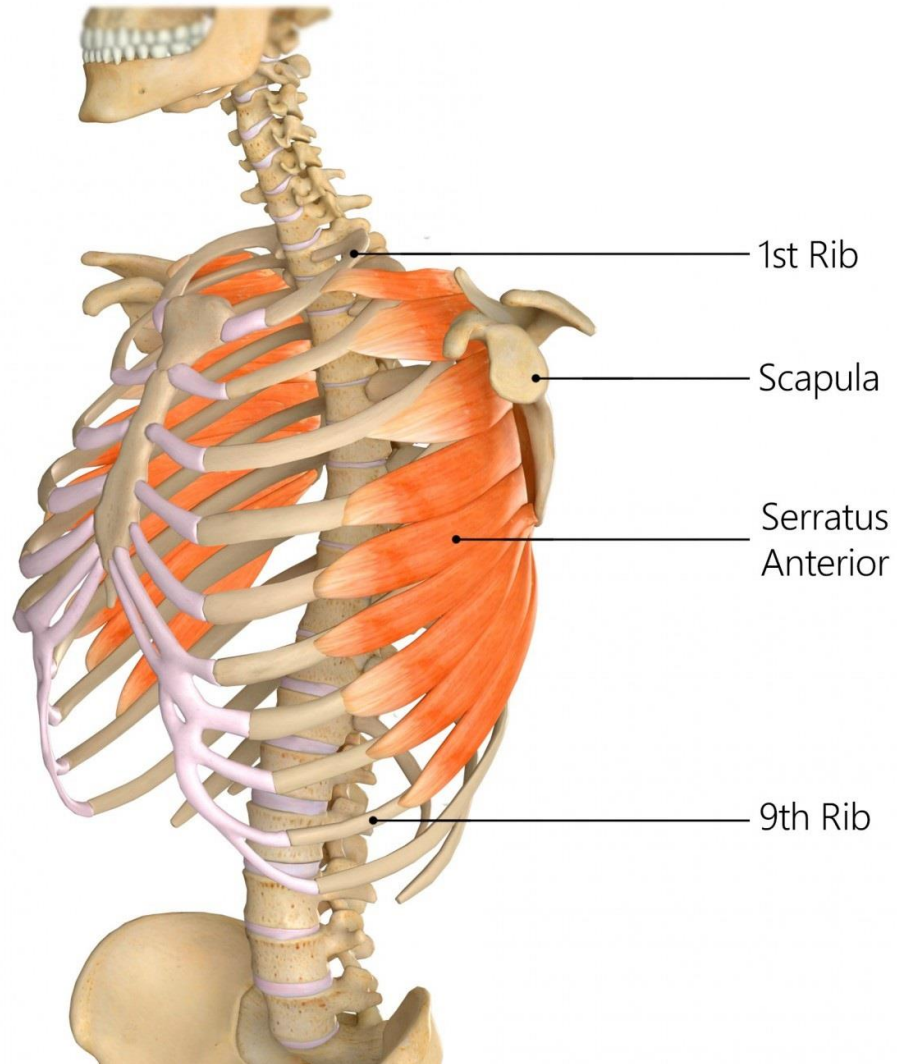
Coracoid process of Scapula

Motion of the muscle

- Shoulder protraction
- Shoulder depression

Dips

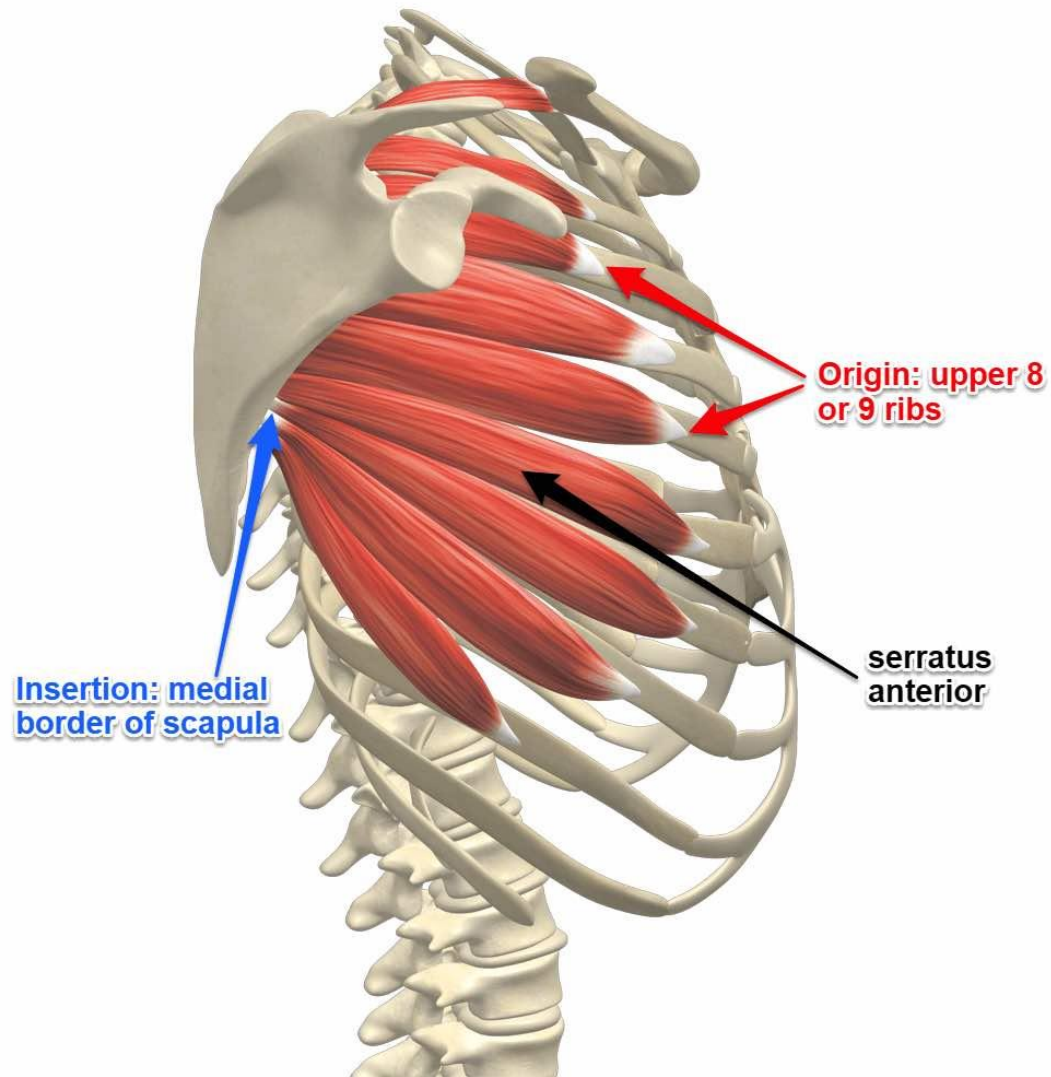
Serratus Anterior



24- Serratus anterior

- العضلة المنشارية
- Origin and Insertion
- بداية العضلة من الضلوع
- نهاية العضلة في لوح الكتف

Serratus anterior



- **Origin :**

1st, 2nd, 3rd, 4th, 5th, 6th,
7th, 8th, & 9th rib

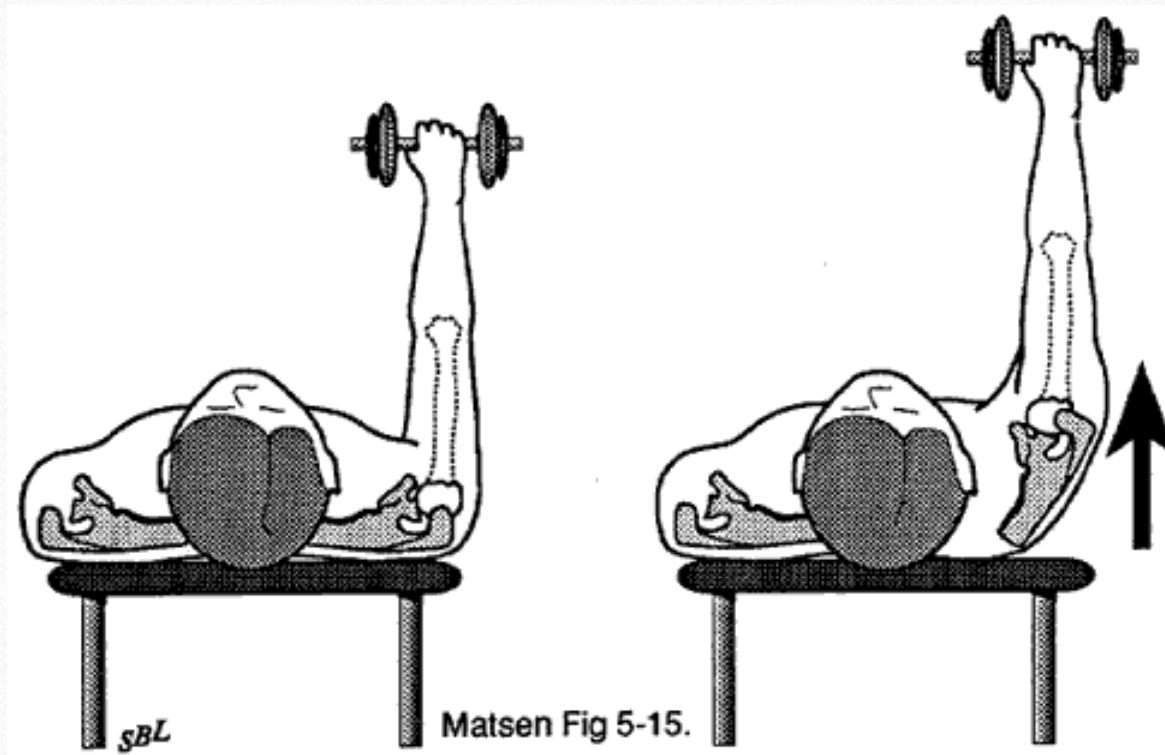
- **Insertion :**

Medial border of **Scapula**

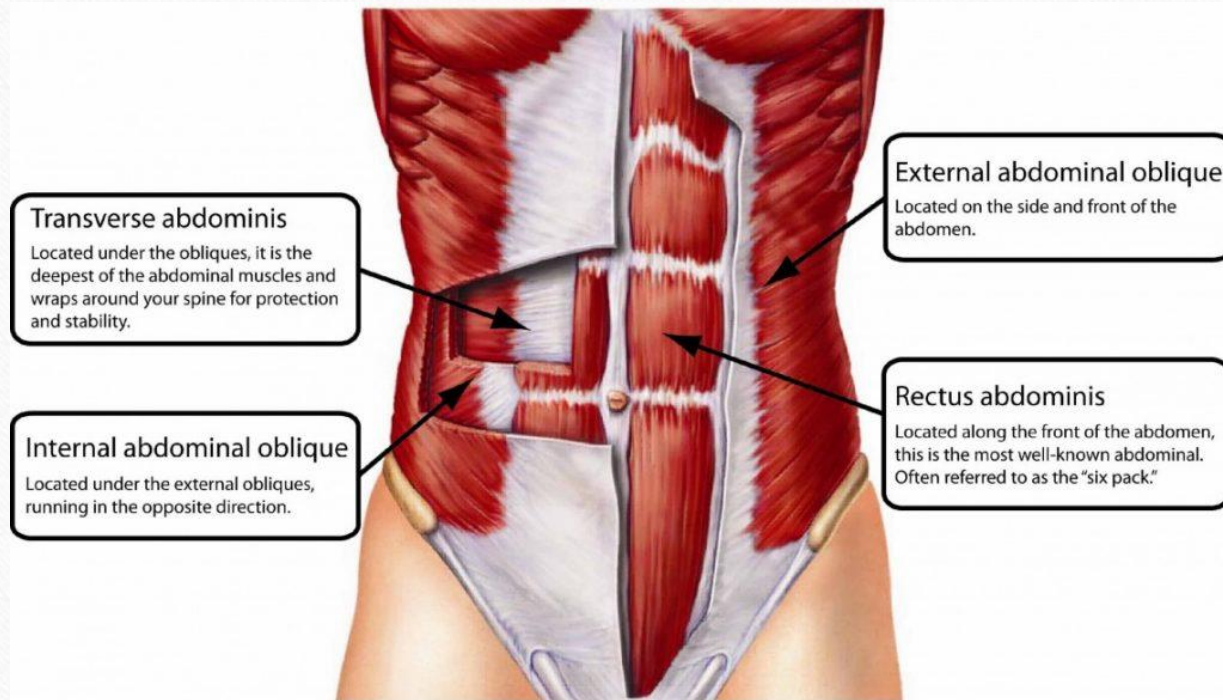
Motion of the muscle

- Shoulder protraction

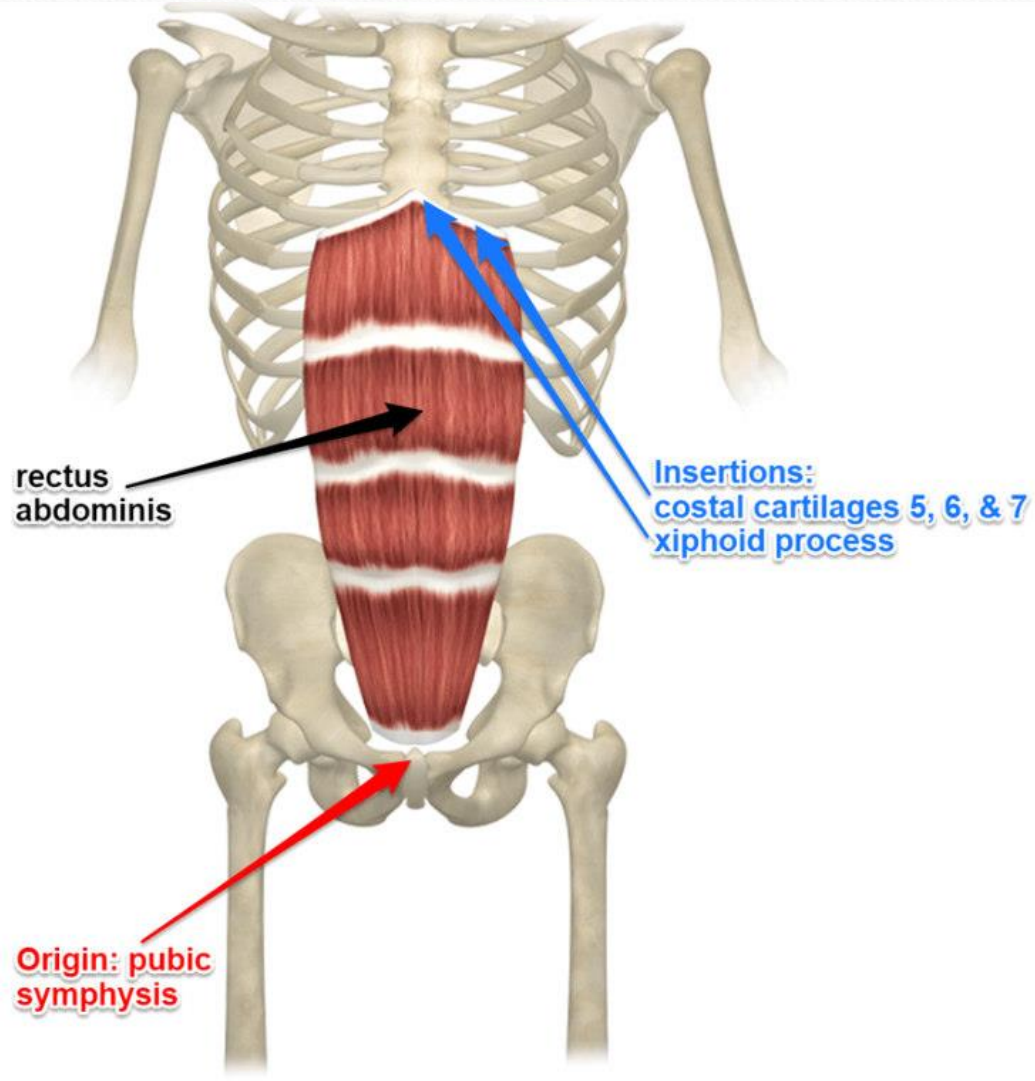
Shoulder protraction



25-26-28- Abdominal muscles



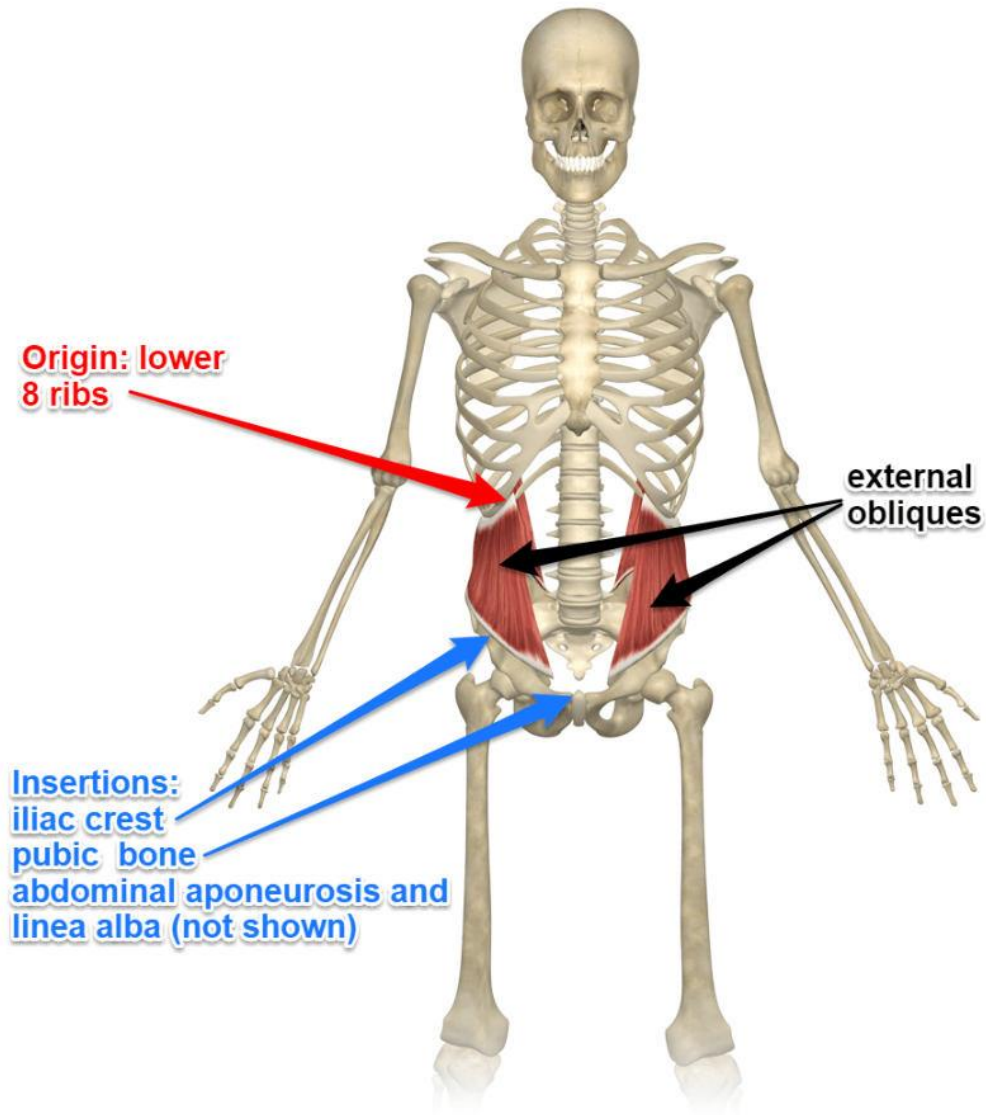
- عضلات البطن تنقسم الى 4 عضلات
 - Origin and Insertion
 - بداية العضلات من الحوض و الضلوع
- نهاية العضلات في الحوض و الضلوع و نهاية عظمة القفص الصدري



Rectus abdominis

- 6 bands muscles
- **Origin :**
Pubic crest of the **Pelvic**
- **Insertion :**
Xiphoid process of the **Sternum**

External obliques



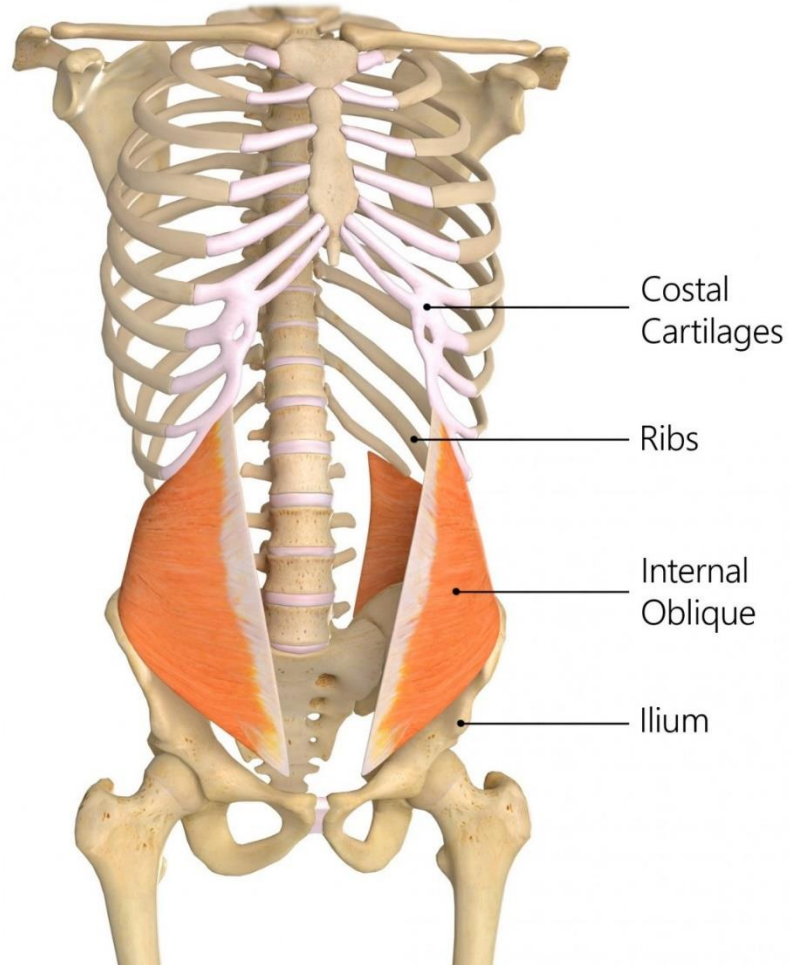
- **Origin :**

5th, 6th, 7th, 8th, 9th, 10th, 11th, & 12th rib

- **Insertion :**

Pubic crest, iliac spine and iliac crest of the pelvic

Internal Oblique



Internal obliques

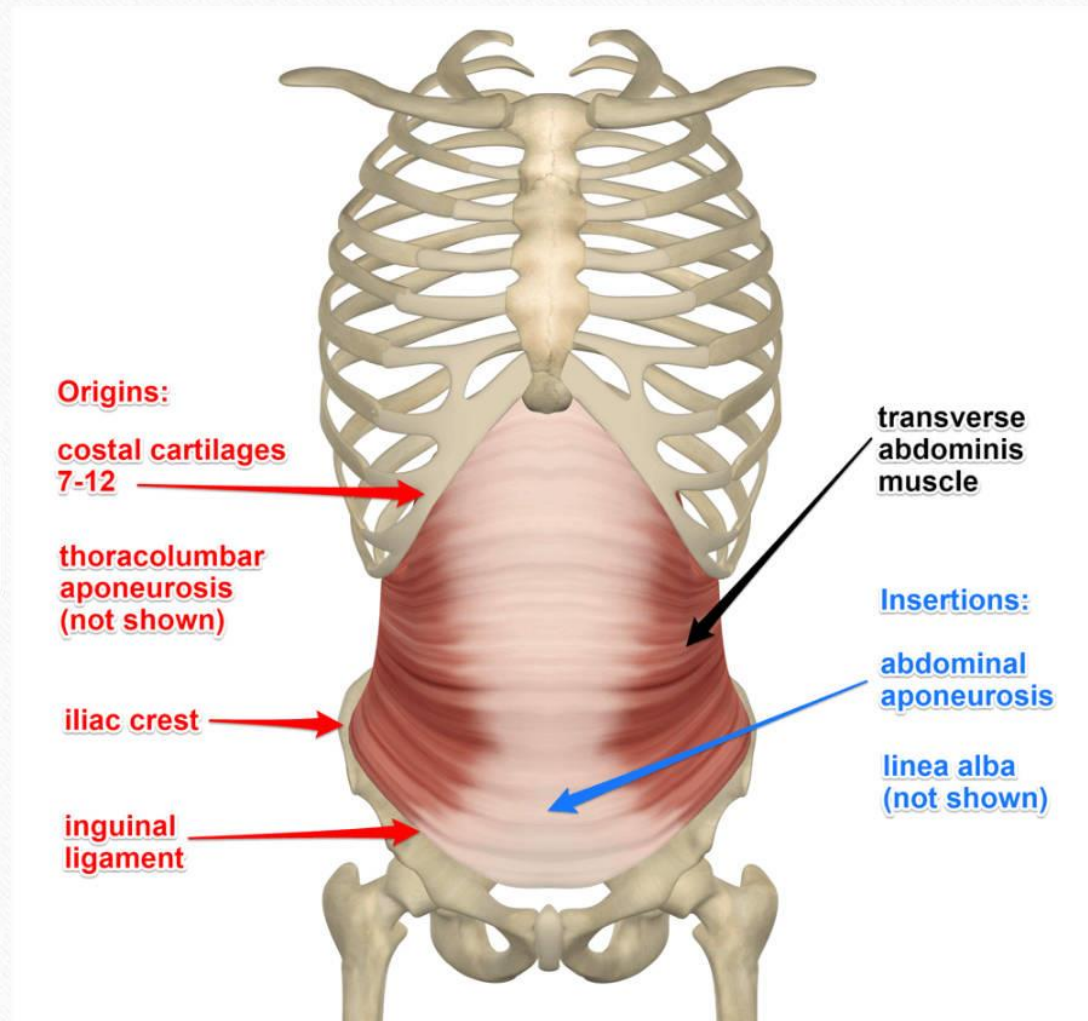
- **Origin :**

Iliac crest of the **pelvic**

- **Insertion :**

10th , 11th , & 12th rib

Transversus abdominis



- **Origin :**

Iliac crest of the **pelvic**

- **Insertion :**

Pubic crest and pectin pubis of the **pelvic**

Motion of the muscle

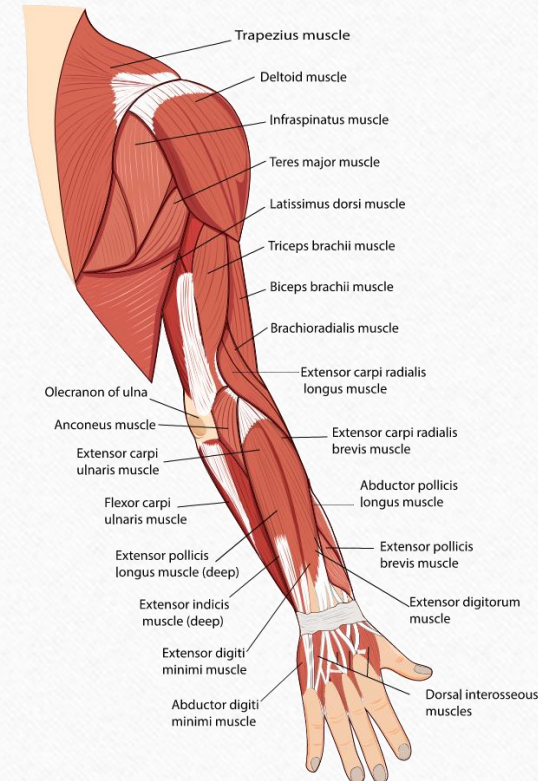
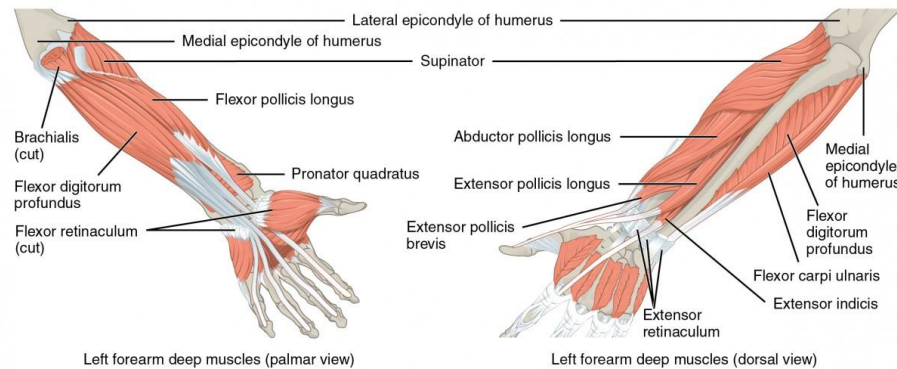
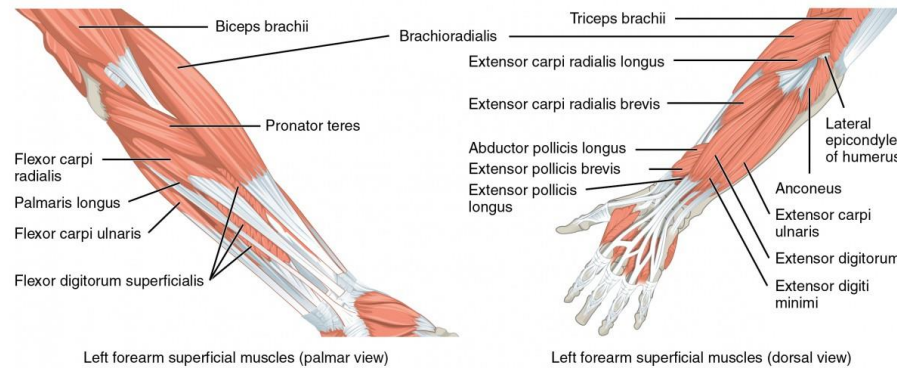
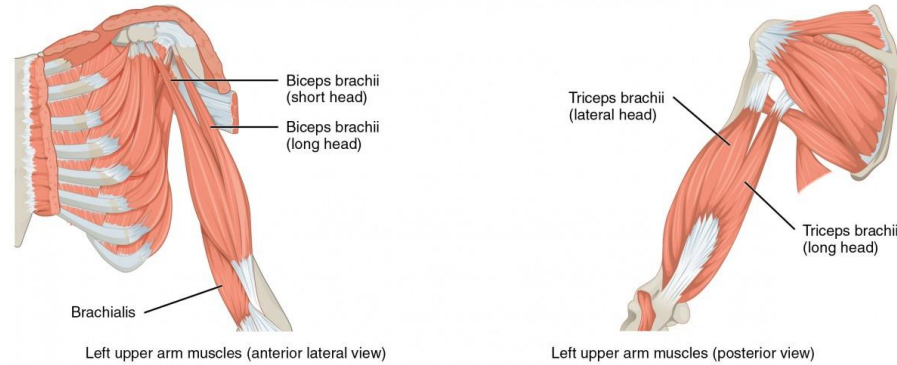
- Trunk flexion
- Trunk lateral flexion
- Trunk rotation

Machine crunches Trunk flexion

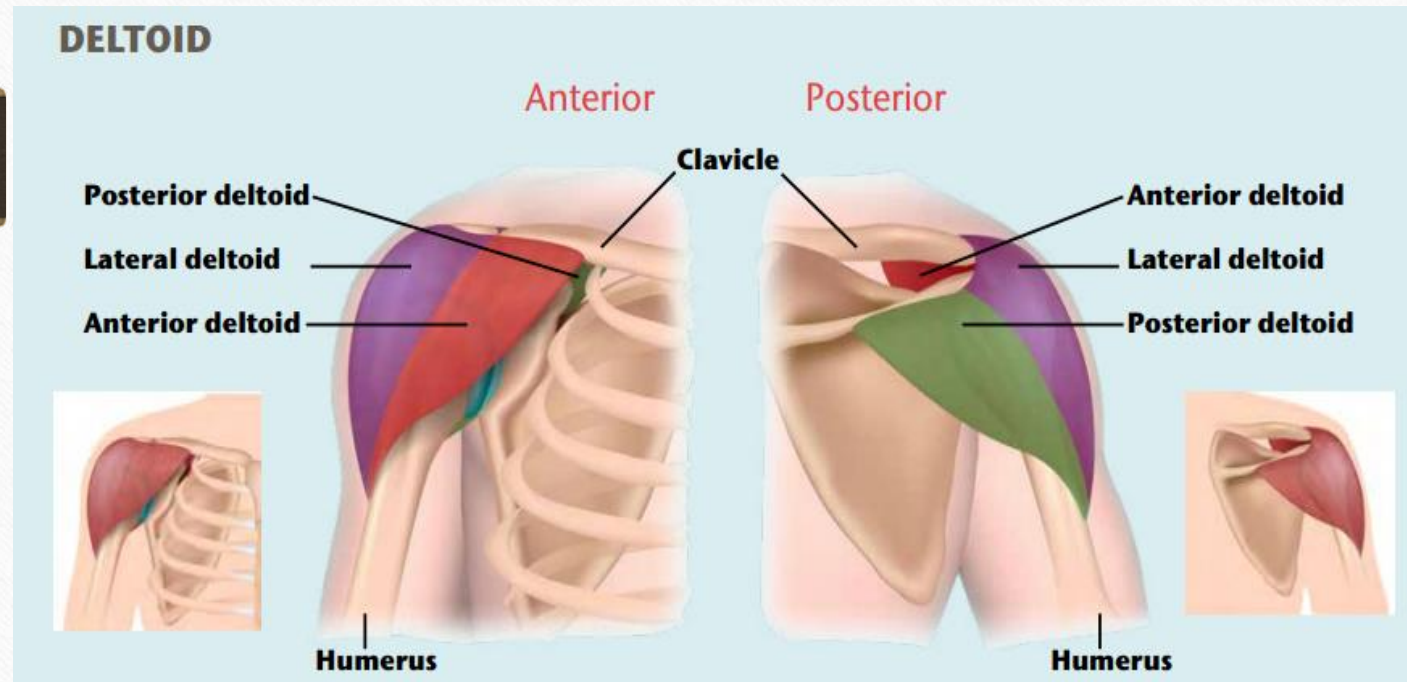
Side plank Trunk lateral flexion

Cable twist Trunk rotation

Upper limb (Arm muscles)



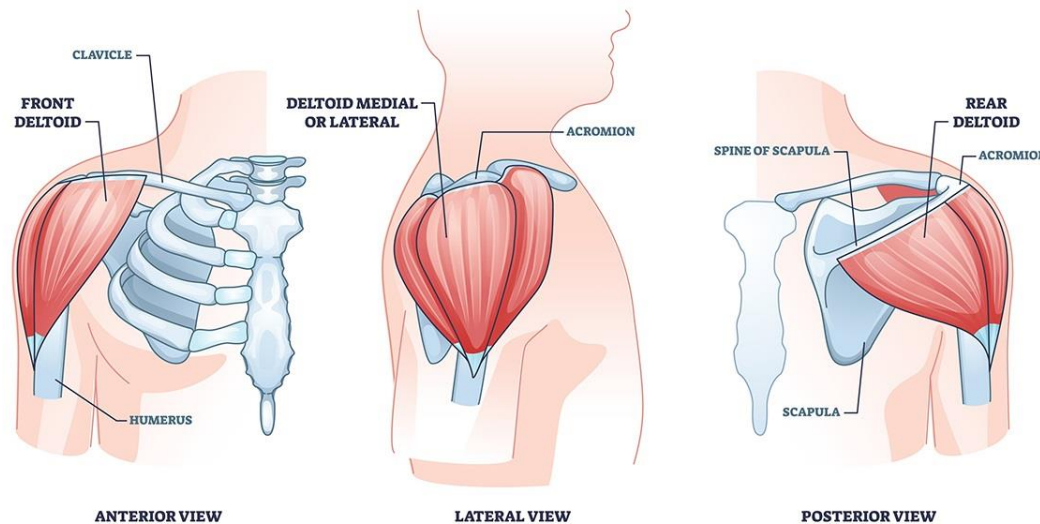
3-19-20- Deltoid muscle:



- عضلة الكتف لها 3 اجزاء
- الجزء الامامى و الجانبى والخلفى
- Origin and Insertion
- البداية من لوح الكتف و التركوة
- النهاية فى عظمة الدراع العضضية

Deltoid muscle: Anterior, Medial, & posterior parts

Deltoid Muscles



Origin :

the anterior border and upper surface of the lateral third of the **clavicle**, **acromion**, spine of the **scapula**

Insertion : deltoid tuberosity of **Humerus**

Anterior deltoid

- Arm medial rotation
- Shoulder transverse
adduction
- Shoulder flexion

Front raises

Medial deltoid

- Shoulder abduction

Lateral raises

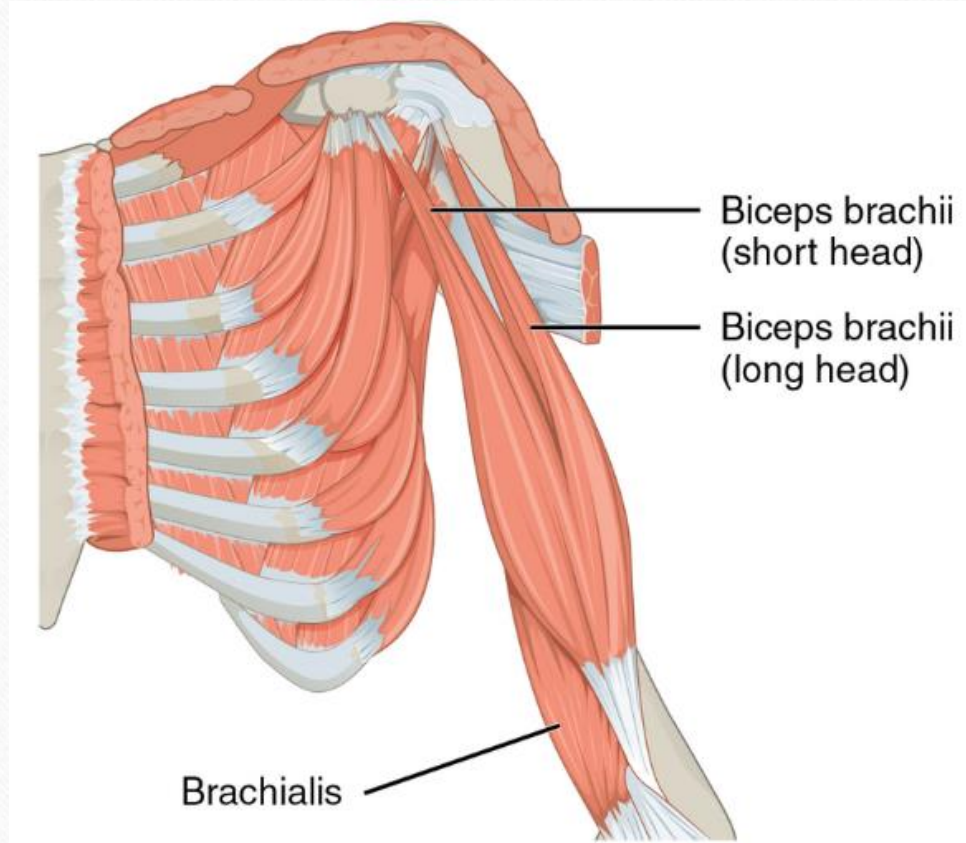
Posterior deltoid

- Arm lateral rotation
- Shoulder transverse abduction
- Shoulder extension & hyperextension

Face Pulls with External Rotation

Bent-over High Row

23- Biceps brachii



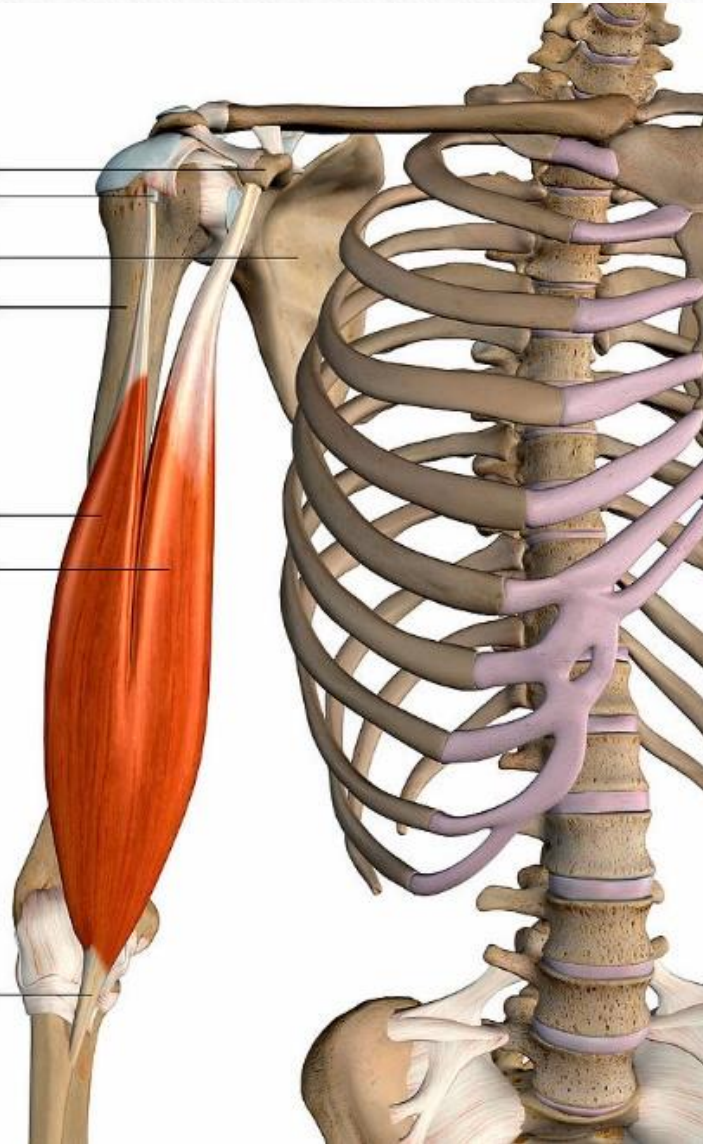
- عضلة الباي تنقسم الى عضلتين
 - عضلة الراس الطويلة
 - عضلة الراس الصغيرة
- Origin and Insertion
- بداية العضلة من لوح الكتف
- نهاية العضلة في عظمة الساعد

Biceps Brachii Muscles
Anterior View

Coracoid Process
Intertubercular Tendon Sheath
Scapula
Humerus

Biceps Brachii Long Head
Biceps Brachii Short Head

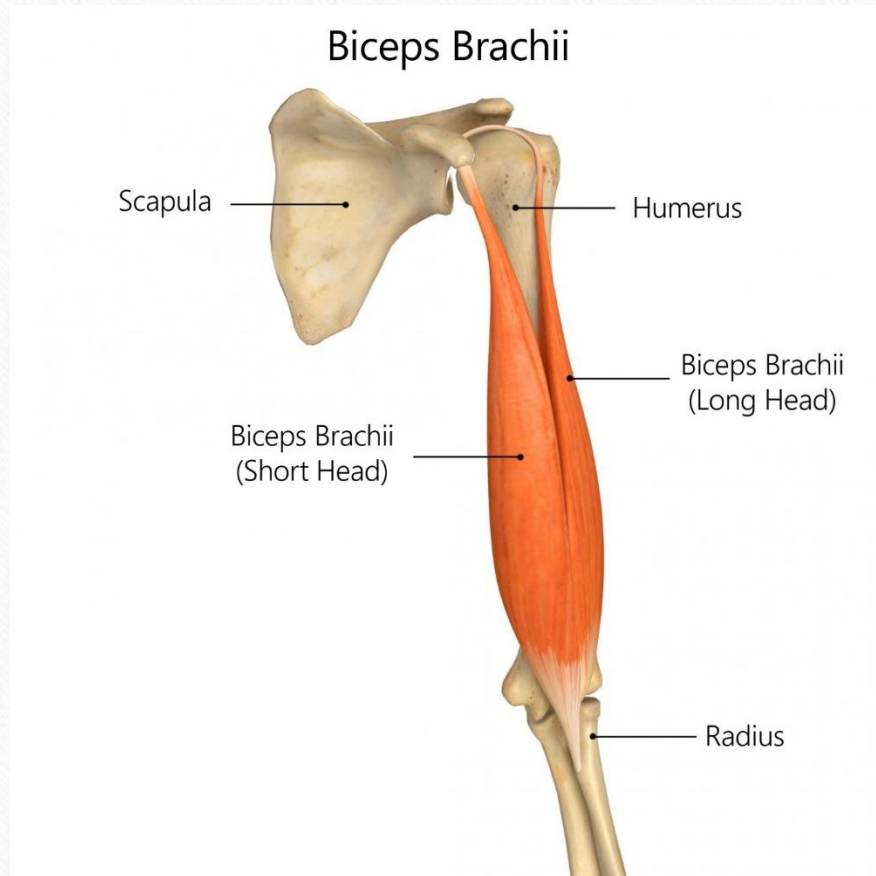
Biceps Brachii Tendon



Long head of the Biceps brachii

- **Origin :**
Tubercle of **scapula**
- **Insertion :**
Radial tuberosity of **radius**

Short head of the Biceps brachii



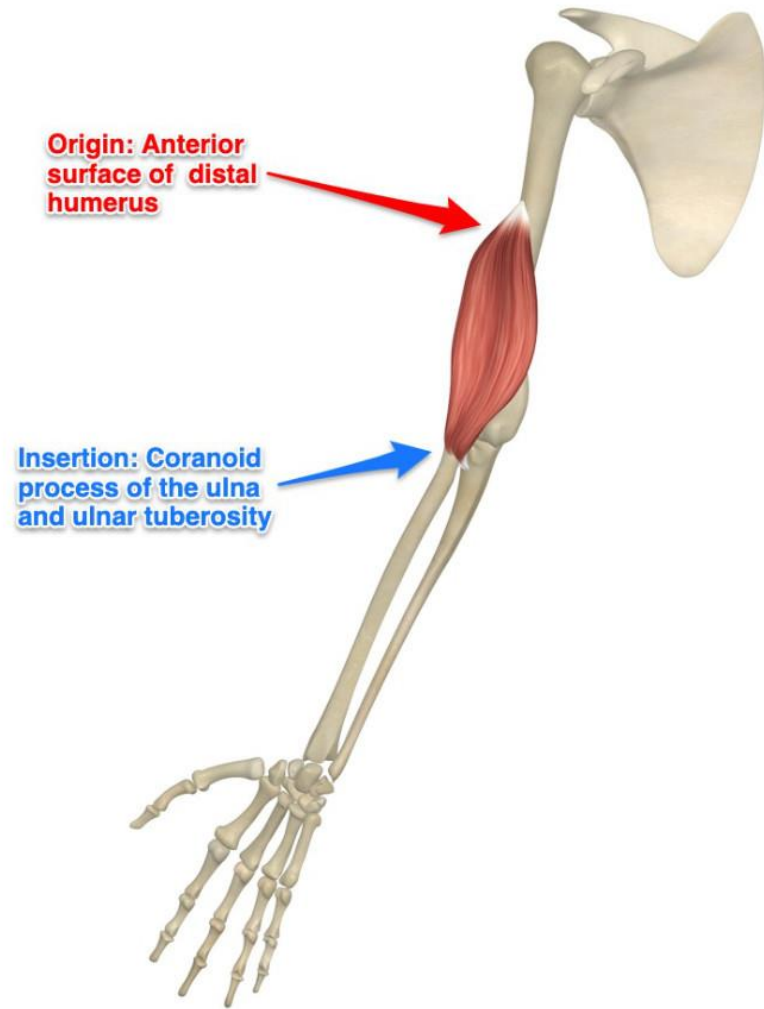
- **Origin :**

Coracoid process of **Scapula**

- **Insertion :**

Radial tuberosity of **radius**

Brachialis



- Origin :

Humerus

- Insertion :

Ulna

Biceps long and short head

- Elbow flexion
- Shoulder flexion
- Forearm
supination

Long head have 1 more action

- Shoulder abduction

Brachialis

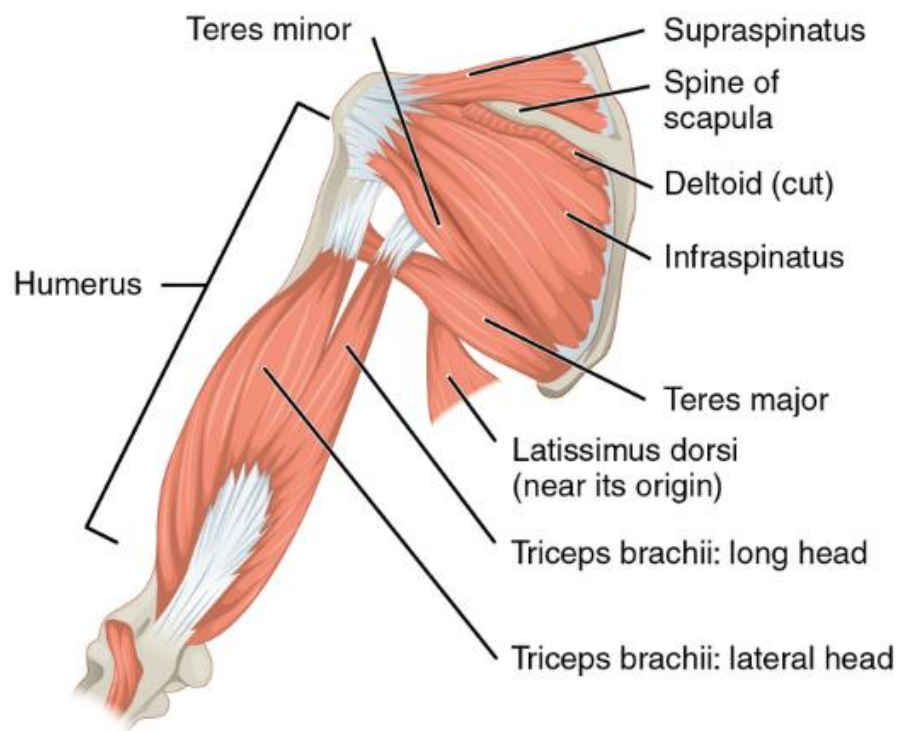
- Elbow flexion

Preacher curls (short head)

Incline biceps curls (Long head)

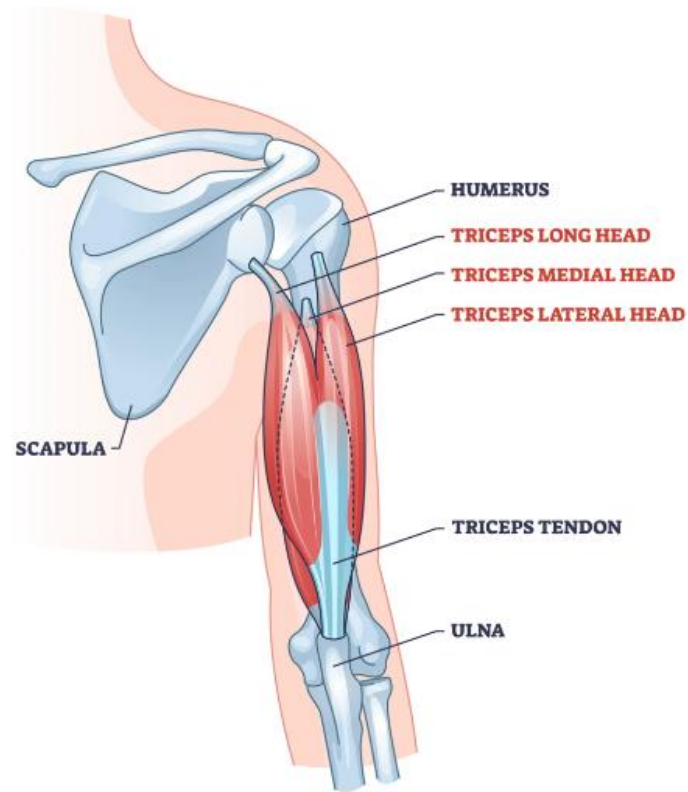
Hammer Curls (Dumbbell)

5-7- Triceps brachii



- عضلة التراى تتكون من 3 عضلات
- Origin and Insertion
- الراس الطويلة بدايتها من لوح الكتف
- الراس جانبية بدايتها من عظمة الدراع العضضية
- العضلة الوسطى بدايتها من عظمة الدراع العضضية
- نهاية العضلات الثلاثة فى الكوع عند عظمة الالنا

Triceps brachii long, short, & medial head



- Origin :

Humerus and Scapula

- Insertion :

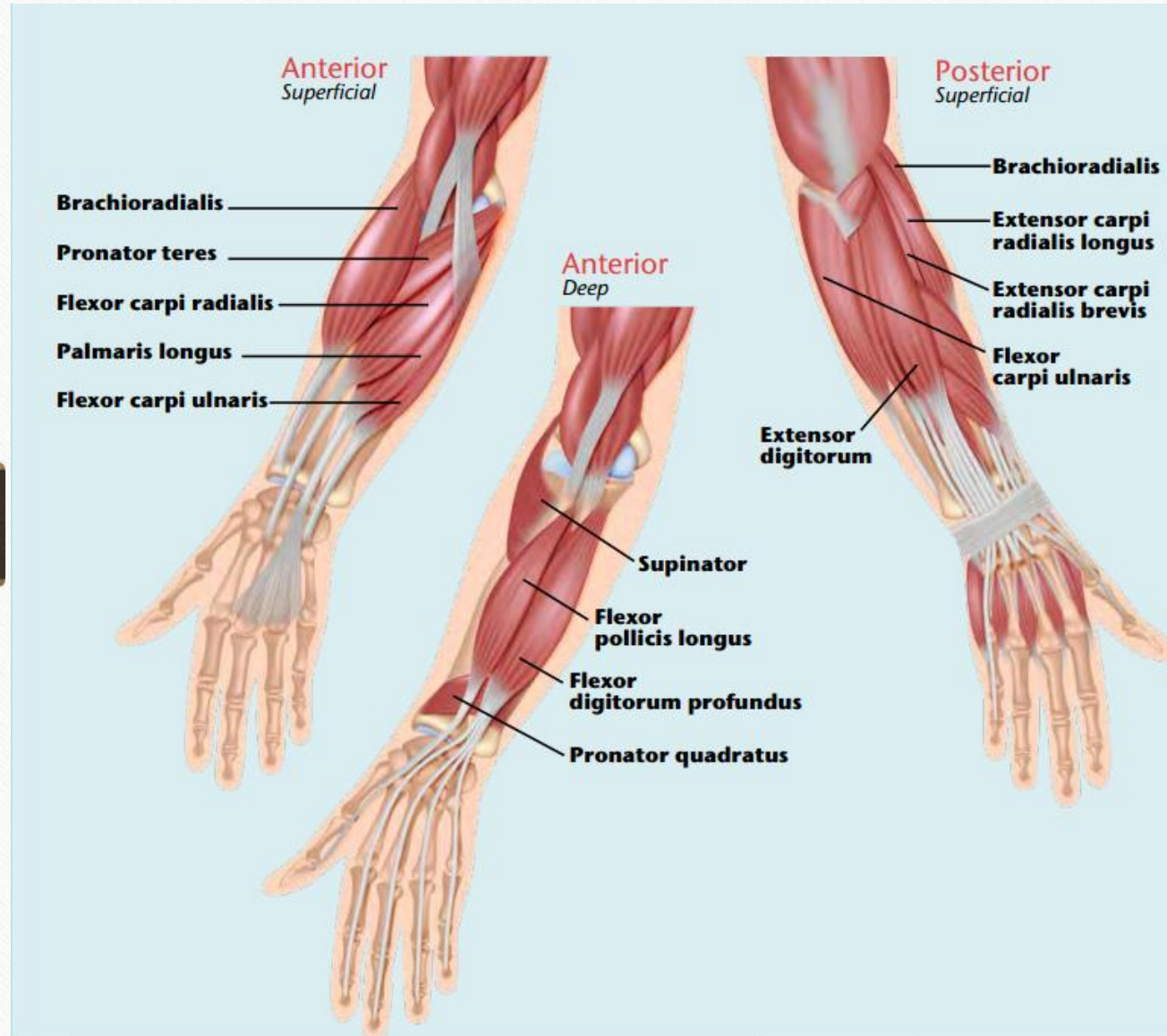
Ulna

Triceps

- Elbow Extension
- Shoulder Extension
& Hyperextension
(Long head)

Triceps Push-down (Cable)

Triceps Extension - Overhead (Dumbbell)



27-29- Wrist muscles

- عضلات الساعد يوجد منها السطحي و العميق
- تتكون من العضلات في جزء الامامي و عضلات في الجزء الخلفي
- Origin and Insertion
- بدايتها من عظمة الزراع و الساعد
- نهايتها في عظمة الساعد و الاصابع و عظم اليد

Wrist

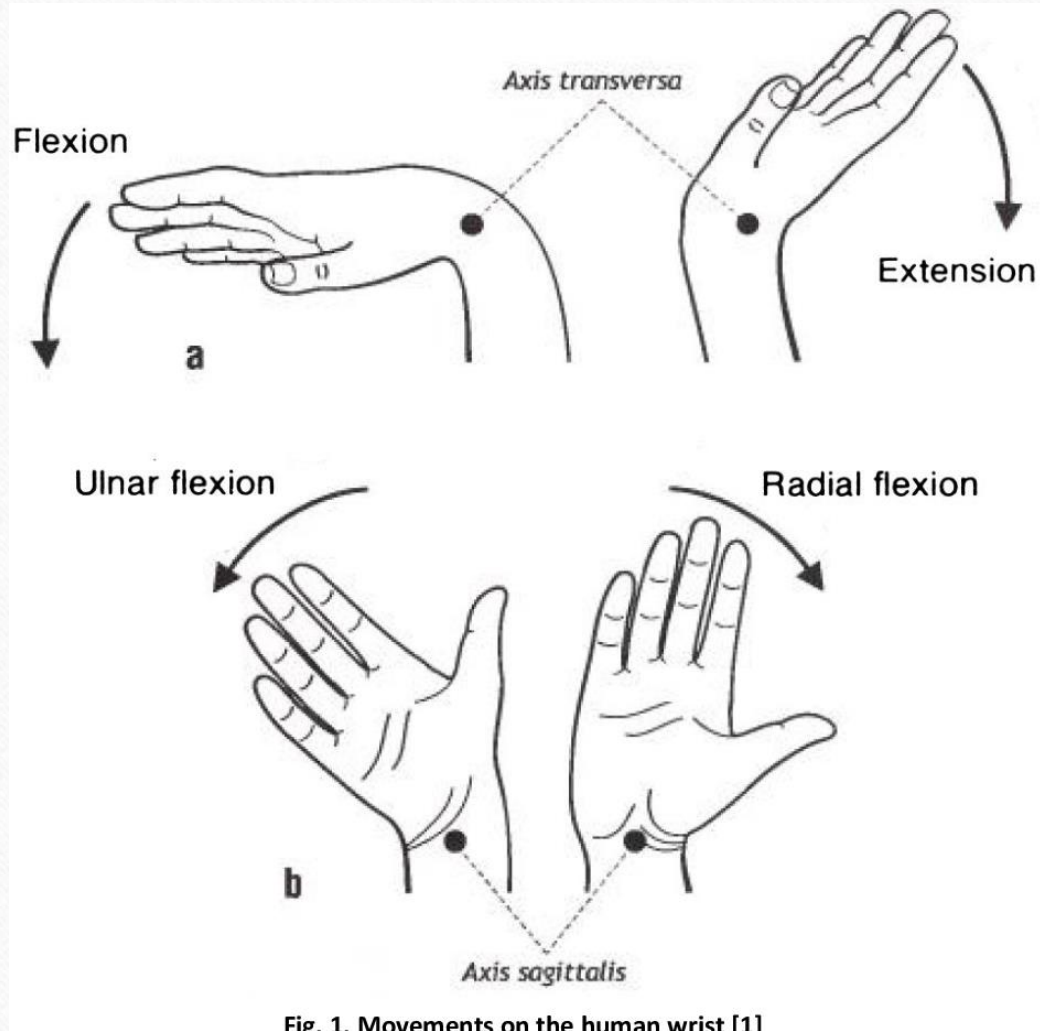
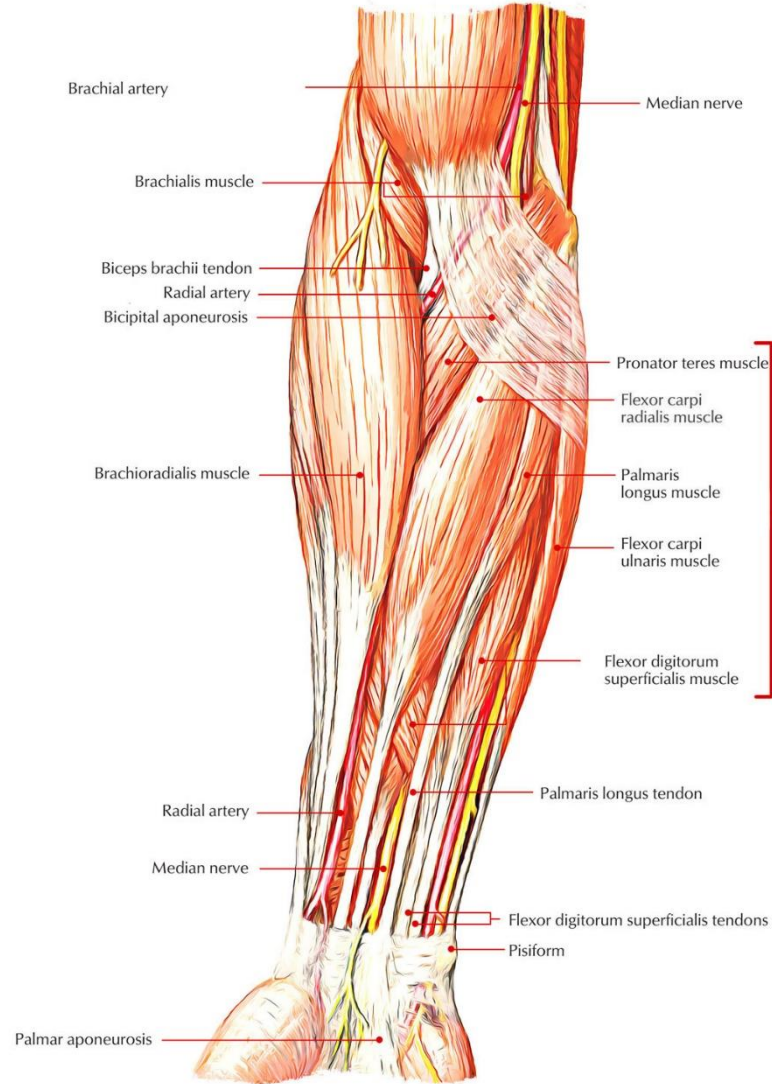


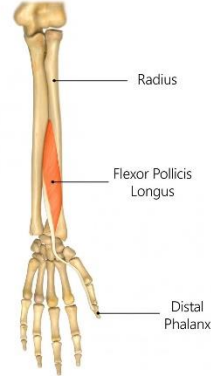
Fig. 1. Movements on the human wrist [1]

- تعمل عضلات الساعد على هذه الحركات الأساسية
- تعمل ايضا على حركة اليد لليمين و للشمال
- تعمل على حركة الصواعب فى اتجاهات مختلفة
- تعمل على ثنى الاصابع

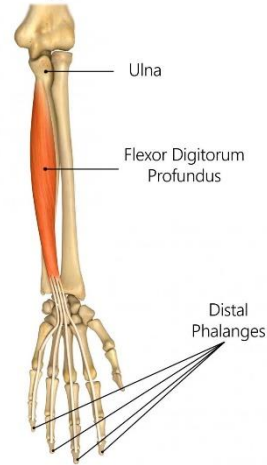
Wrist flexor muscles



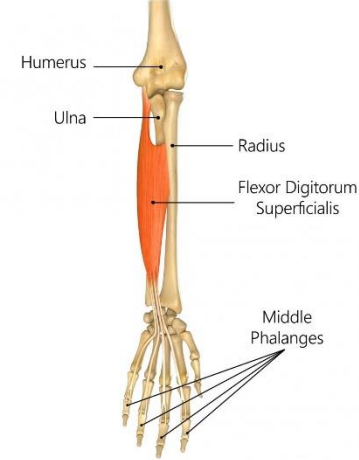
Flexor Pollicis Longus



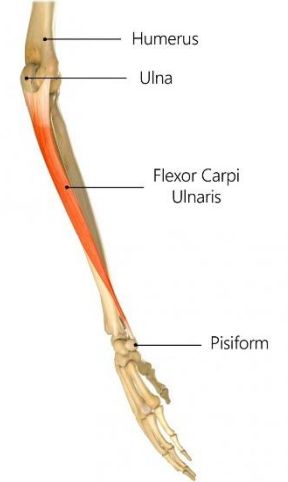
Flexor Digitorum Profundus



Flexor Digitorum Superficialis

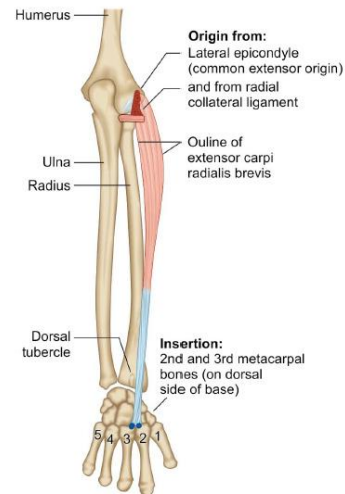


Flexor Carpi Ulnaris

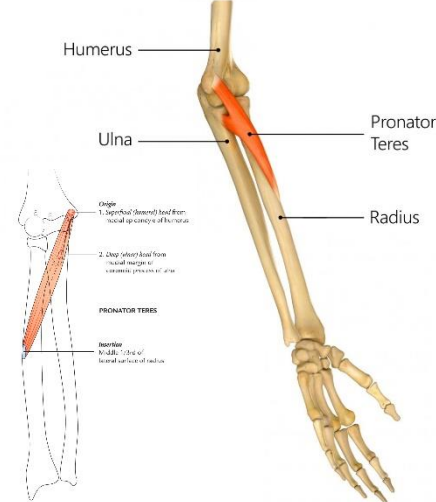


Superficial flexor muscles

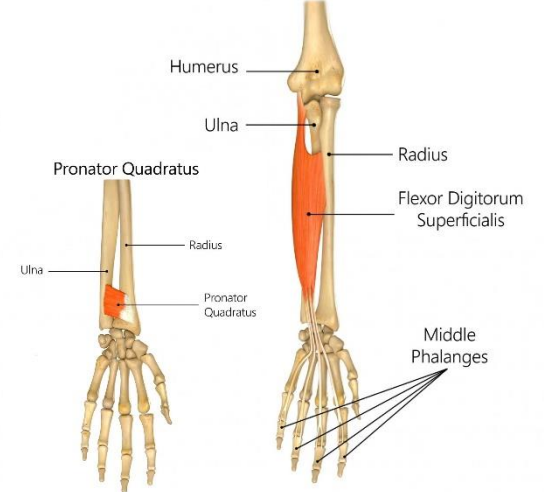
Extensor carpi radialis brevis



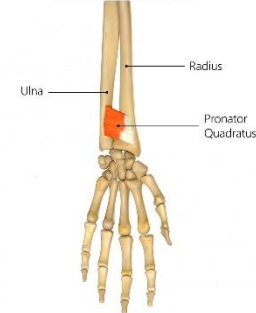
Pronator Teres



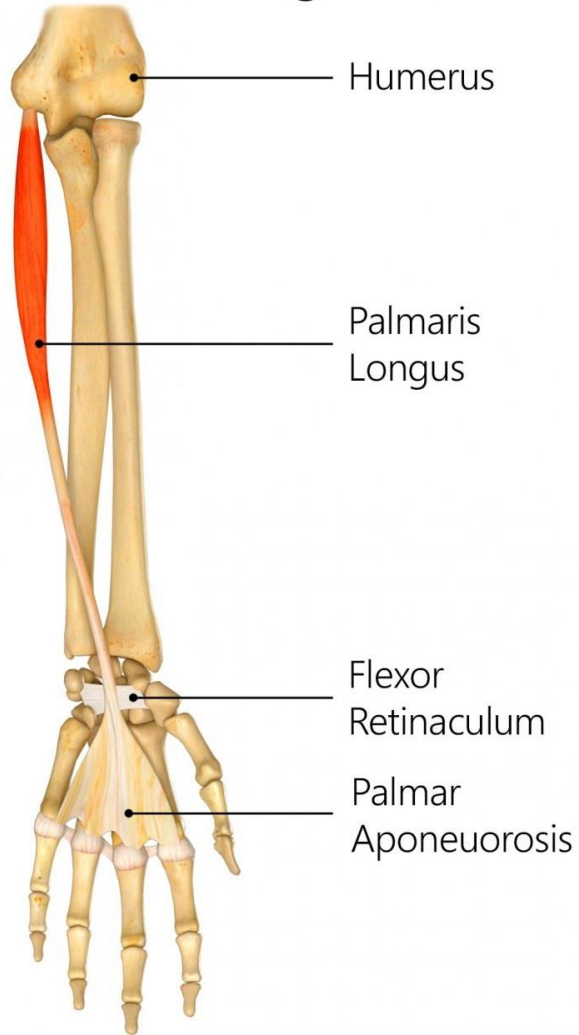
Flexor Digitorum Superficialis



Pronator Quadratus



Palmaris Longus



29- Palmaris longus

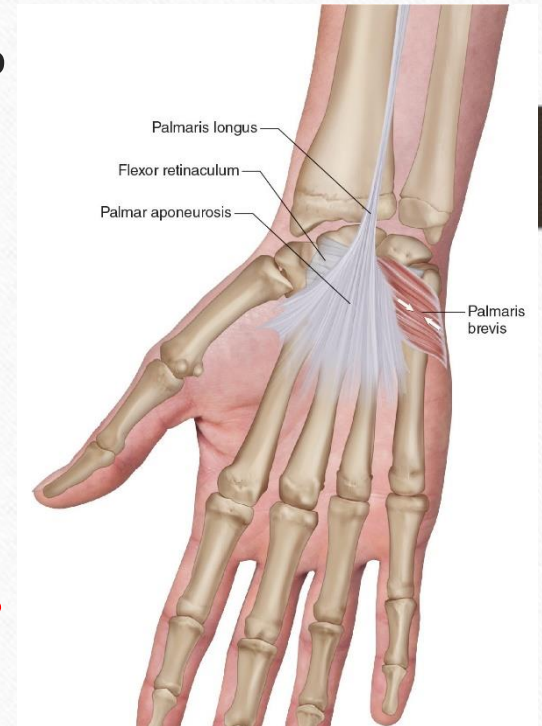
- **Not found in 15%**

- **Origin :**

Humerus

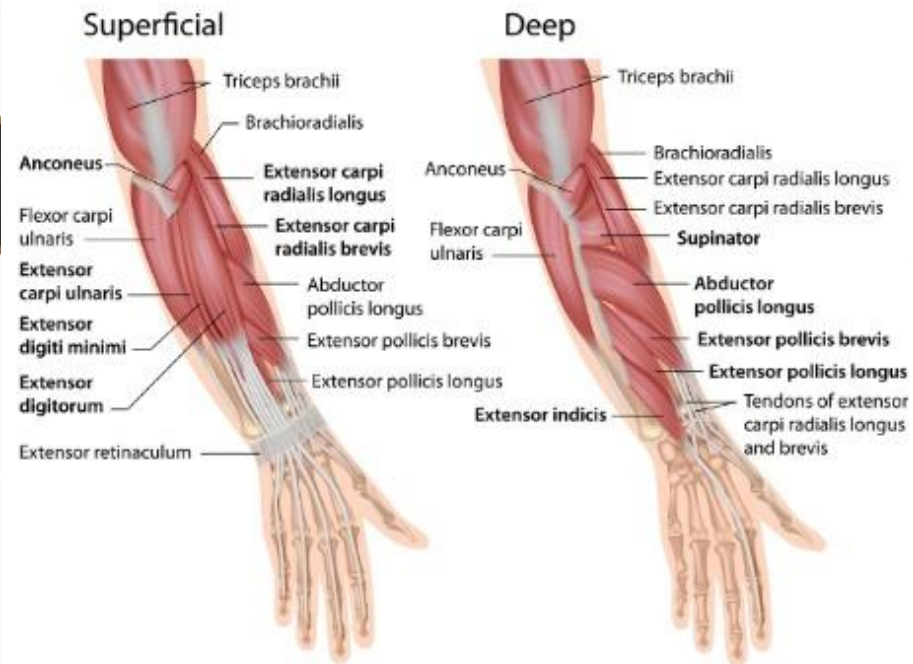
- **Insertion :**

Palmar Aponeurosis

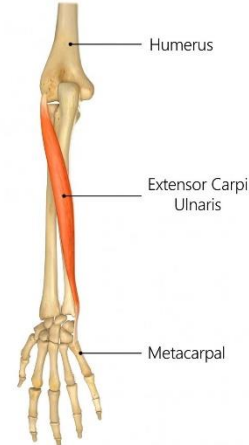


Wrist extensor muscles

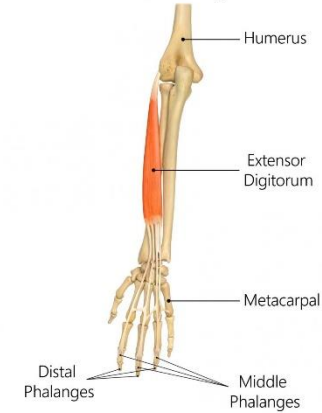
Muscles of the Forearm (right arm, posterior compartment)



Extensor Carpi Ulnaris



Extensor Digitorum



Supinator

Extensor Digitorum

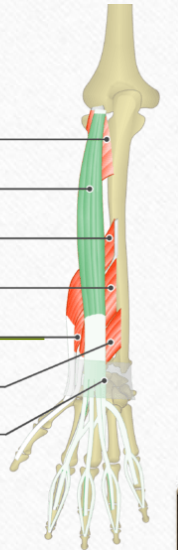
Abductor Pollicis Longus

Extensor Pollicis Longus

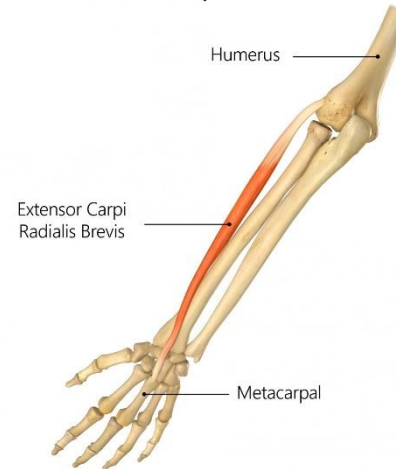
Extensor Pollicis Brevis

Extensor Indicis

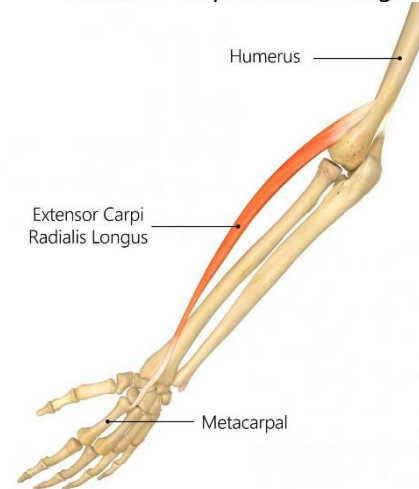
Extensor Retinaculum



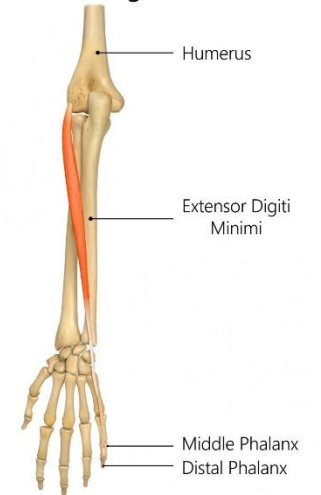
Extensor Carpi Radialis Brevis



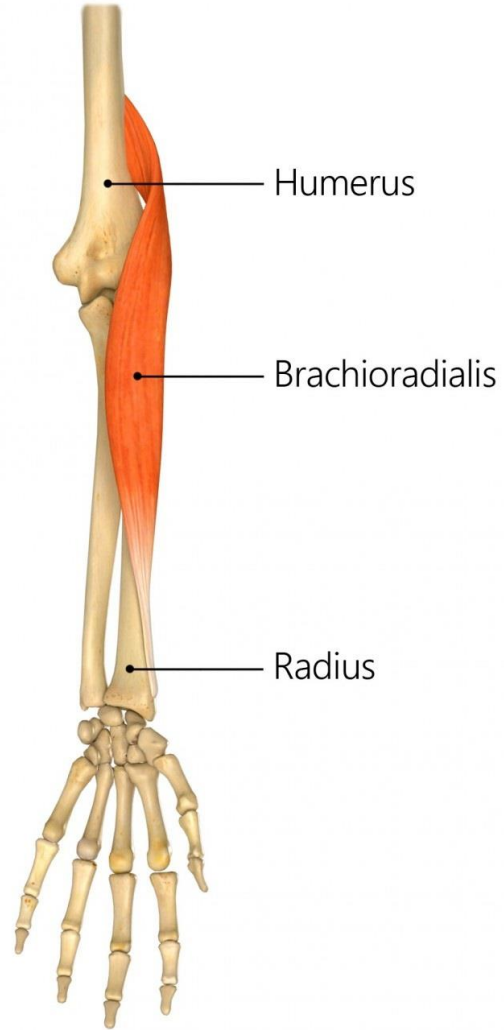
Extensor Carpi Radialis Longus



Extensor Digiti Minimi



Brachioradialis



27- Brachioradialis

- العضلة دى بتعمل فلكشن مع انها فى عضلات الاكستنشن
- **Origin :**
Humerus
- **Insertion :**
Radius

Brachioradialis

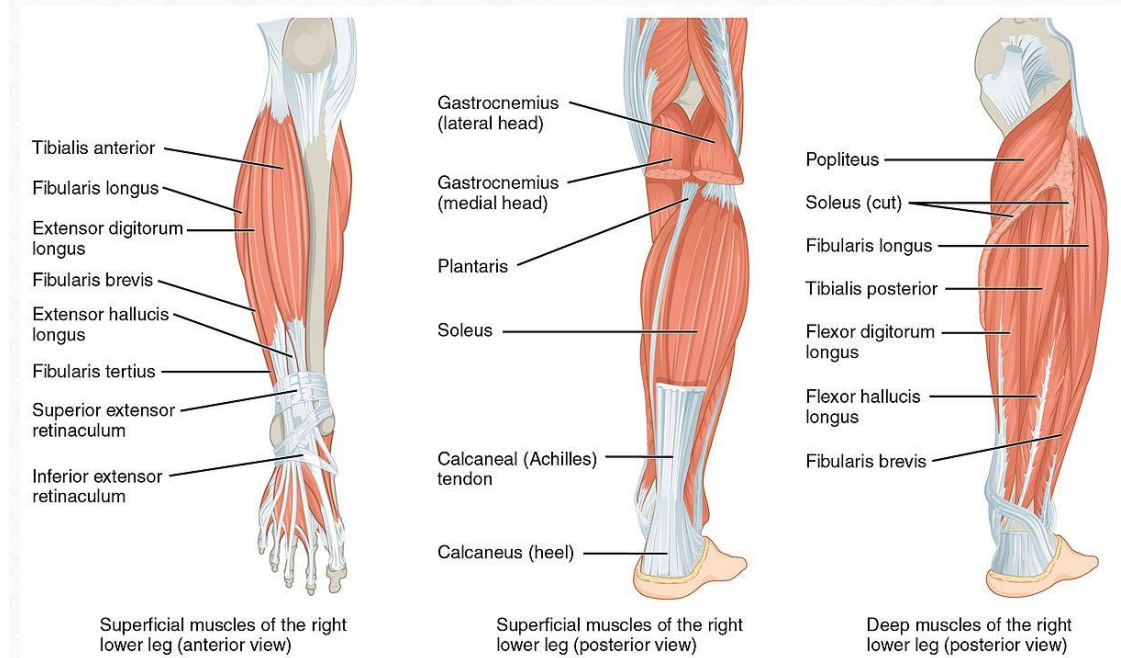
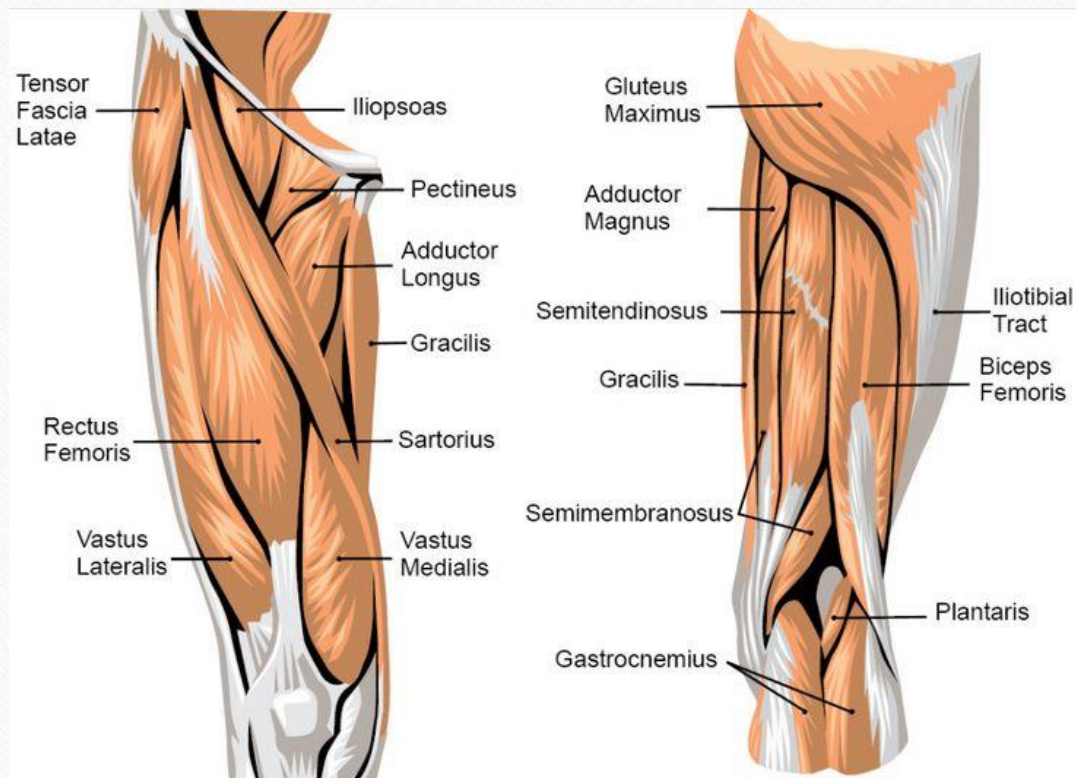
- Elbow flexion

Wrist Curl (Cable)

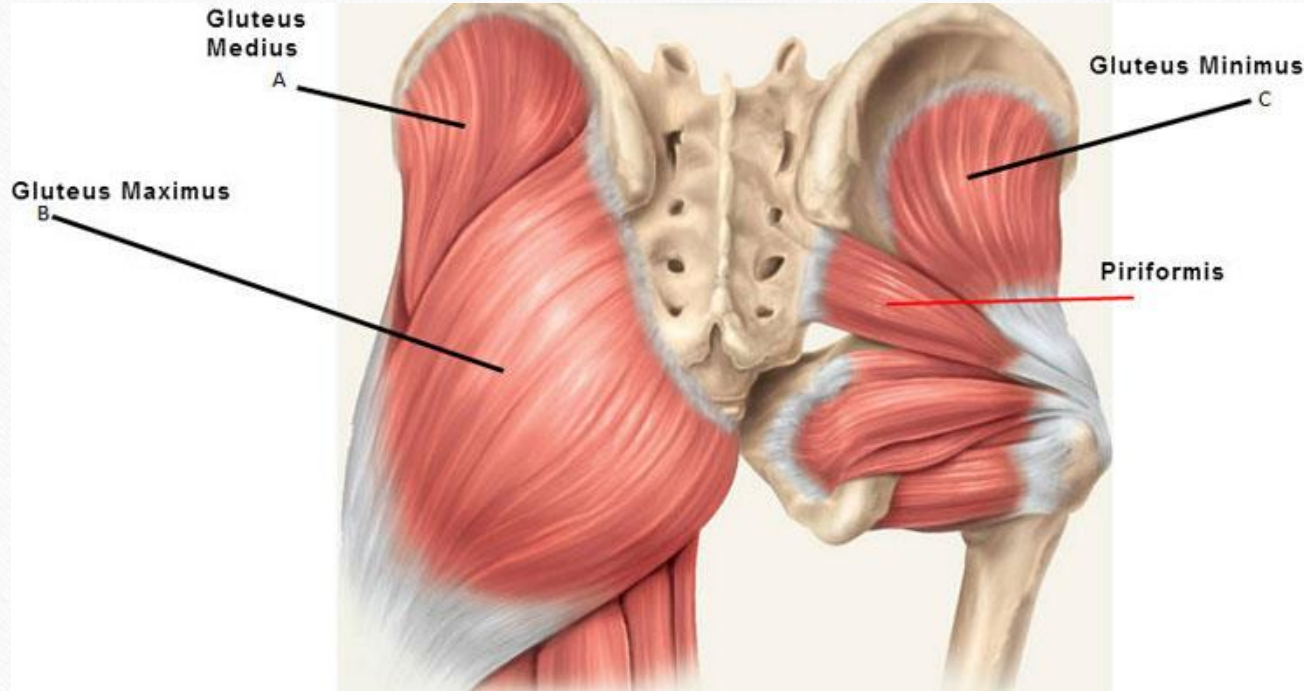
Wrist Extension (Dumbbell)

Hammer Curls (Dumbbell)

Lower limb muscles (Leg muscles)

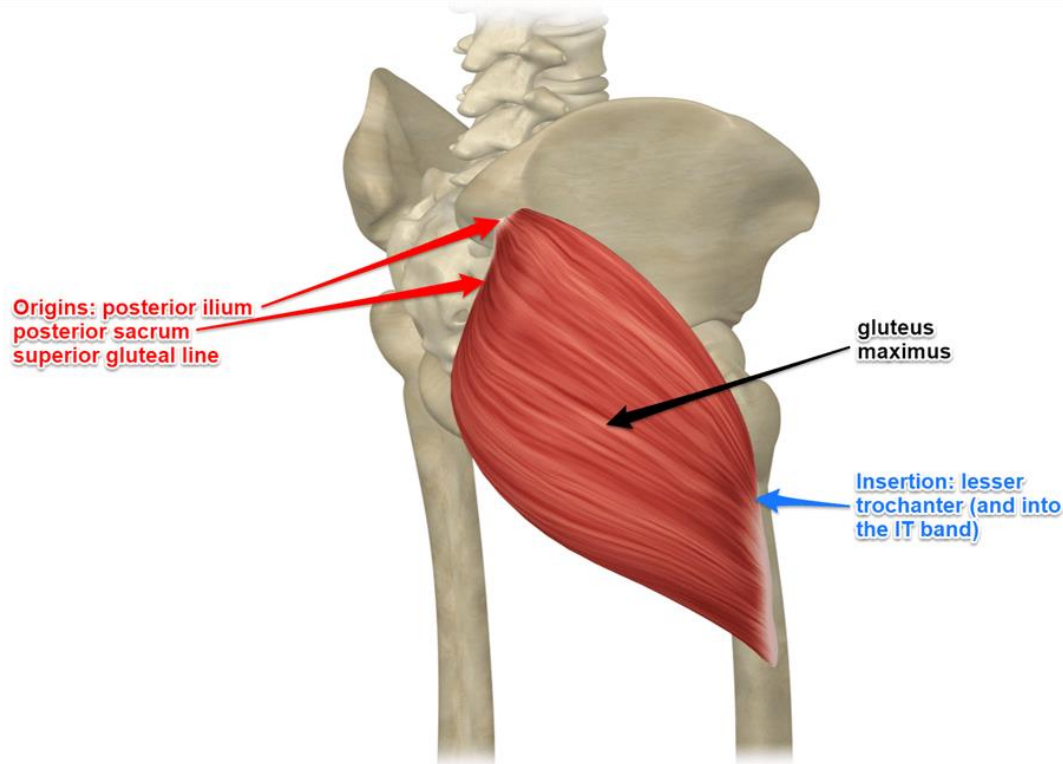


10-11- Gluteus



- تنقسم عضلة المؤخرة الى 3 اجزاء
 - عضلة المؤخرة الكبيرة
 - عضلة المؤخرة الجانبية الوسطى
 - عضلة المؤخرة الصغيرة
 - Origin and Insertion
- بداية العضلة من الحوض و اسفل العمود الفقري
- نهاية العضلة فى عظمة الفخذ

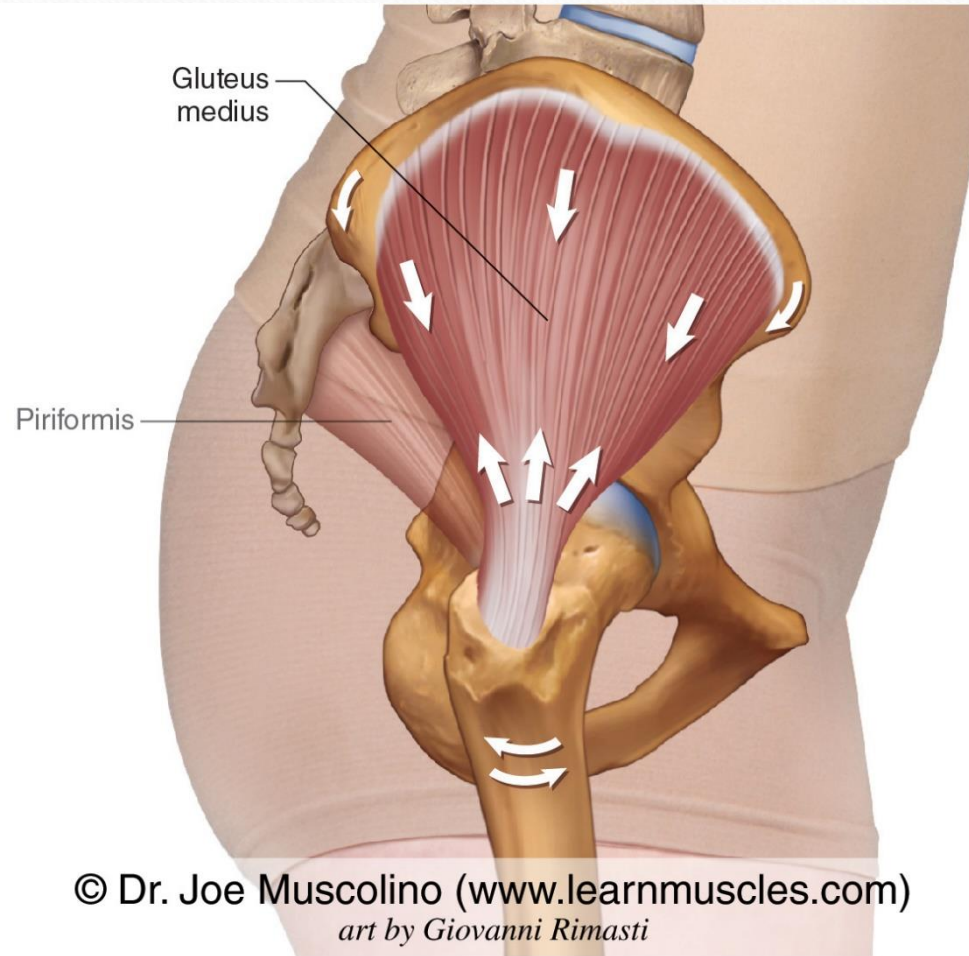
11- Gluteus maximus



- عضلة المؤخرة الكبيرة
- بدايتها في الحوض و اسفل العمود الفقري
- نهايتها في عظمة الفخذ

- **Origin :**
ilium of the Pelvic, Coccyx, and Sacrum (vertebrate)
- **Insertion :**
Femur

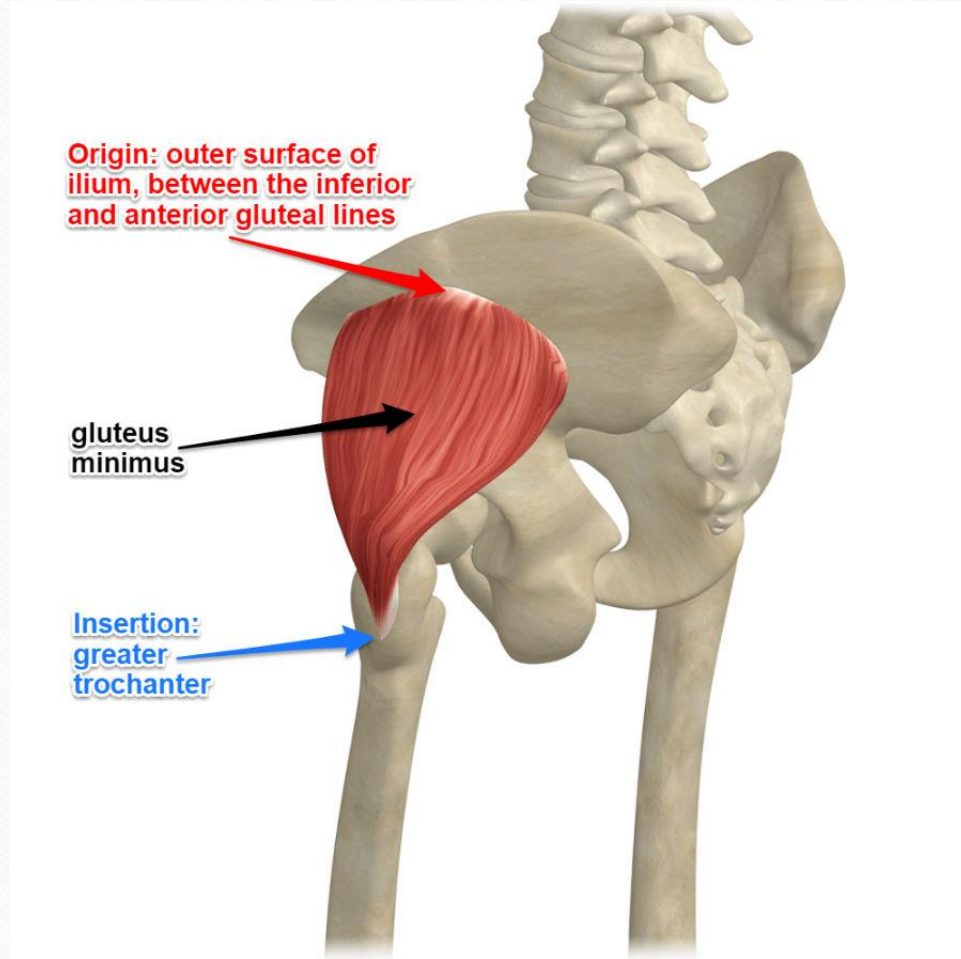
10- Gluteus medius



- عضلة المؤخرة الجانبية
- بدايتها في الحوض
- نهايتها في عظمة الفخذ

- **Origin :**
ilium of the **Pelvic**
- **Insertion :**
Femur

Gluteus minimus



- عضلة المؤخرة الجانبية
- بدايتها في الحوض
- نهايتها في عظمة الفخذ

- Origin :
ilium of the Pelvic
- Insertion :
Femur

Motion of the muscles G Max

- Hip extension
- Hip abduction
- Hip lateral rotation

Motion of the muscles

Gluteus Med & min

- Hip abduction
- Hip medial rotation

Hip extensions

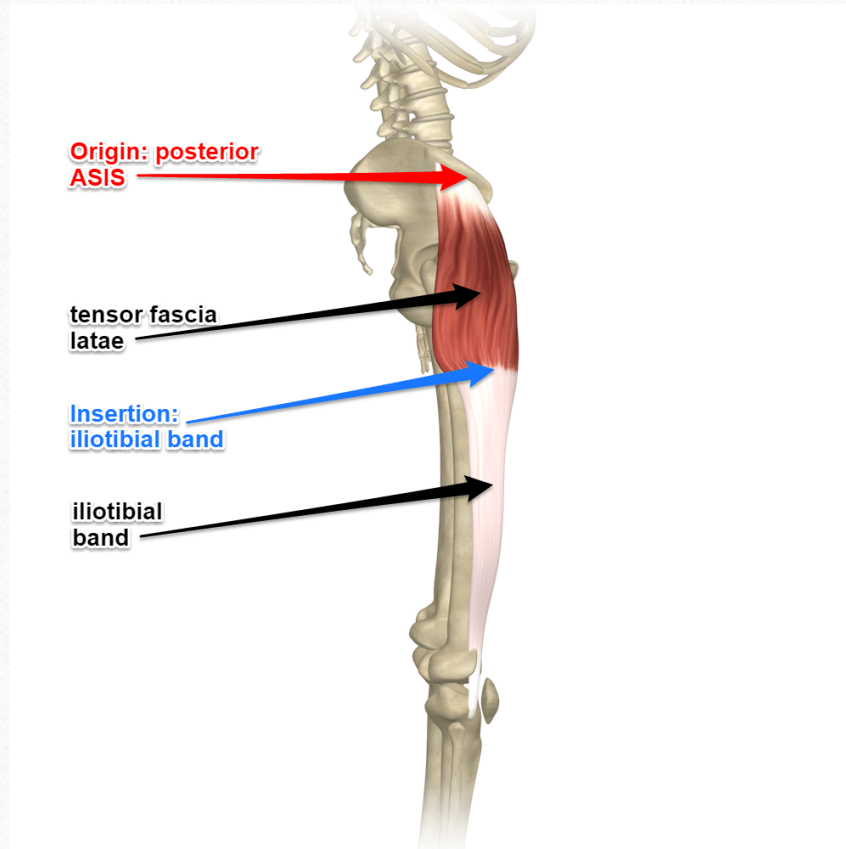
Hip thrust

Lunges

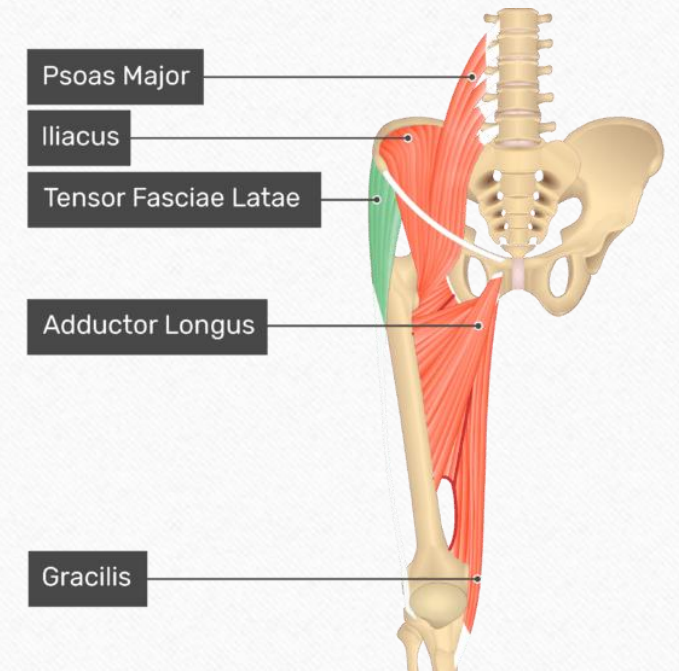
Machine hip abduction

Machine abduction

31- Tensor fasciae latae



- Origin :
Iliac spine of the pelvic
- Insertion :
Tibia



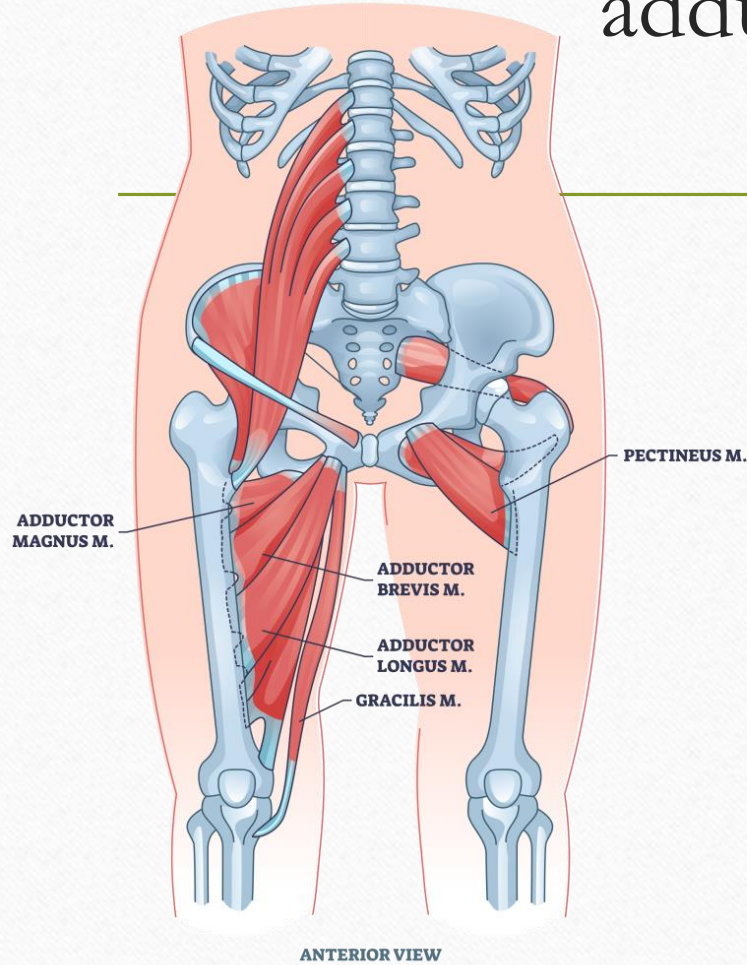
Motion of the muscle

- Hip abduction
- Hip medial rotation
- Hip flexion

Cable hip abduction

Hip flexion

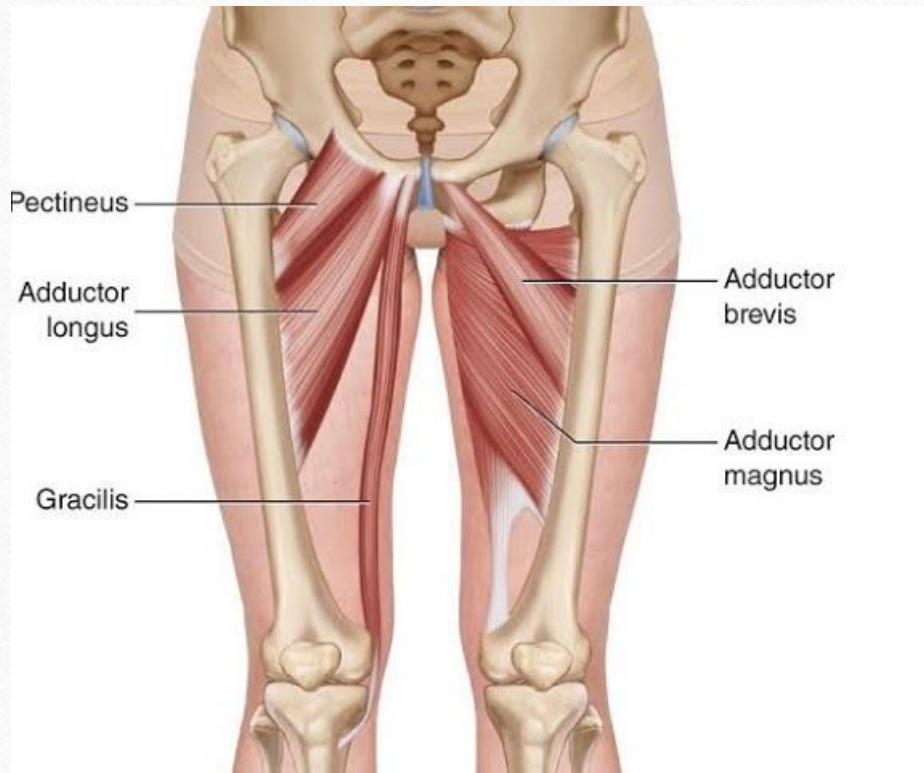
12-30-32-34- Hip adduction muscles adductor longus, adductor brevis, adductor magnus, gracilis, and pectineus.



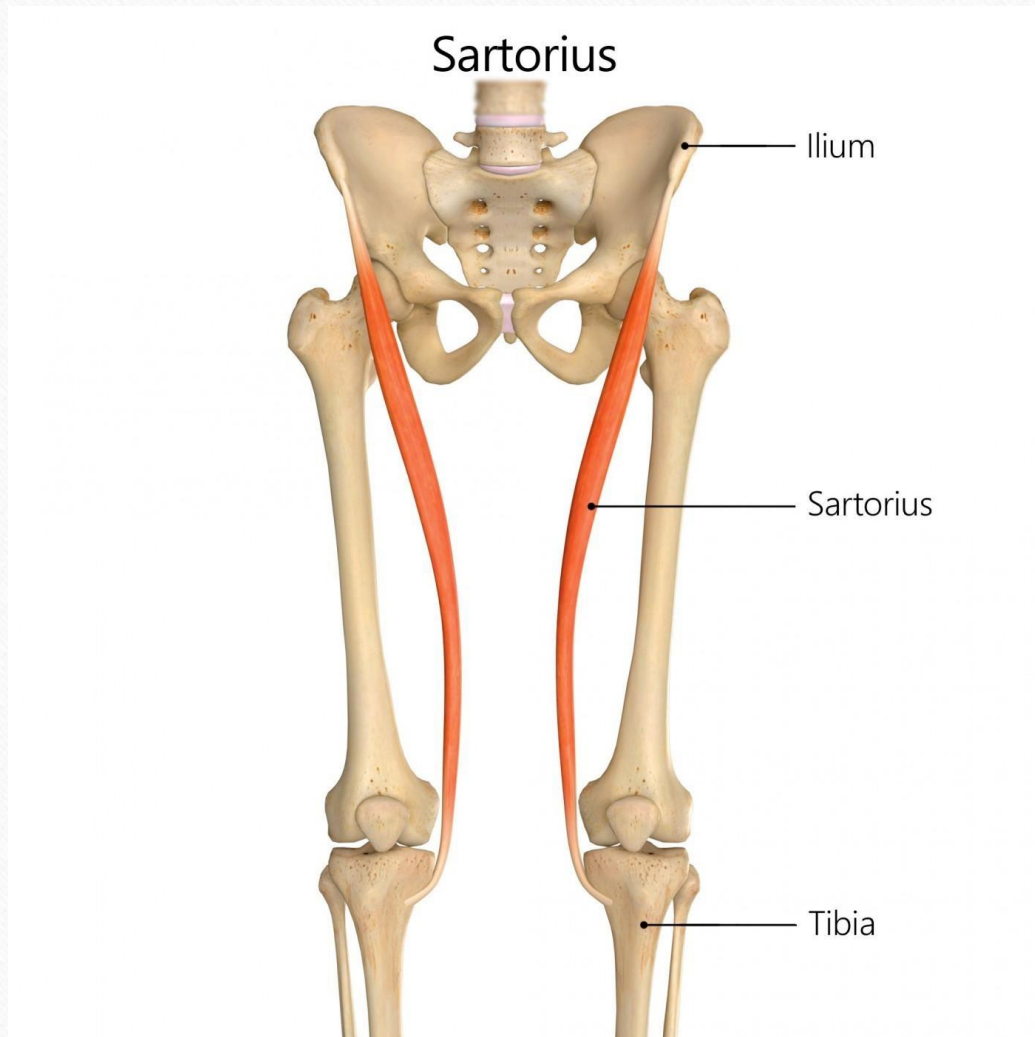
- عضلات الضامة تتكون من 5 عضلات
 - Origin and Insertion
- بداية العضلات خمسة من الحوض
- نهاية العضلات الـ 4 في عظمة الفخذ
- عضلة واحدة فقط الجراسيليس نهايتها في القصبه

12- Hip adduction muscles

adductor longus, adductor brevis, adductor magnus, **gracilis**,
and **pectineus**.



- **Origin :**
pubic and **pubis** of the **pelvic**
- **Insertion :**
Femur (for all muscles except Gracilis)
& **tibia** (for Gracilis muscle)



35- Sartorius muscle

- **Origin :**

Iliac spine of the pelvic

- **Insertion :**

Tibia

Motion of the muscle

- Hip adduction
- Hip flexion

Motion of the muscle

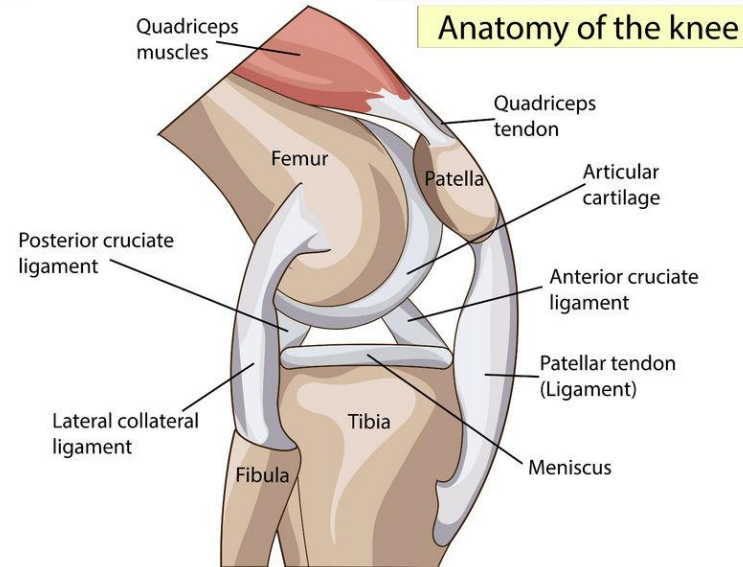
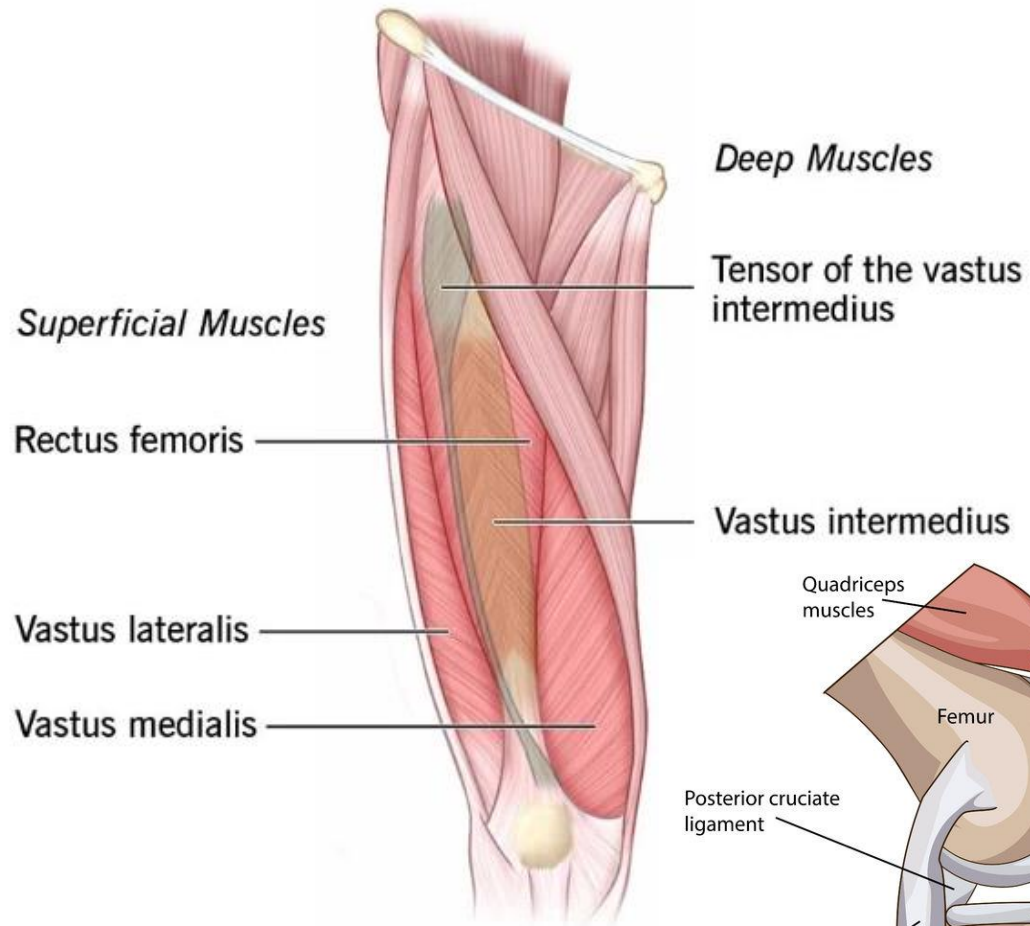
- Hip medial rotation
- Hip extension
- Knee flexion

Machine adduction

Cable adduction

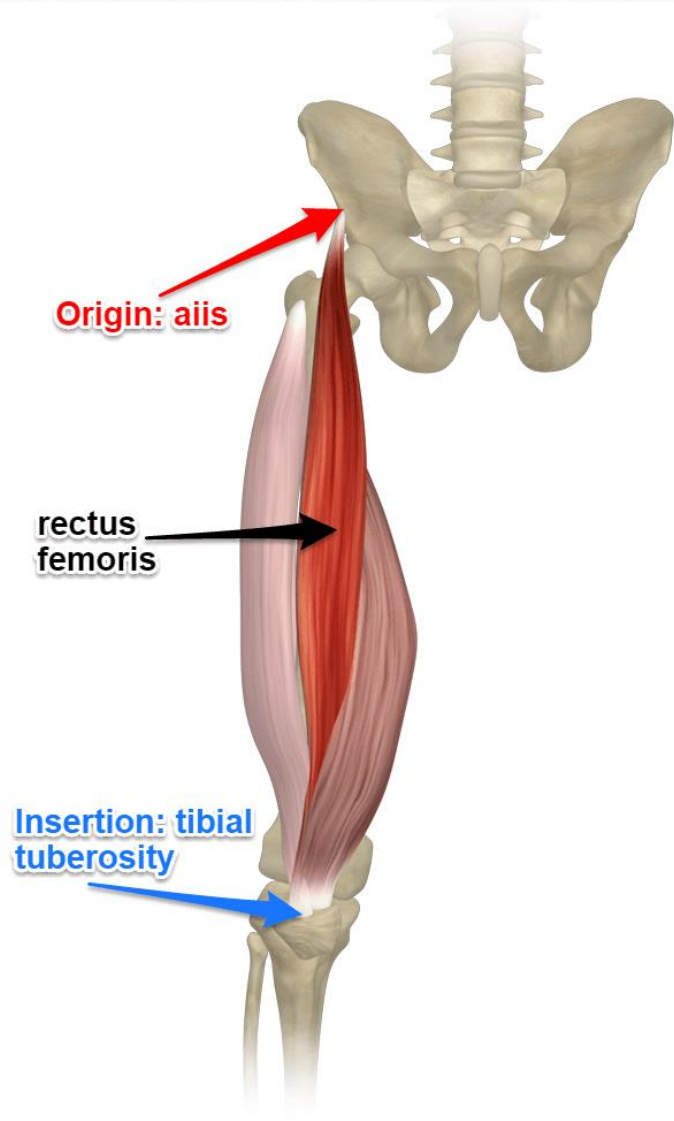
Hip flexion

33-36-37 Quadriceps muscles



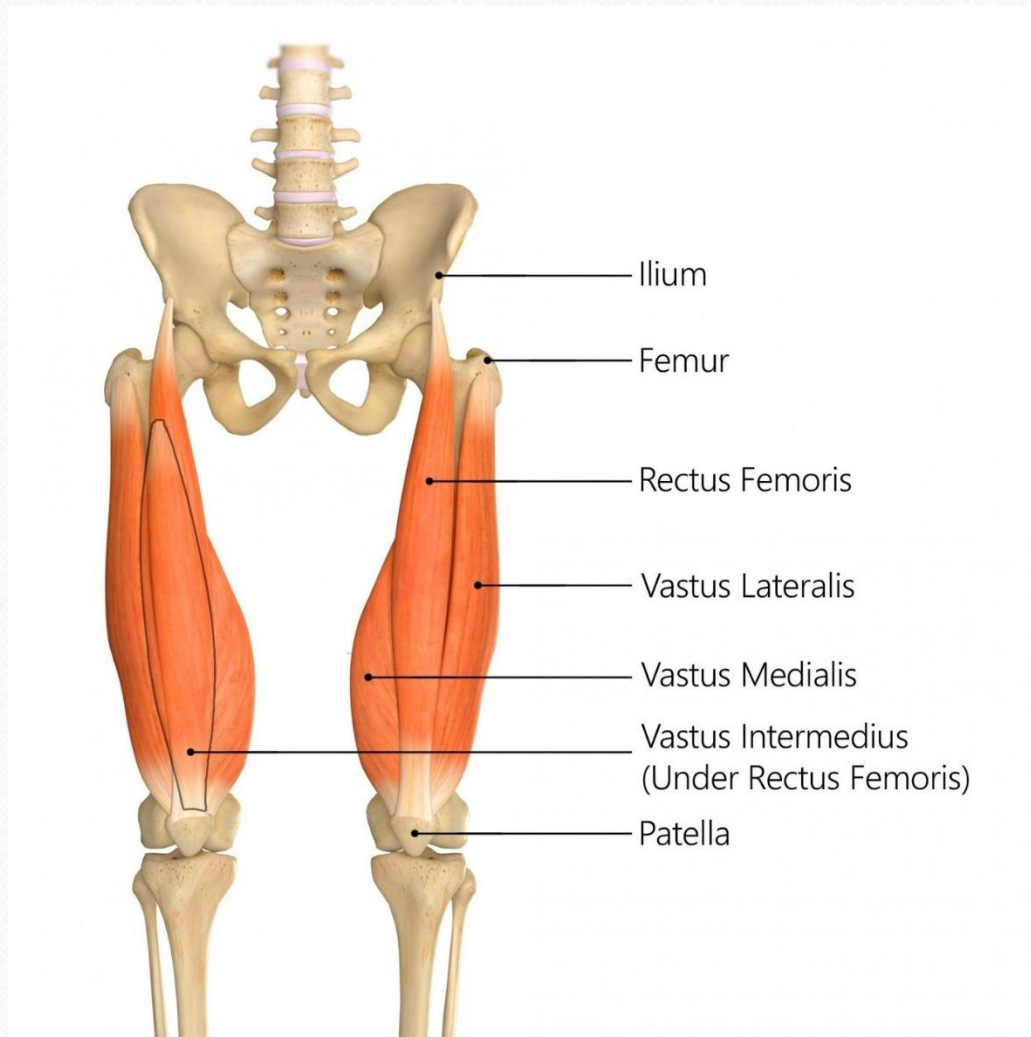
- اماميات الرجل تنقسم الى 4 عضلات
- Origin and Insertion
- بداية العضلات من الحوض
و عظمة الفخذ
- نهاية العضلات في قصبة الرجل

Rectus femoris



- **Origin :**
Groove of Ilium and iliac spine of the pelvic
- **Insertion :**
Patella & Tibia

Vastus medialis

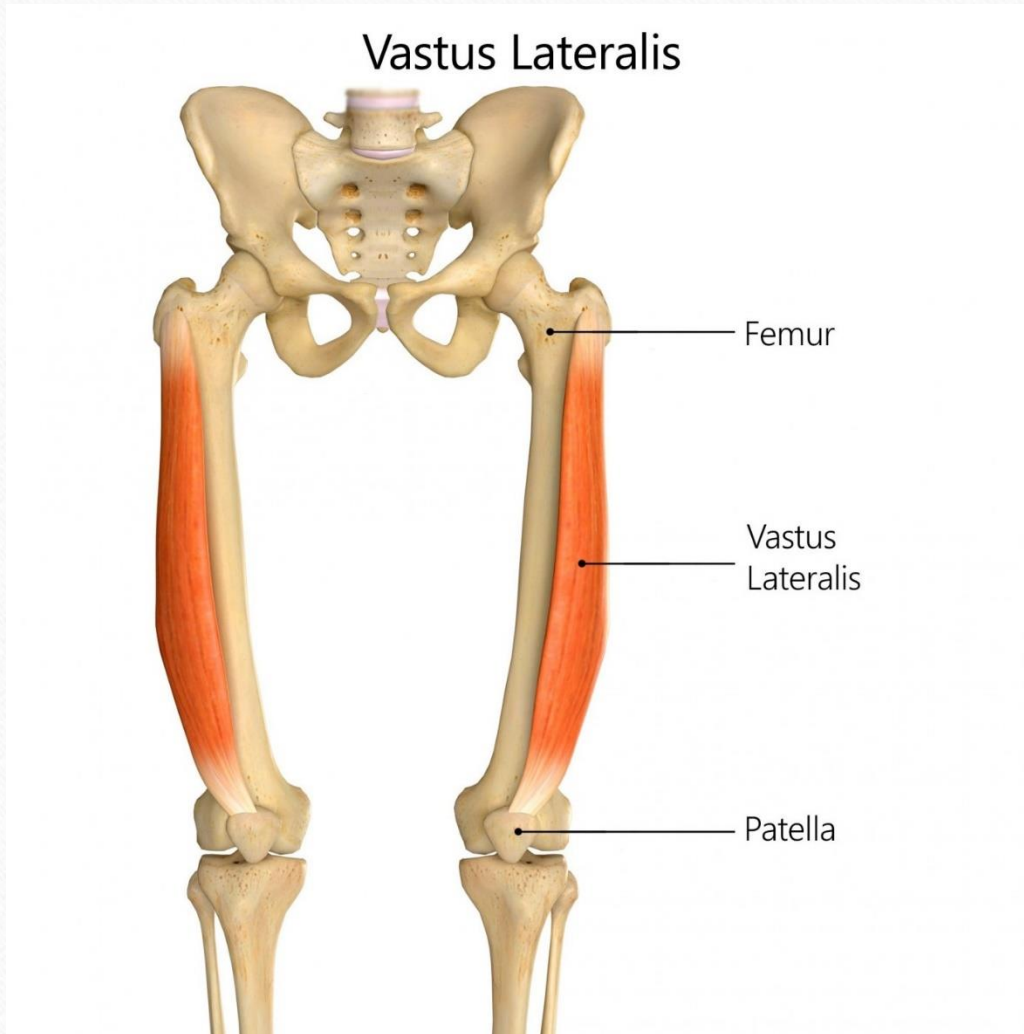


- **Origin :**

Femur

- **Insertion :**

Patella & Tibia



Vastus Lateralis

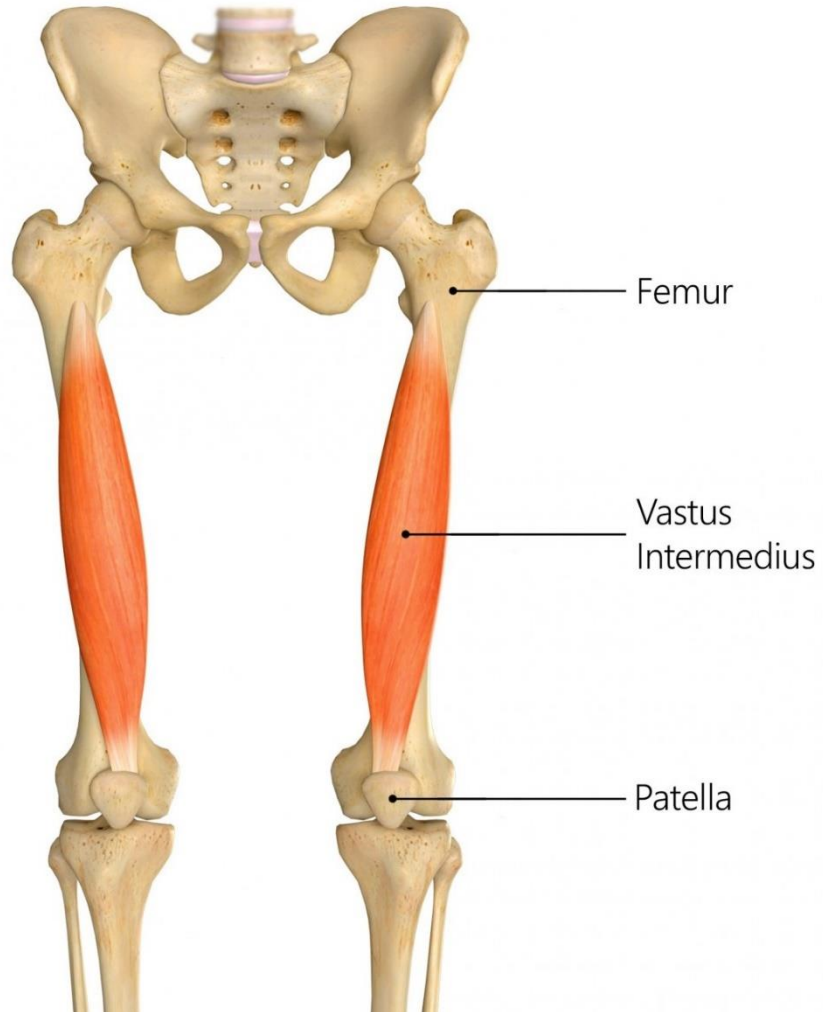
- **Origin :**

Femur

- **Insertion :**

Patella & Tibia

Vastus Intermedius



Vastus intermedius

- **Origin :**

Femur

- **Insertion :**

Patella & Tibia

Motion of the muscle

- Knee extension
- Hip flexion

Leg extension

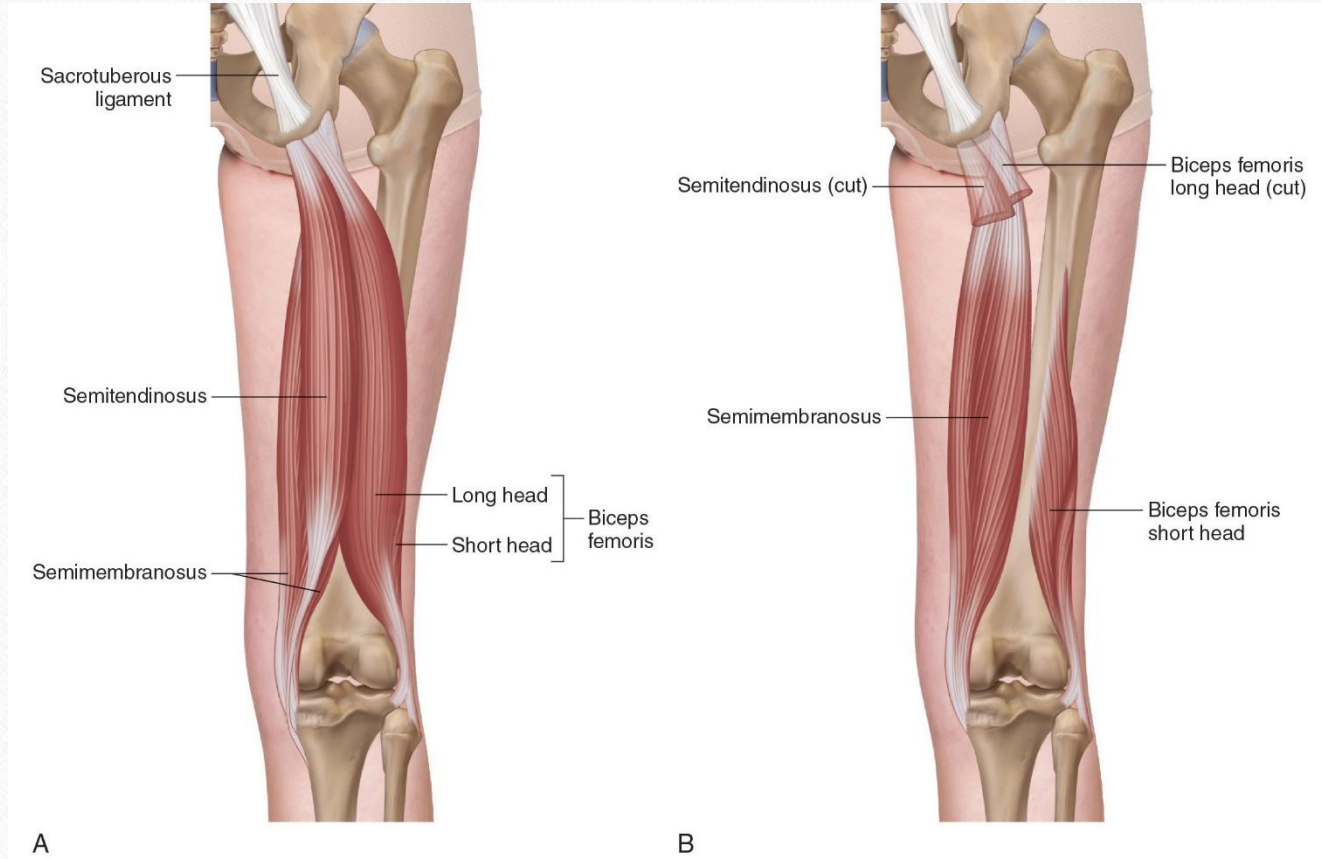
Squats

Leg presses

Lunges

Box Marches (hip extensions)

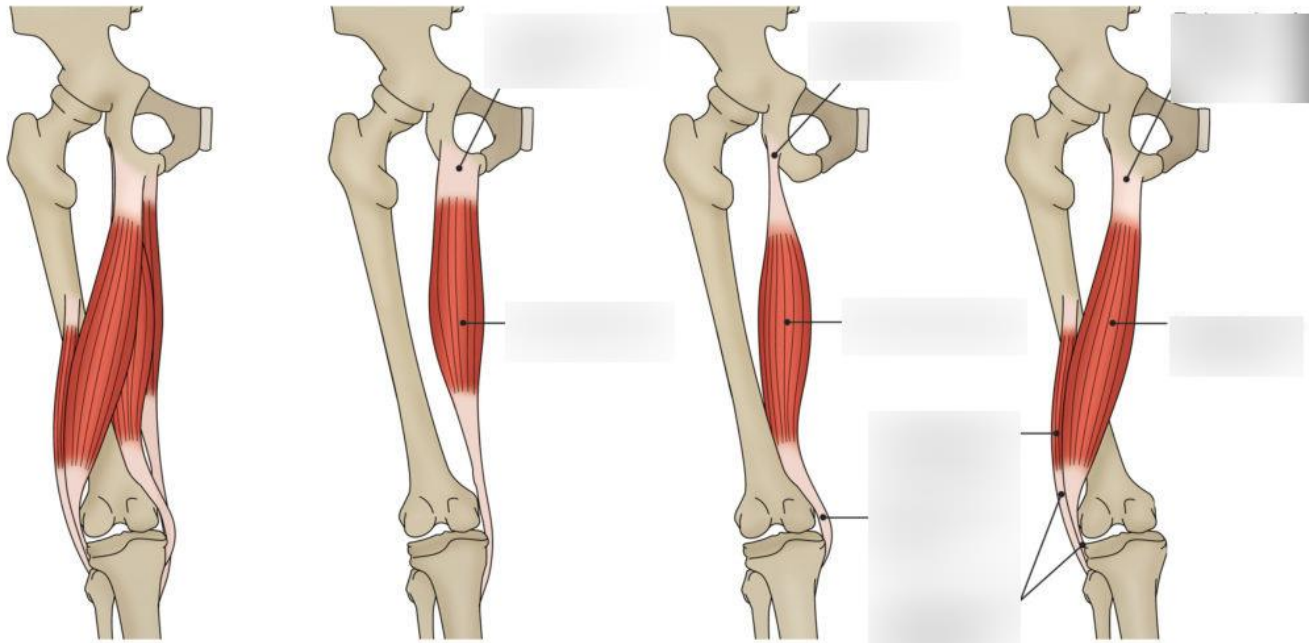
13-14- Hamstring muscles



- عضلات الخلفية (خلفيات الرجل)
 - تتكون من 4 عضلات
 - Origin and Insertion
- بداية ال 3 العضلات من الحوض و عضلة الباي الصغيرة فقط بدايتها من عظمة الفخذ
- نهاية عضلات الباي في الفيبولا و باقى العضلات نهايتها في القصبة

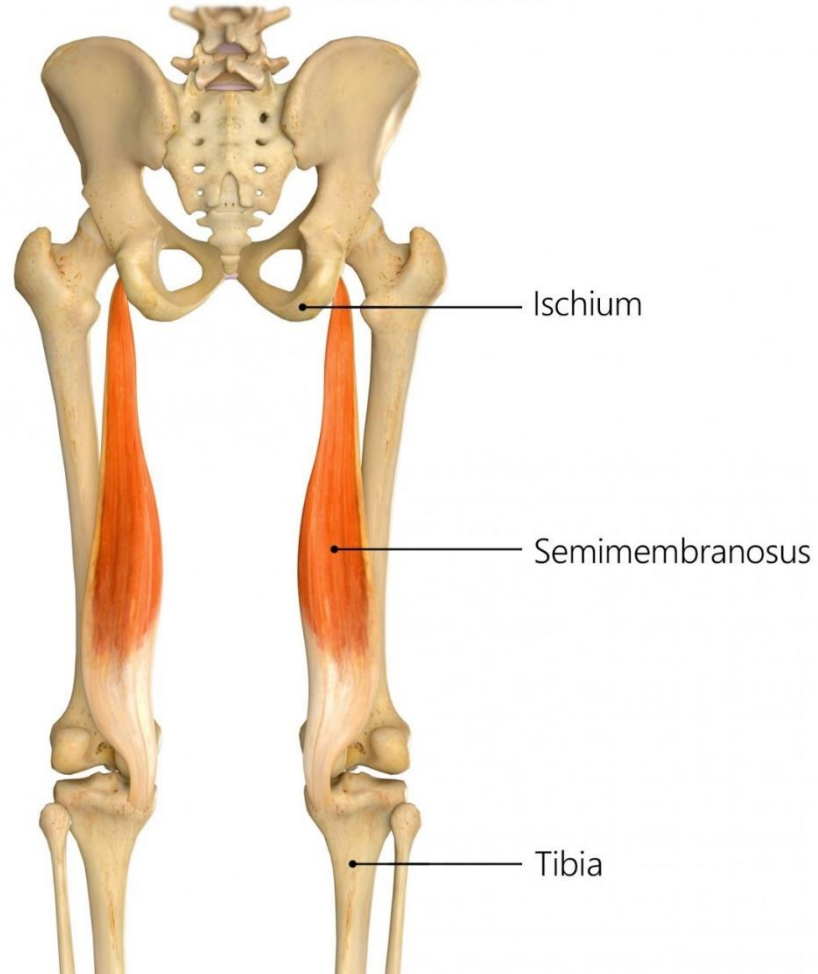
Hamstring: Semimembranosus, Semitendinosus, Biceps femoris - Long head, Biceps femoris – Short head.

Hamstring Group Posterior view



- **Origin :**
Pelvic (for Semi and long biceps)
& **femur** (for short biceps)
- **Insertion :**
Tibia (for Semi)
& **fibula** (for Biceps)

Semimembranosus



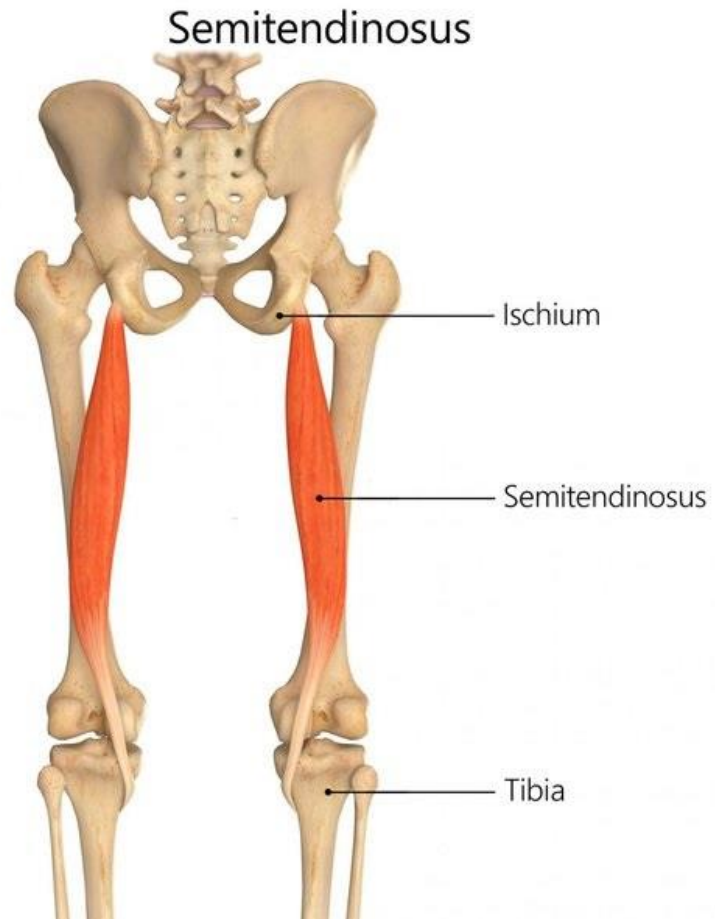
Semimembranosus

- **Origin :**

Pelvic

- **Insertion :**

Tibia



Semitendinosus

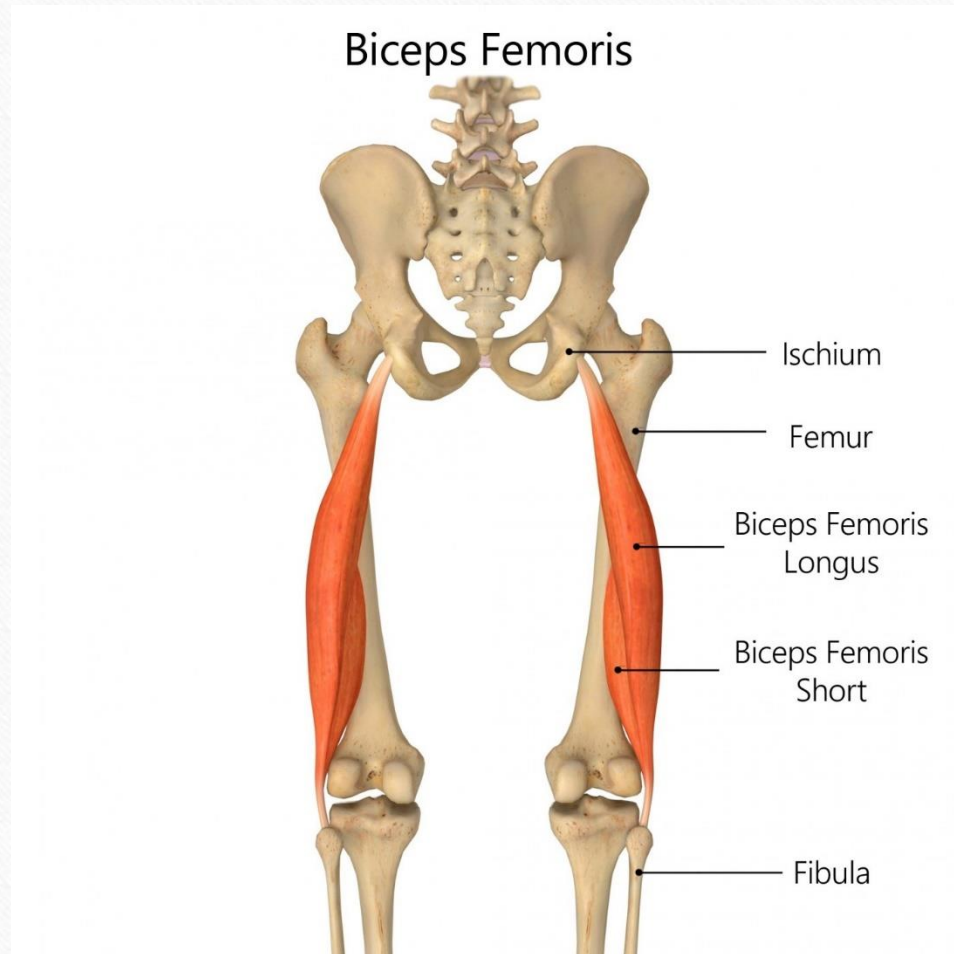
- **Origin :**

Pelvic

- **Insertion :**

Tibia

Biceps femoris – Long & short head



- **Origin :**
Pelvic (for long biceps)
& **femur** (for short biceps)
- **Insertion :**
fibula (for Biceps long & short)

Motion of the muscle

- Hip extension
- Knee Flexion

Motion of the muscle

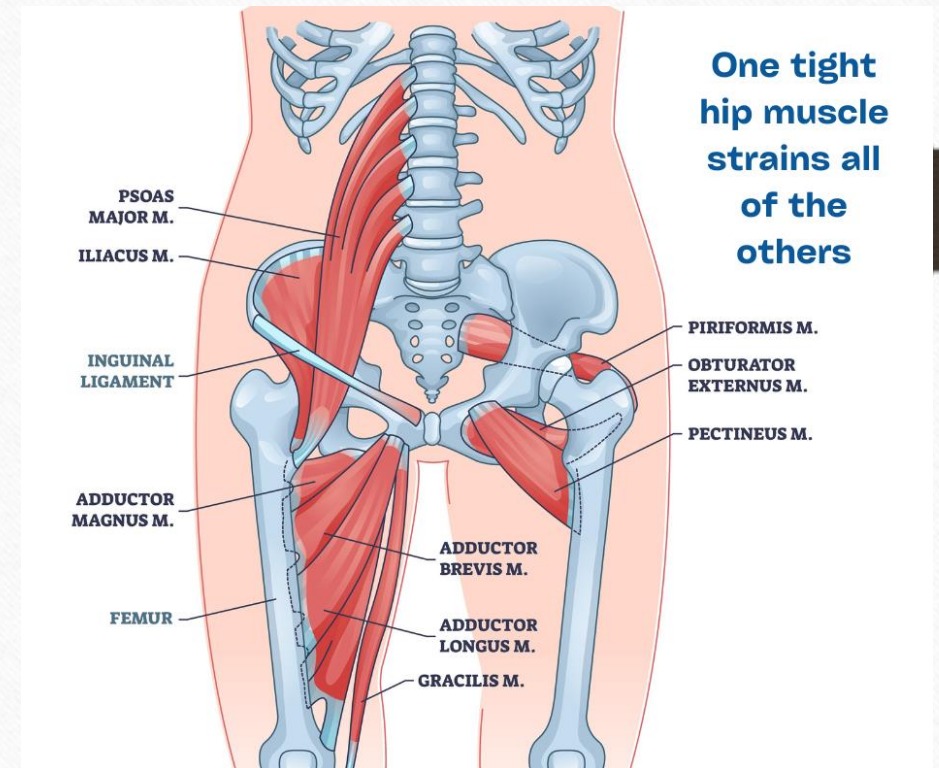
- Knee lateral rotation (L)
- Knee medial rotation (Semi)

Leg Curls

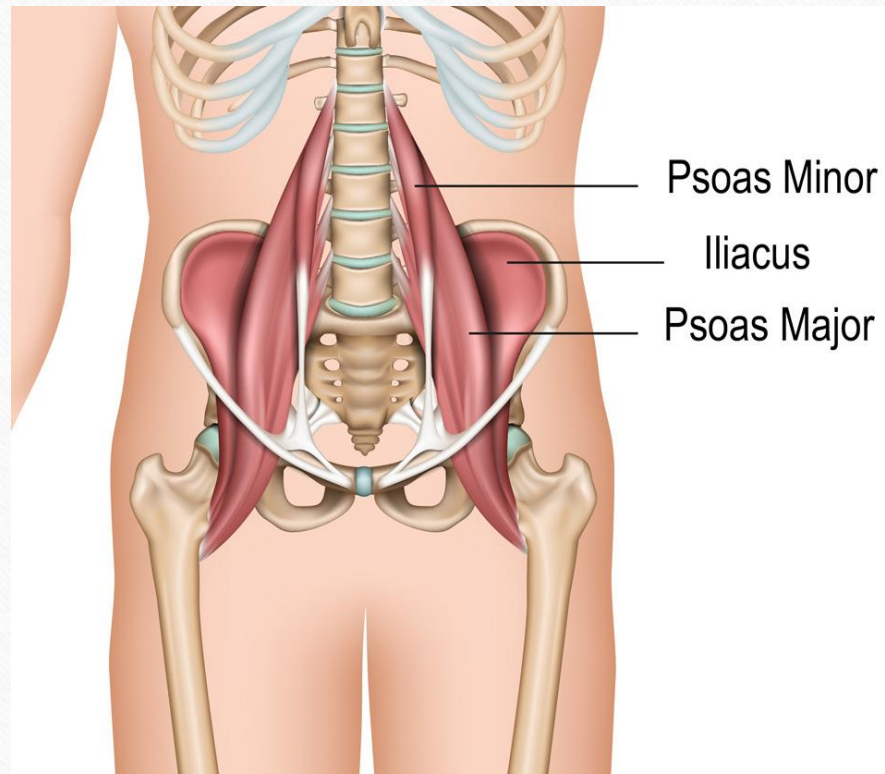
Leg extension

Hip Flexor muscles

- Adductor longus muscle
- Adductor brevis muscle
- Adductor magnus muscle
- Pectineus muscle
- Gracilis muscle
- Rectus femoris muscle
- Sartorius
- Tensor fasciae latae muscle
- Iliacus muscle
- Iliopsoas
- Psoas major muscle

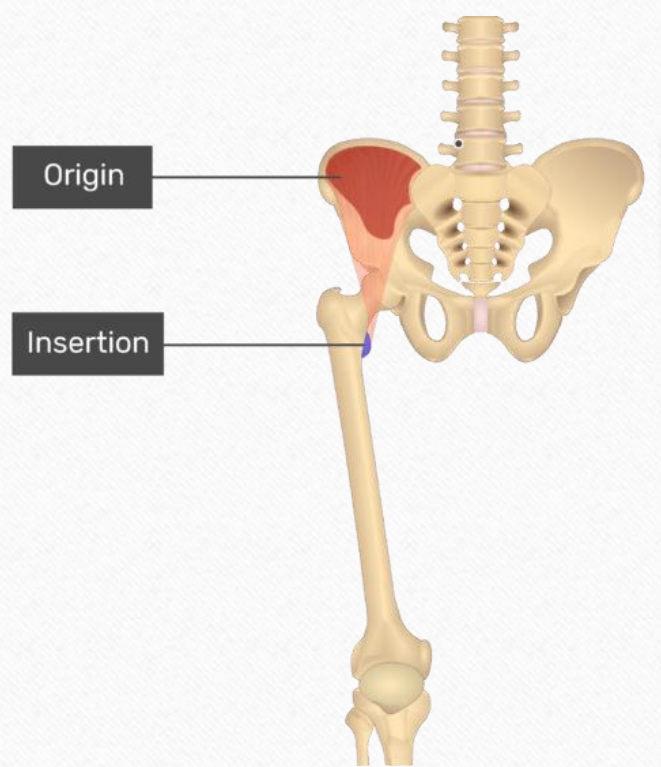


Iliacus muscle & Psoas major muscle



Origin: bodies of the vertebrae from T12- L4

Insertion: lesser trochanter



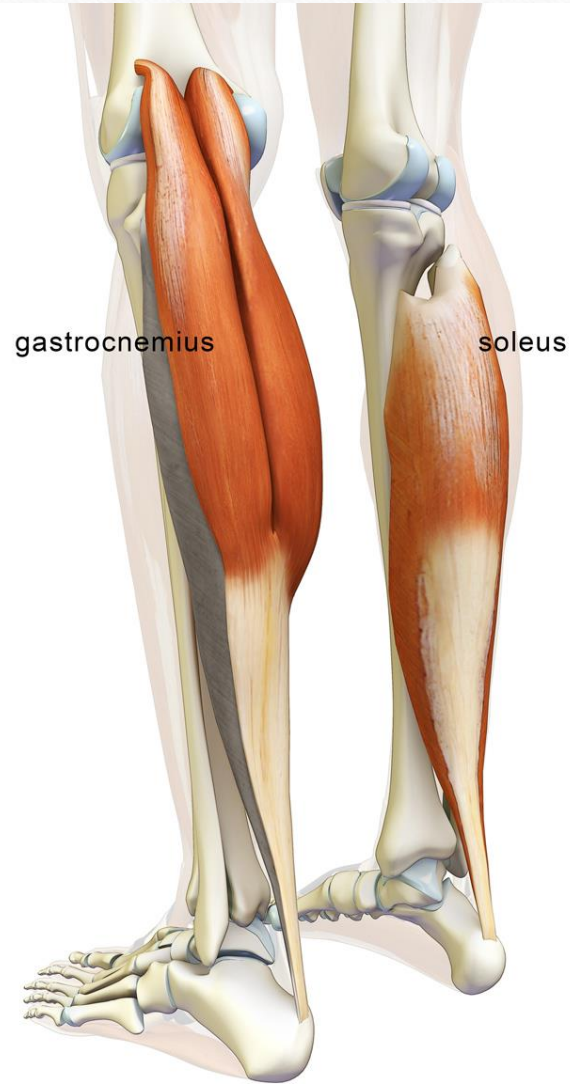
Motion of the muscle

- Hip flexion

Motion of the muscle

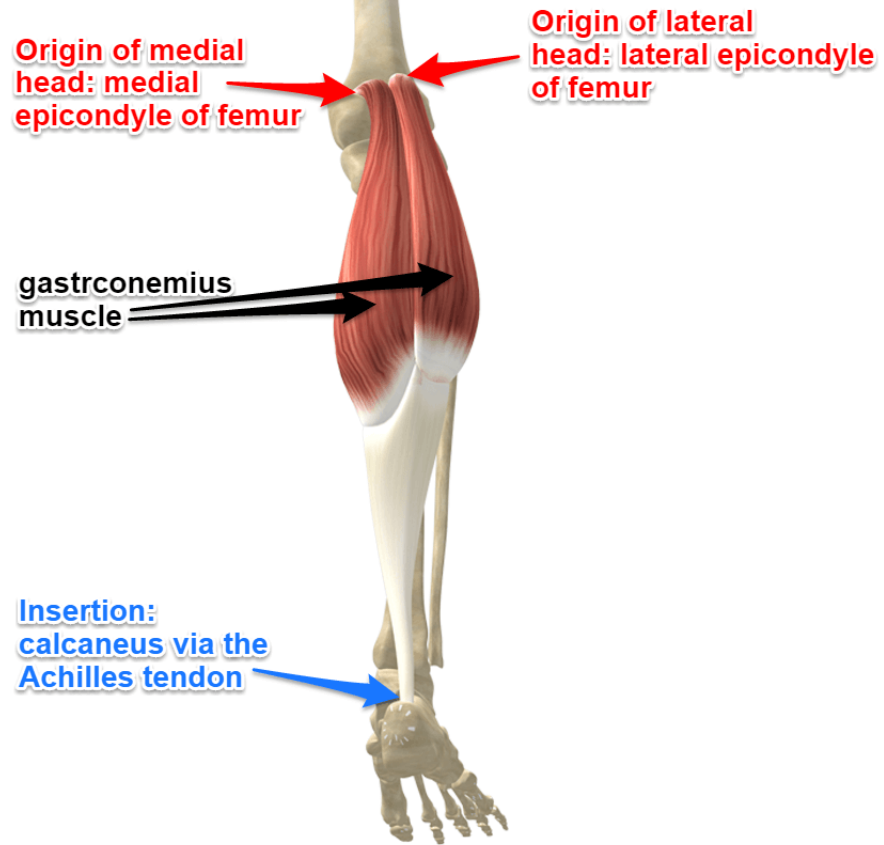
- Trunk flexion
- Trunk Lateral flexion

15-16- Calf muscles



- عضلات السمانة تنقسم الى عضلتين
- Origin and Insertion
- بداية العضلات فى عظمة الفخذ و القصبة
- نهاية العضلات فى كعب الرجل

15- Gastrocnemius



- Origin :

Femur

- Insertion :

Calcaneus

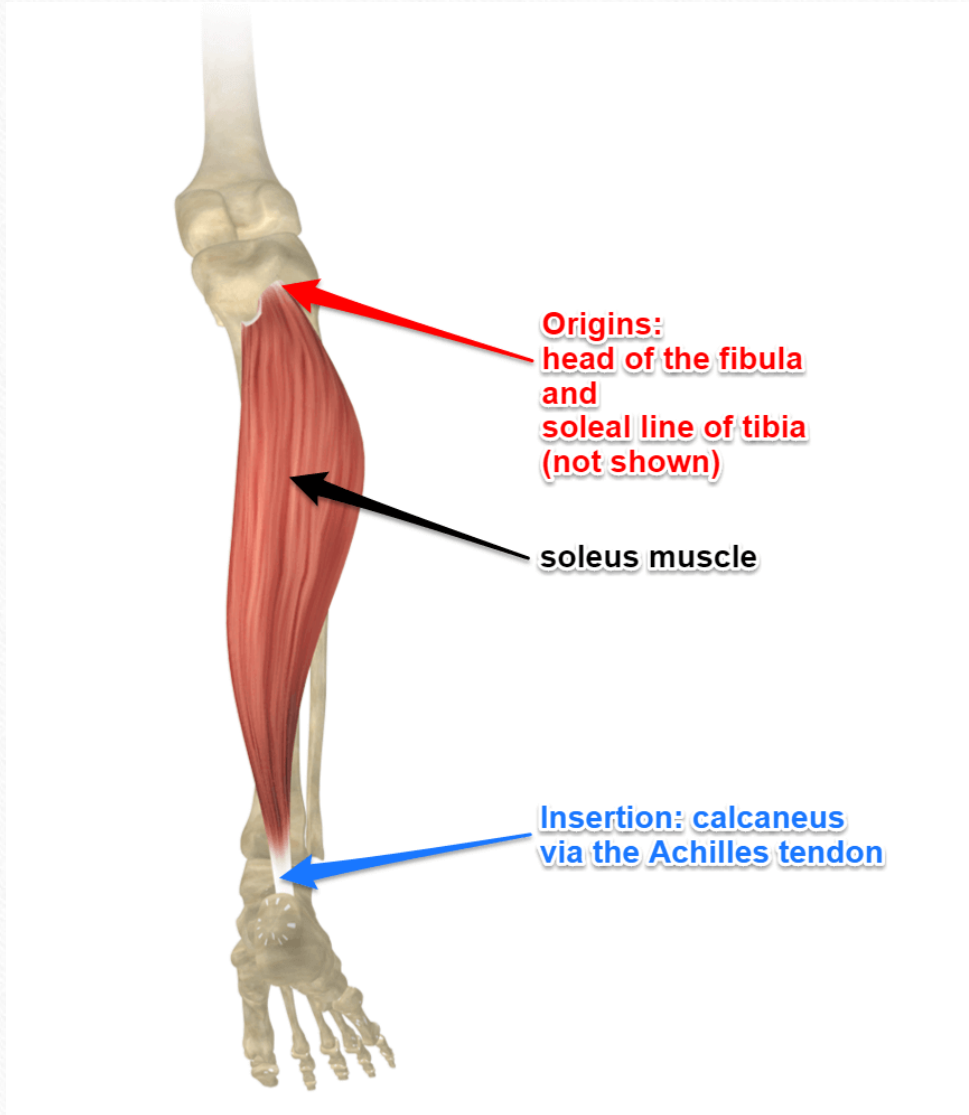
Motion of the muscle

- Foot plantarflexion
- Knee flexion

Standing calf raises

Calf raises on leg presses

16- Soleus



- Origin :
Tibia & fibula
- Insertion :
Calcaneus

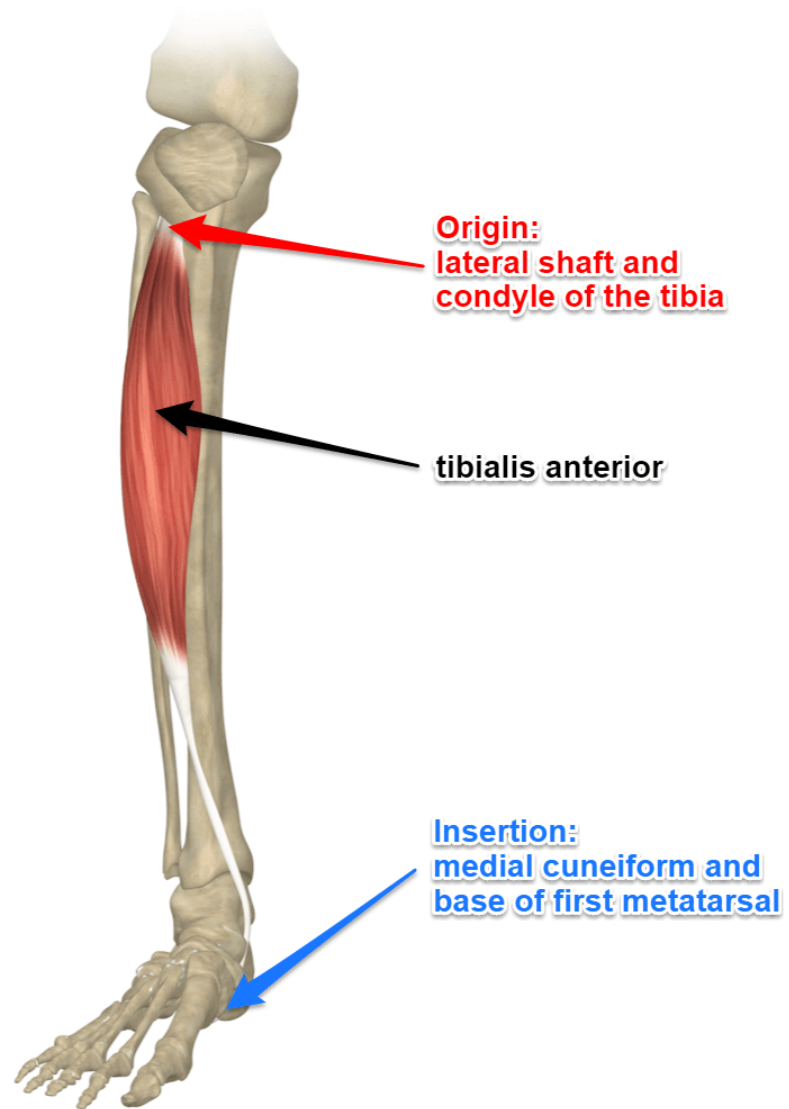
Motion of the muscle

- Foot plantarflexion

Seated calf raises machine

Standing calf raises

39- Tibialis anterior



- Origin :

Tibia

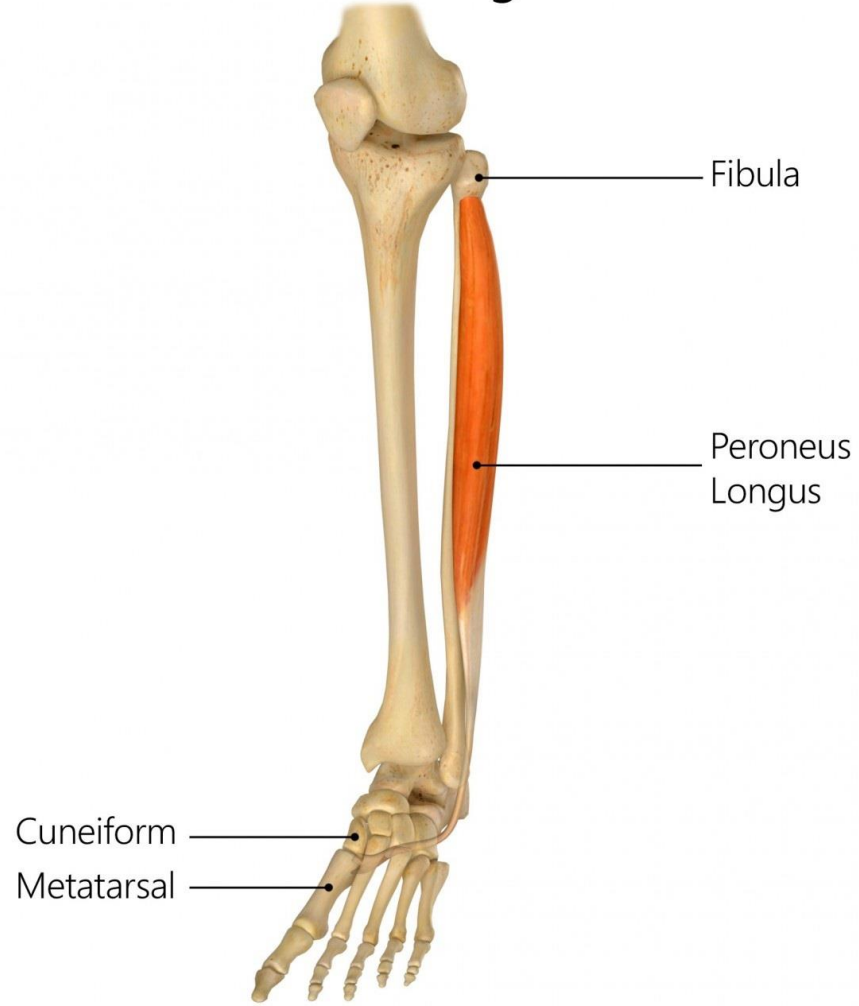
- Insertion :

Metatarsal bone & cuneiform bone

Motion of the muscle

- Foot Dorsiflexion
- Foot Inversion

Peroneus Longus



38- peroneus longus (Fibularis longus)

- **Origin :**

Fibula

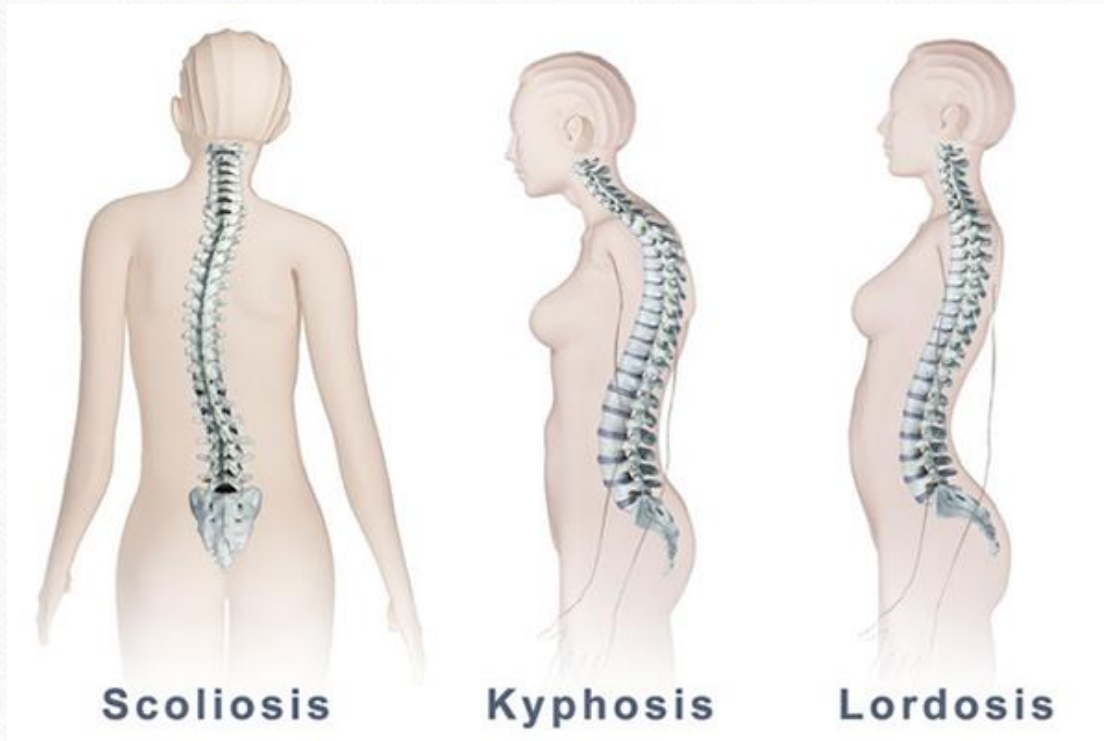
- **Insertion :**

Metatarsal & cuneiform bone

Motion of the muscle

- Foot Eversion

Posture deviations and injuries



- (kyphosis, Scoliosis, Lordosis, scapular winging, & APT
- Muscle imbalance

Kyphosis

WHAT IS KYPHOSIS



typical



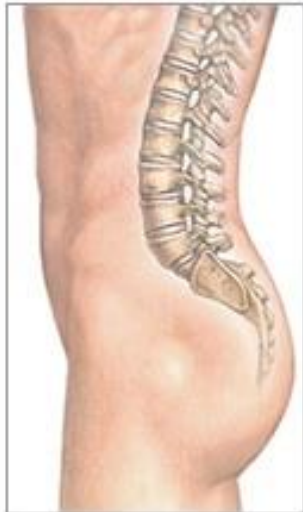
kyphosis

- Kyphosis

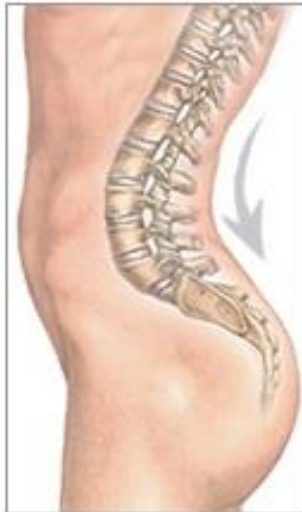
- ضعف عضلات الظهر
- مع قوة عضلات الصدر الامامية

Lordosis

Normal spine



Lordosis of the spine

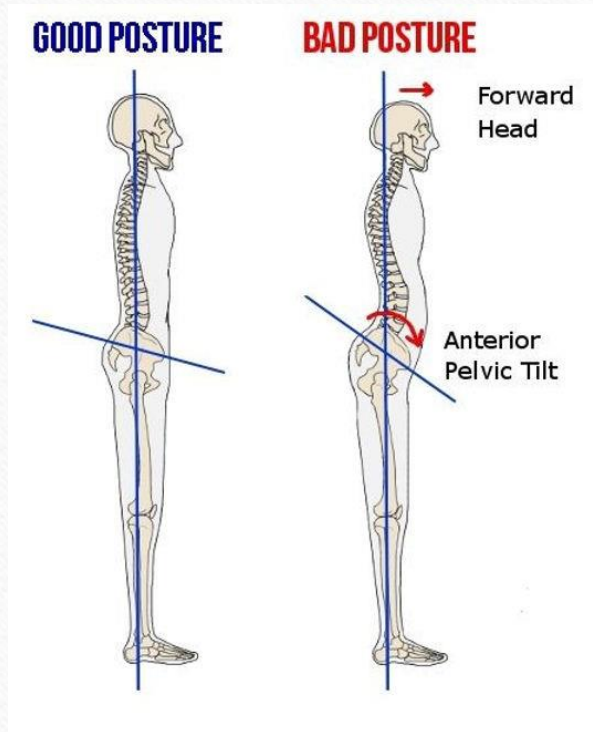


Exaggerated
lumbar
curve

- Lordosis

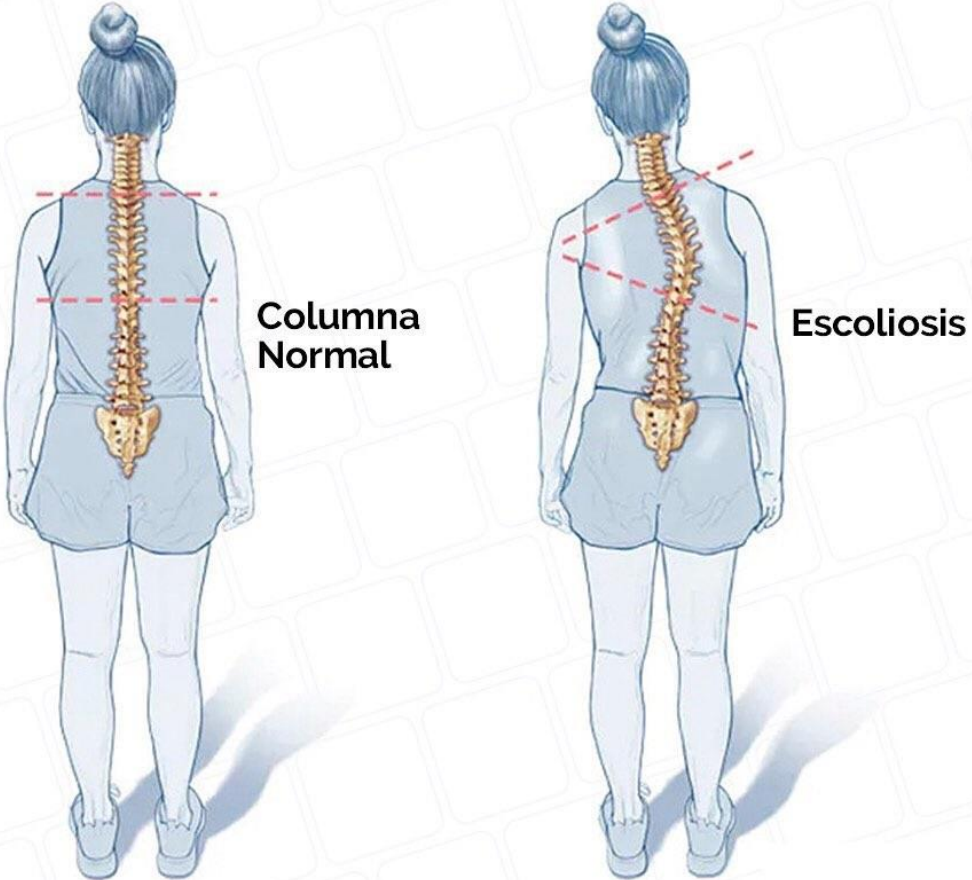
- ضعف في عضلات البطن و المؤخرة
- شد في عضلات الحوض و الظهر

Anterior pelvic tilt (ATP)



- APT is a common cause of lordosis
- ضعف فى عضلات البطن و المؤخرة
- شد فى عضلات الحوض و الظهر
- انقلاب الحوض

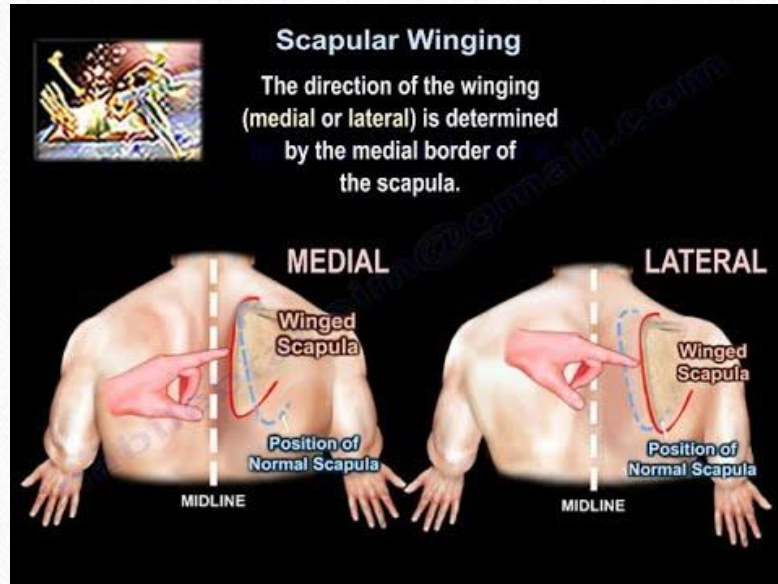
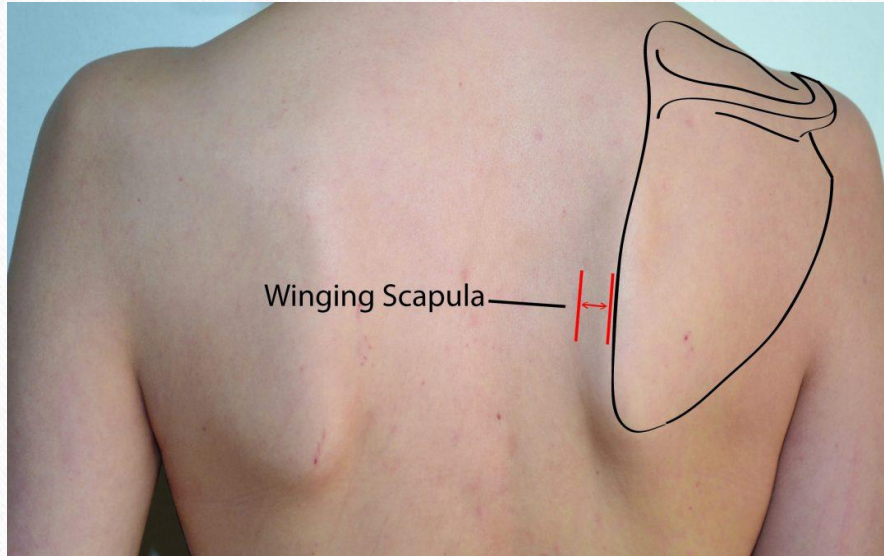
TRATAMIENTOS ESCOLIOSIS



Scoliosis

- Scoliosis
 - قد تكون مشاكل وراثية
- ممكن ان تكون بسبب الجلوس الخاطى
- ضعف فى عضلات اليمين عن الشمال

Scapular winging



- Scapular winging
 - بسبب ضعف فى العضلة المنشارية
 - Serratus anterior muscle
 - و مع عمل العضلات الاخرى يحدث التجنح
 - Trapezius & Rhomboid

Practicing (GYM)



Case study (Design a workout plan)

Target all the muscle groups

- 26 years old man
- 84 Kg, 180 Cm
- 10 years at gym not regular (Advanced fitness level)
- Inbody
- Photos

Thank you ♥